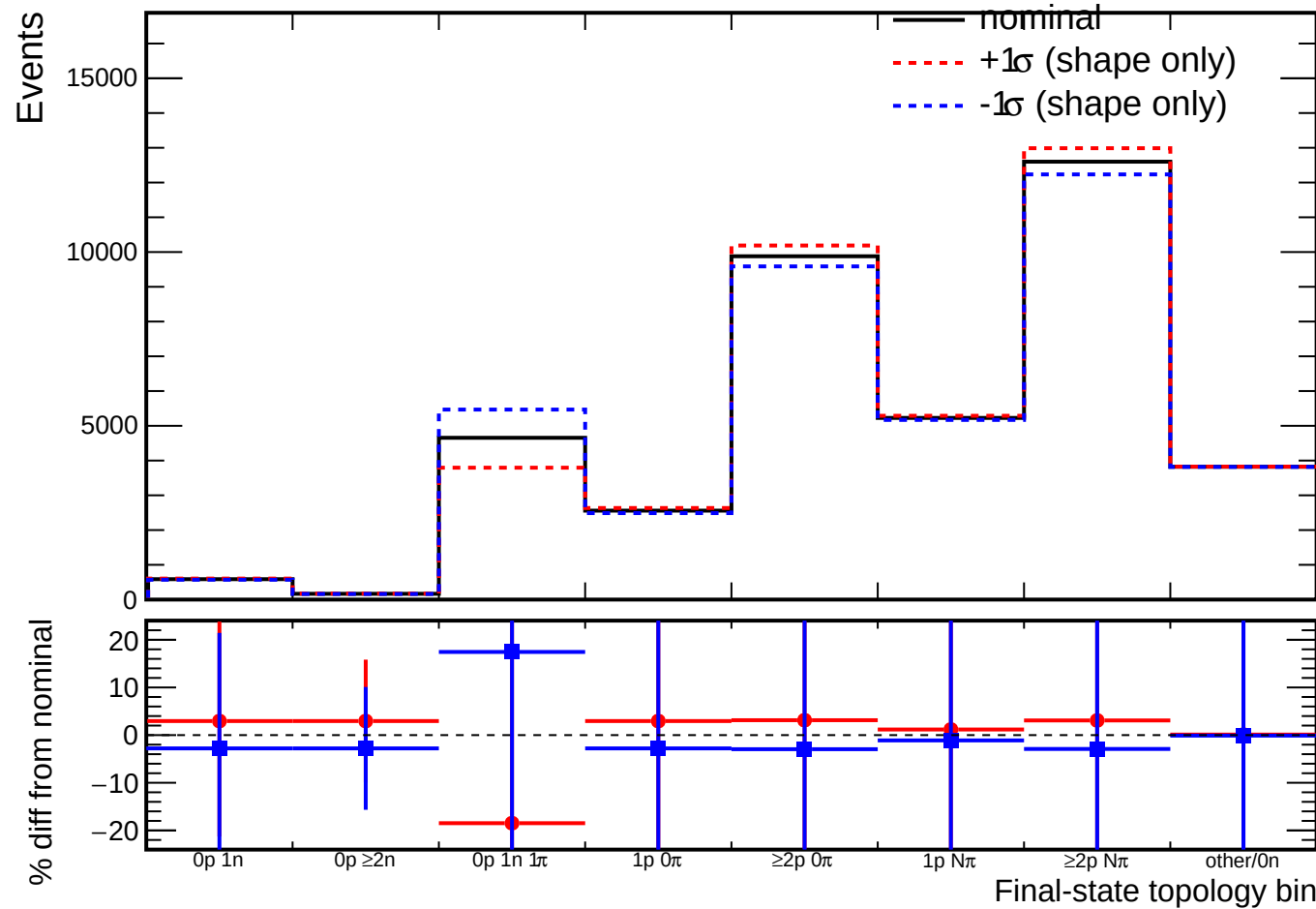
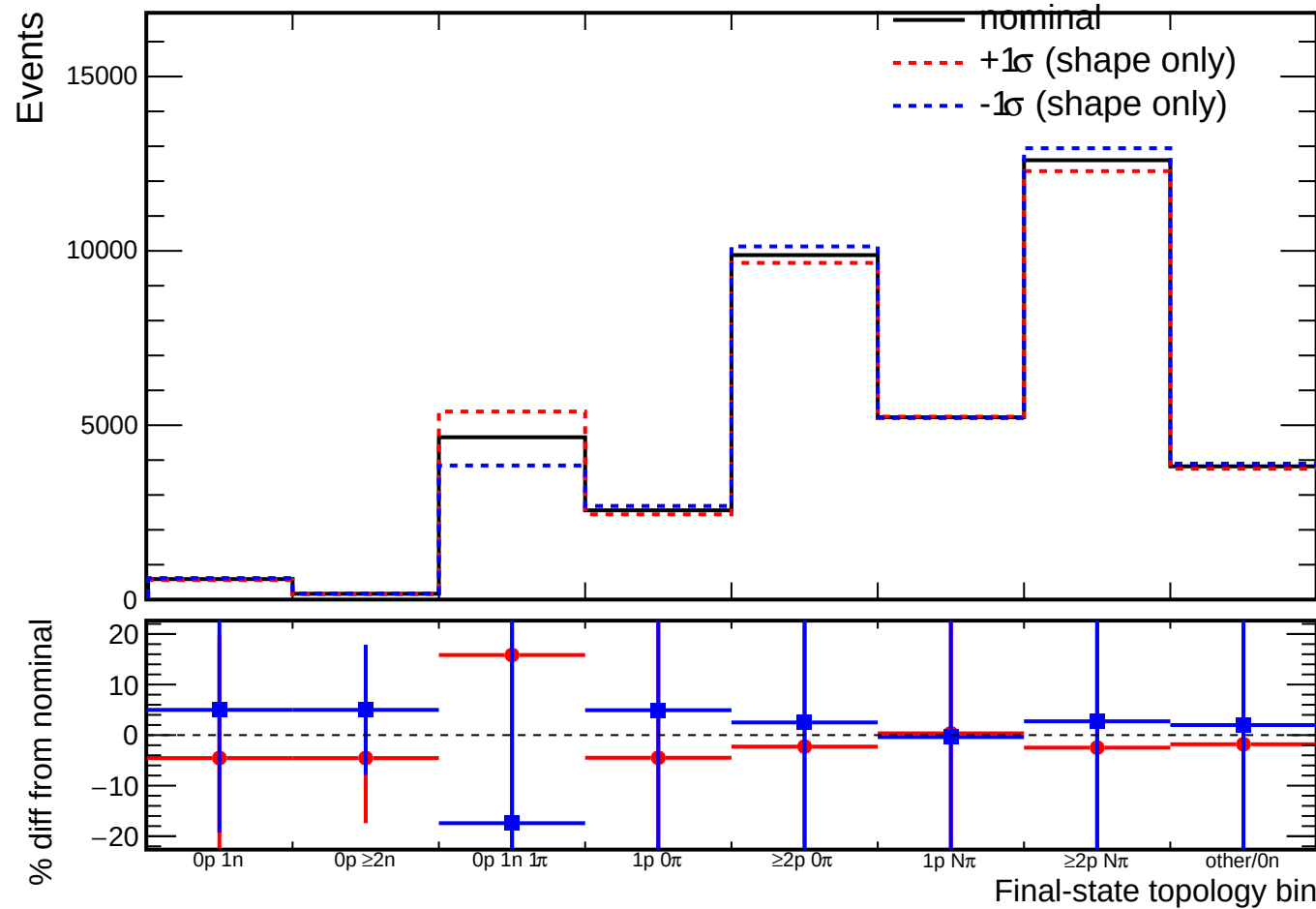


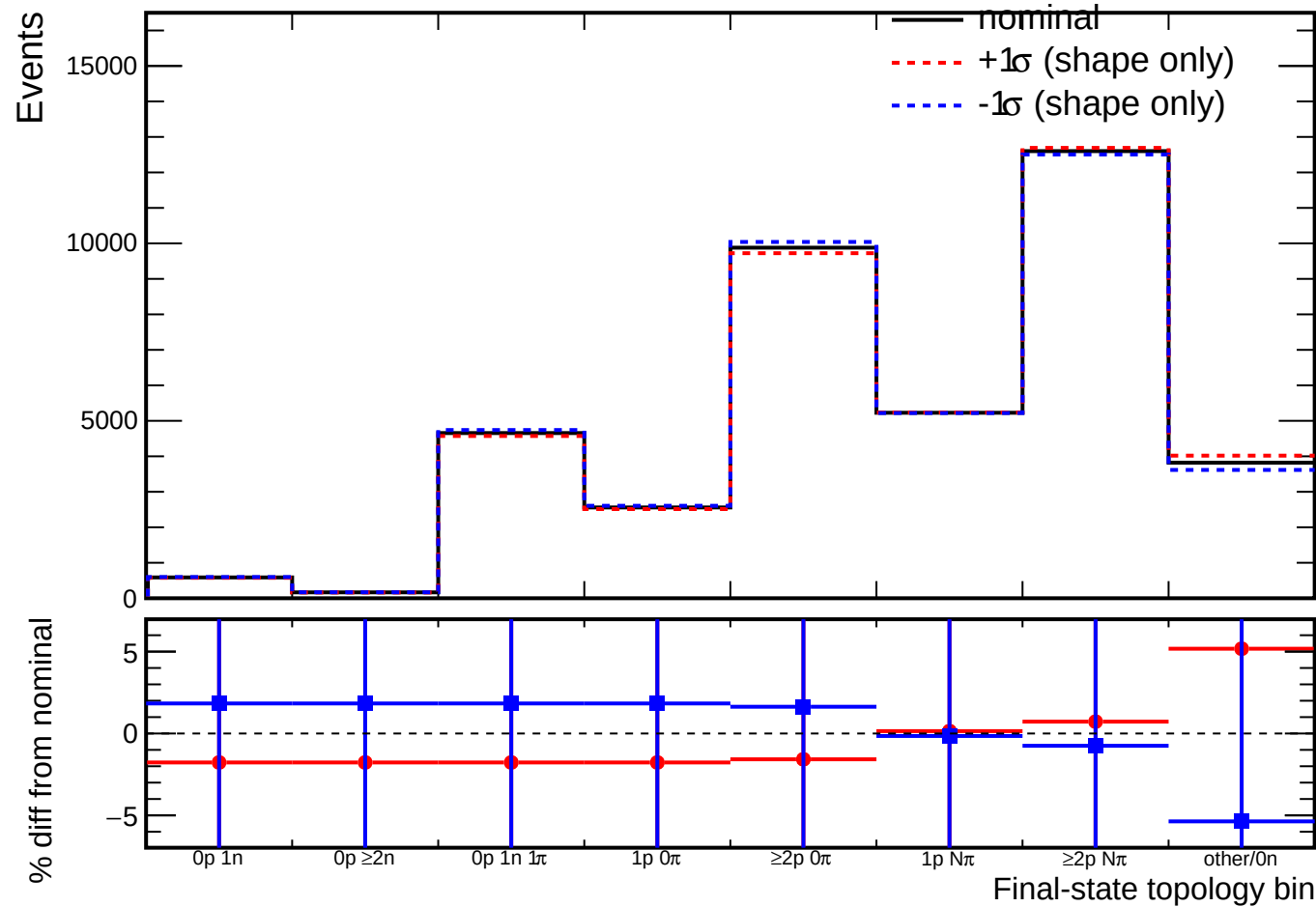
FHC [DIS]: DISNuHadronQ1Syst (shape only) $\chi^2_{\text{shape},+1\sigma}=184.18$ $\chi^2_{\text{shape},-1\sigma}=164.23$ $\Delta\langle x \rangle_{+1\sigma}=+0.0610$ $\Delta\langle x \rangle_{-1\sigma}=-0.0576$
 $\chi^2_{\text{raw},+1\sigma}=206.22$ $\chi^2_{\text{raw},-1\sigma}=206.22$ rate: -2.87% / +2.87% $\Delta\sigma^2$: -0.1230 / +0.1093



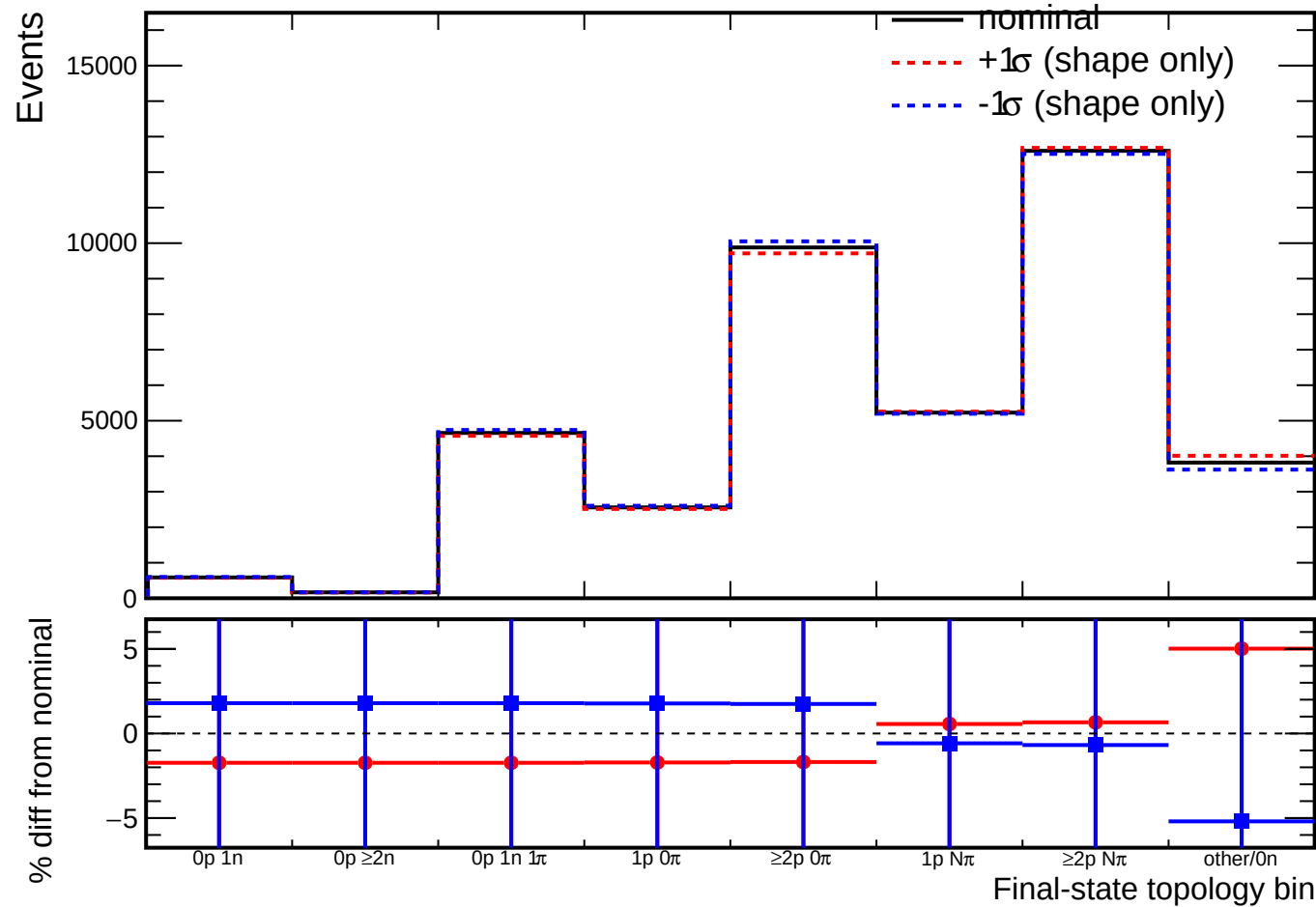
FHC [DIS]: DISvnCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=137.70$ $\chi^2_{\text{shape},-1\sigma}=166.55$ $\Delta\langle x \rangle_{+1\sigma}=-0.0518$ $\Delta\langle x \rangle_{-1\sigma}=+0.0570$
 $\chi^2_{\text{raw},+1\sigma}=240.27$ $\chi^2_{\text{raw},-1\sigma}=240.27$ rate: +4.75% / -4.75% $\Delta\sigma^2$: +0.0809 / -0.0952



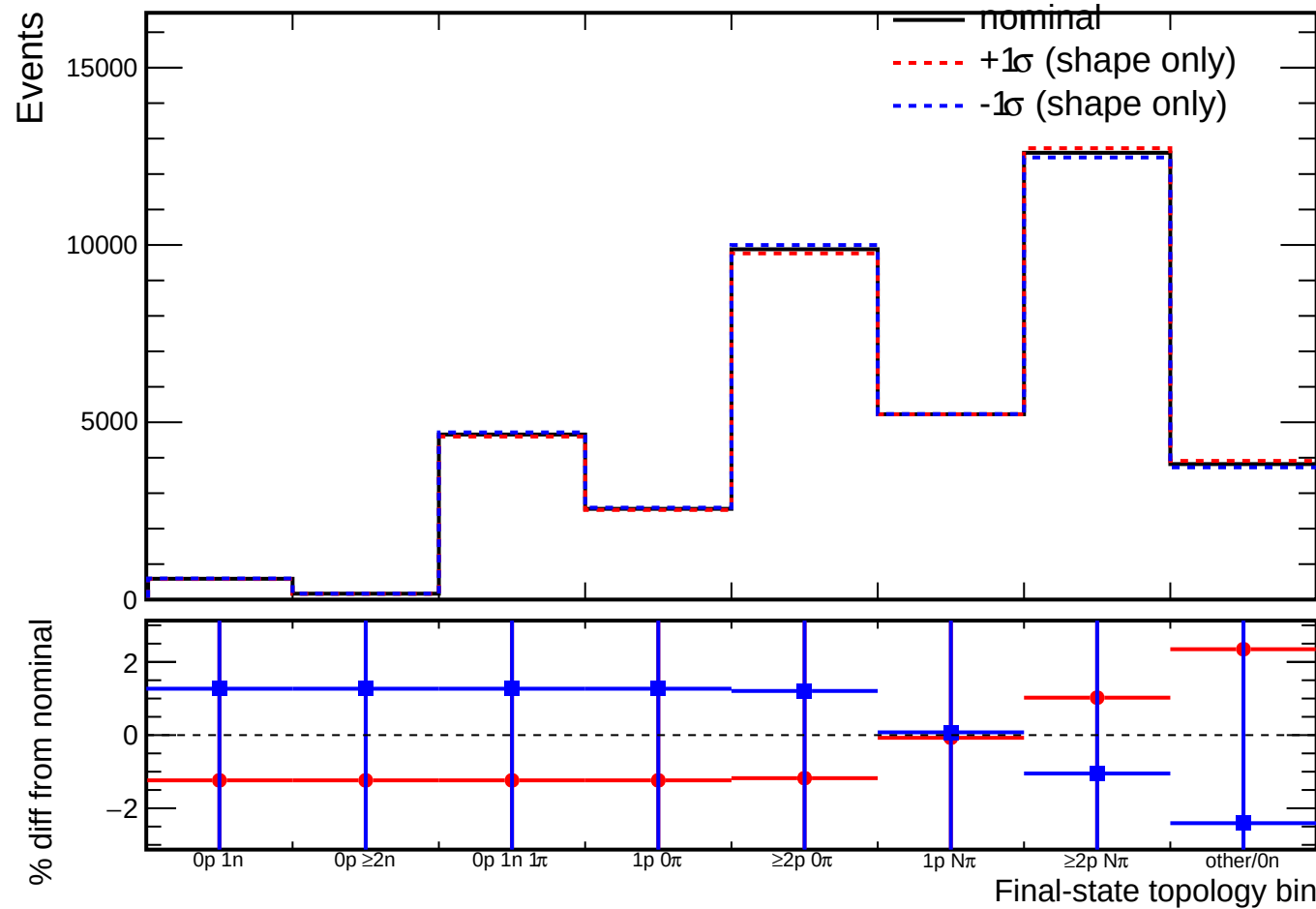
FHC [DIS]: DISvpCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=15.86$ $\chi^2_{\text{shape},-1\sigma}=17.04$ $\Delta\langle x \rangle_{+1\sigma}=+0.0264$ $\Delta\langle x \rangle_{-1\sigma}=-0.0274$
 $\chi^2_{\text{raw},+1\sigma}=29.24$ $\chi^2_{\text{raw},-1\sigma}=29.24$ rate: +1.80% / -1.80% $\Delta\sigma^2$: +0.0031 / -0.0047



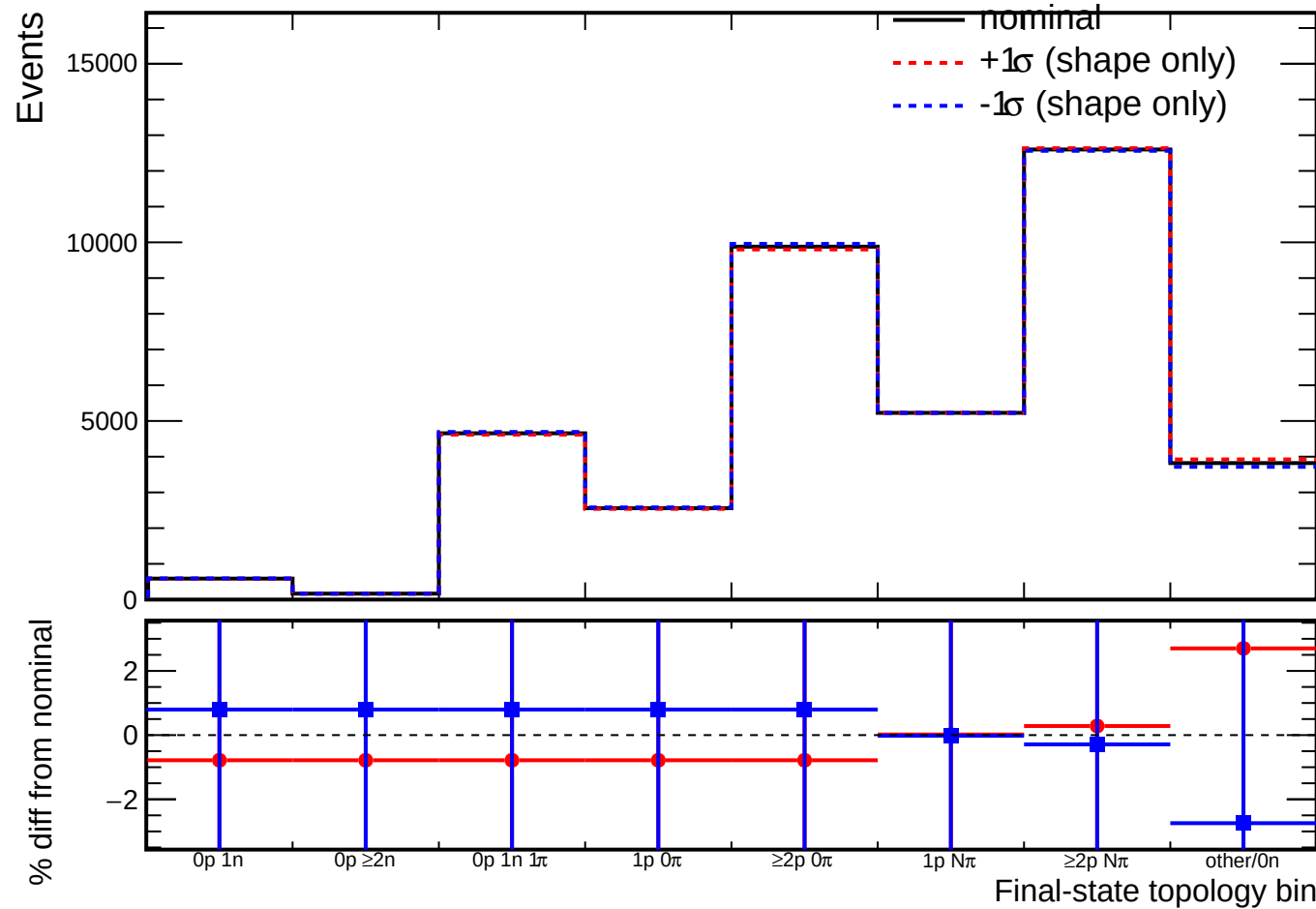
FHC [DIS]: DISvnCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=15.51$ $\chi^2_{\text{shape},-1\sigma}=16.64$ $\Delta\langle x \rangle_{+1\sigma}=+0.0260$ $\Delta\langle x \rangle_{-1\sigma}=-0.0269$
 $\chi^2_{\text{raw},+1\sigma}=28.34$ $\chi^2_{\text{raw},-1\sigma}=28.34$ rate: +1.76% / -1.76% $\Delta\sigma^2$: +0.0024 / -0.0039



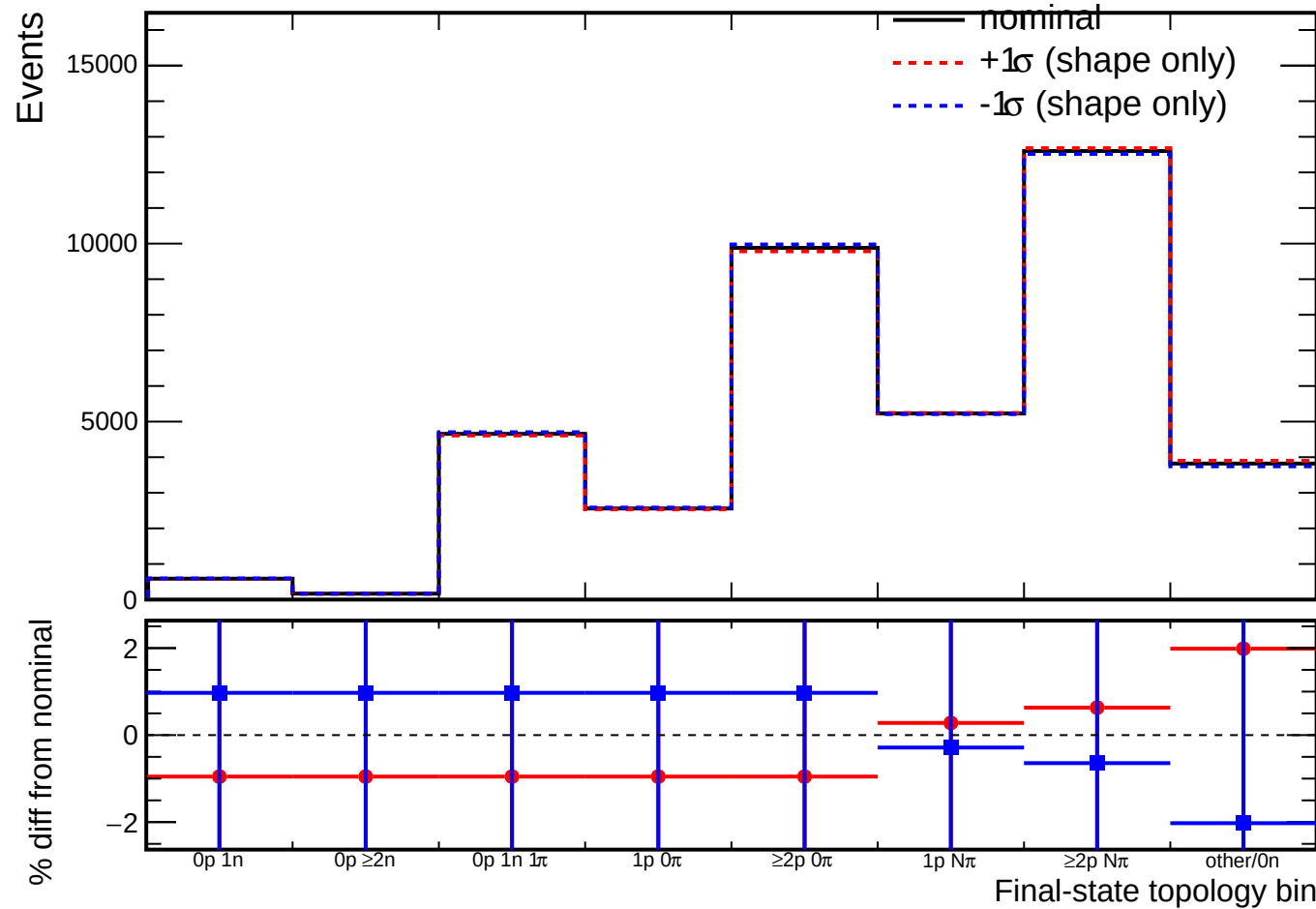
FHC [DIS]: DISvpCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=6.02$ $\chi^2_{\text{shape},-1\sigma}=6.32$ $\Delta\langle x \rangle_{+1\sigma}=+0.0179$ $\Delta\langle x \rangle_{-1\sigma}=-0.0183$
 $\chi^2_{\text{raw},+1\sigma}=12.36$ $\chi^2_{\text{raw},-1\sigma}=12.36$ rate: +1.25% / -1.25% $\Delta\sigma^2$: -0.0015 / +0.0009



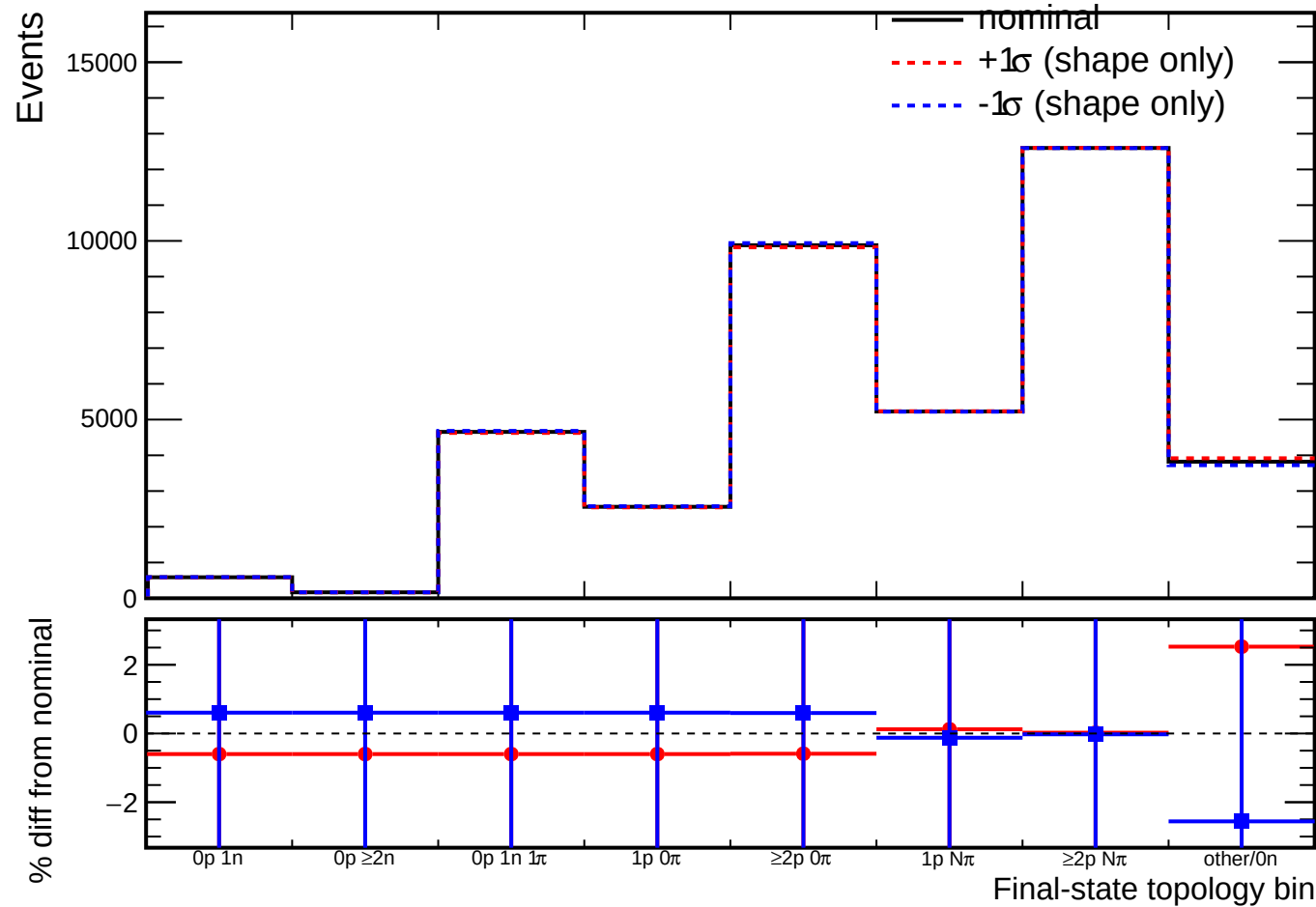
FHC [DIS]: DISvpNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=3.98$ $\chi^2_{\text{shape},-1\sigma}=4.11$ $\Delta\langle x \rangle_{+1\sigma}=+0.0126$ $\Delta\langle x \rangle_{-1\sigma}=-0.0128$
 $\chi^2_{\text{raw},+1\sigma}=6.51$ $\chi^2_{\text{raw},-1\sigma}=6.51$ rate: +0.79% / -0.79% $\Delta\sigma^2$: +0.0033 / -0.0037



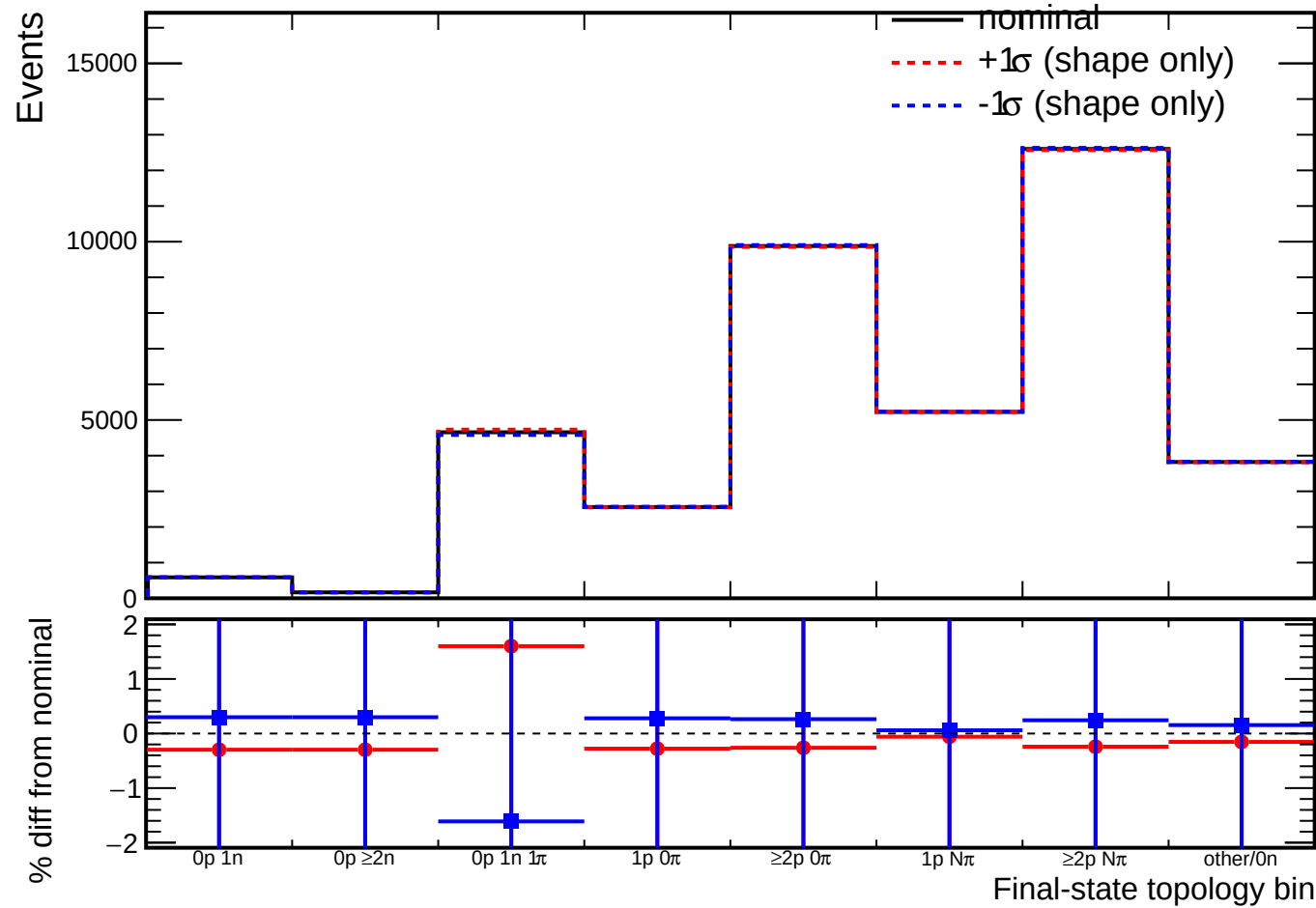
FHC [DIS]: DISvNCC3pi_2020 (shape only) $\chi^2_{\text{shape}, +1\sigma} = 3.67$ $\chi^2_{\text{shape}, -1\sigma} = 3.82$ $\Delta\langle x \rangle_{+1\sigma} = +0.0137$ $\Delta\langle x \rangle_{-1\sigma} = -0.0140$
 $\chi^2_{\text{raw}, +1\sigma} = 7.41$ $\chi^2_{\text{raw}, -1\sigma} = 7.41$ rate: +0.96% / -0.96% $\Delta\sigma^2$: -0.0011 / +0.0007



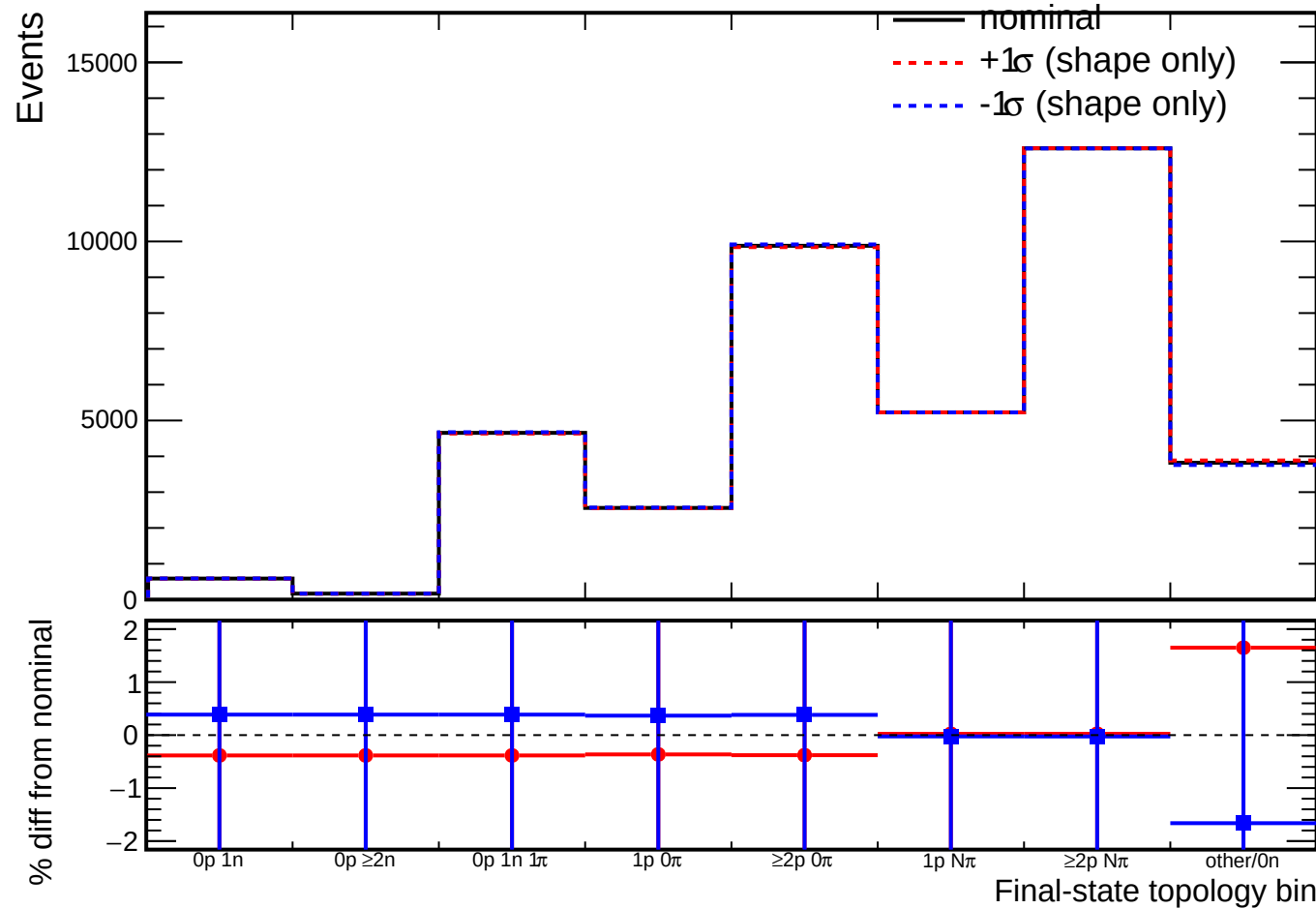
FHC [DIS]: DISvpNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=3.08$ $\chi^2_{\text{shape},-1\sigma}=3.15$ $\Delta\langle x\rangle_{+1\sigma}=+0.0099$ $\Delta\langle x\rangle_{-1\sigma}=-0.0100$
 $\chi^2_{\text{raw},+1\sigma}=4.55$ $\chi^2_{\text{raw},-1\sigma}=4.55$ rate: +0.60% / -0.60% $\Delta\sigma^2$: +0.0039 / -0.0042



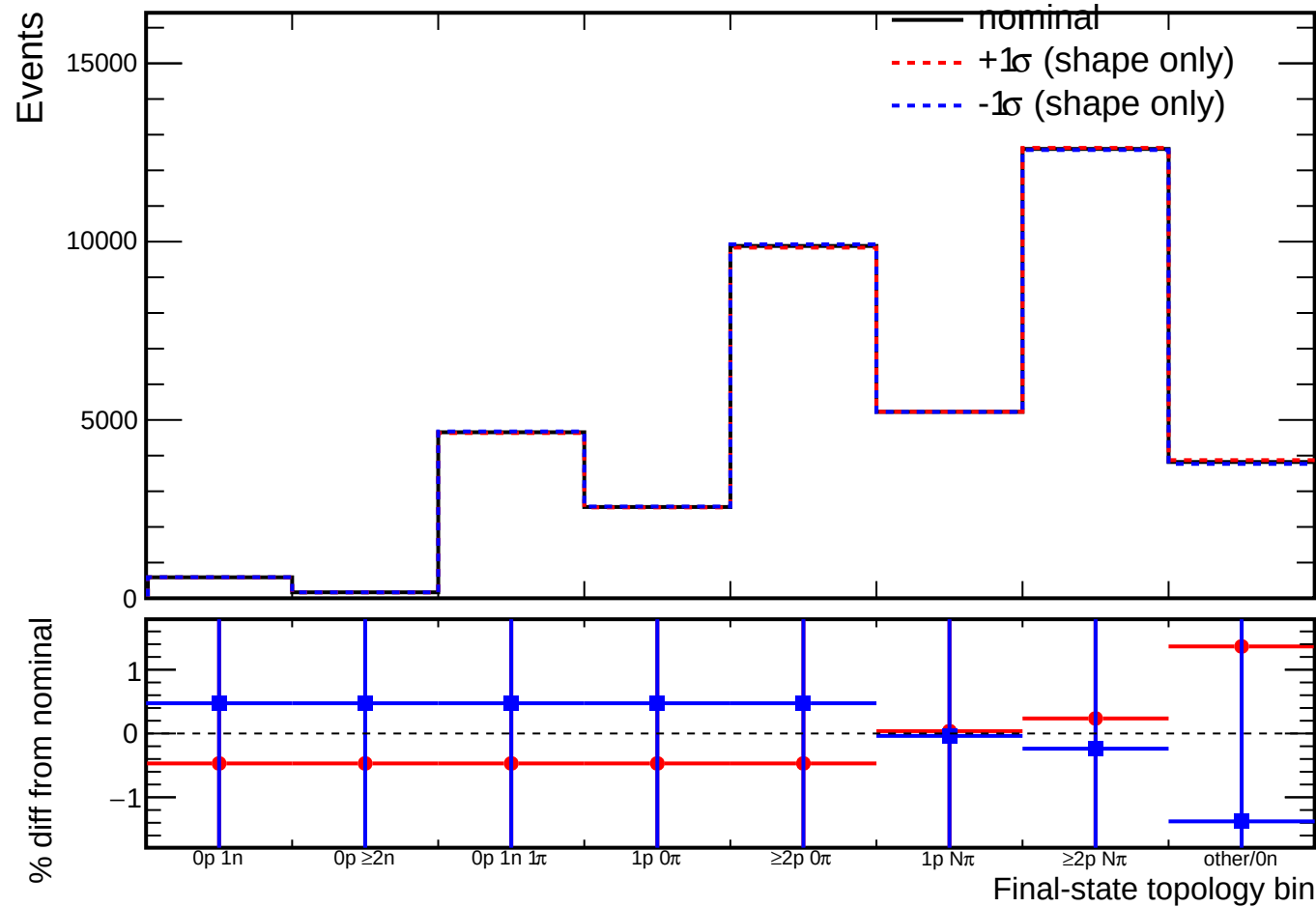
FHC [DIS]: DISvpNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=1.37$ $\chi^2_{\text{shape},-1\sigma}=1.39$ $\Delta\langle x\rangle_{+1\sigma}=-0.0054$ $\Delta\langle x\rangle_{-1\sigma}=+0.0055$
 $\chi^2_{\text{raw},+1\sigma}=1.73$ $\chi^2_{\text{raw},-1\sigma}=1.73$ rate: +0.30% / -0.30% $\Delta\sigma^2$: +0.0095 / -0.0096



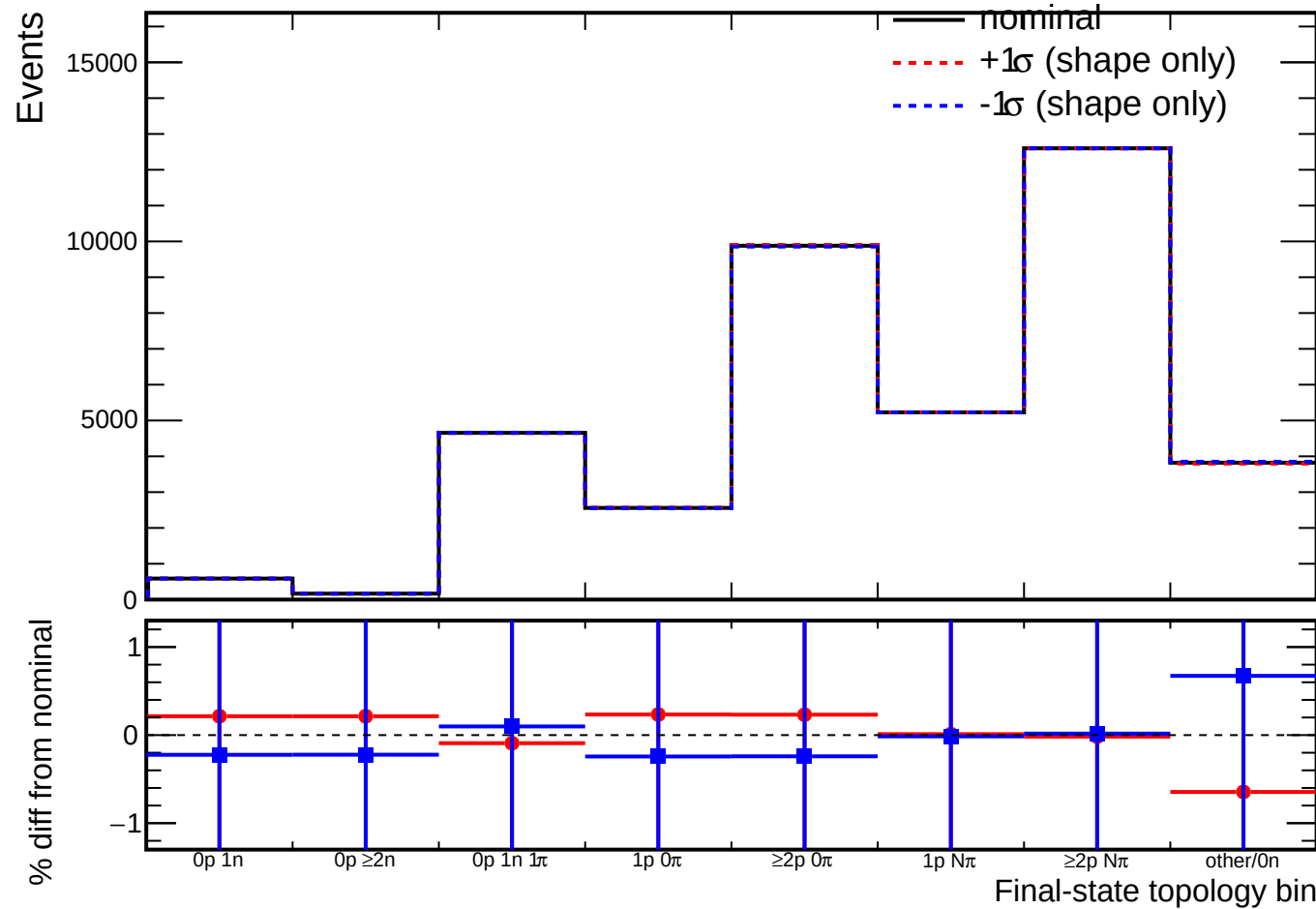
FHC [DIS]: DISvNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=1.30$ $\chi^2_{\text{shape},-1\sigma}=1.32$ $\Delta\langle x \rangle_{+1\sigma}=+0.0064$ $\Delta\langle x \rangle_{-1\sigma}=-0.0064$
 $\chi^2_{\text{raw},+1\sigma}=1.89$ $\chi^2_{\text{raw},-1\sigma}=1.89$ rate: +0.39% / -0.39% $\Delta\sigma^2$: +0.0028 / -0.0029



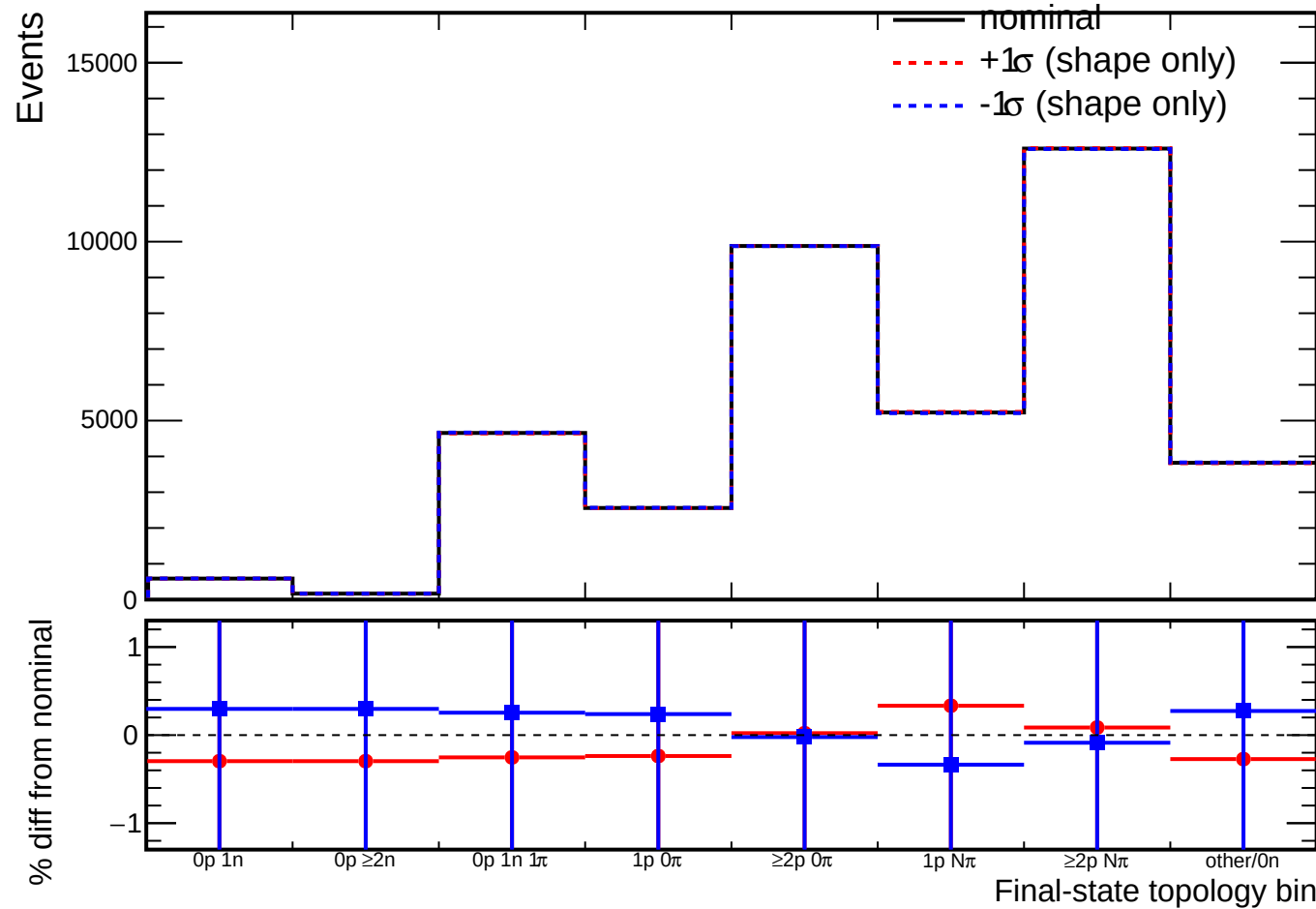
FHC [DIS]: DISvNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=1.18$ $\chi^2_{\text{shape},-1\sigma}=1.20$ $\Delta\langle x\rangle_{+1\sigma}=+0.0073$ $\Delta\langle x\rangle_{-1\sigma}=-0.0073$
 $\chi^2_{\text{raw},+1\sigma}=2.07$ $\chi^2_{\text{raw},-1\sigma}=2.07$ rate: +0.47% / -0.47% $\Delta\sigma^2$: +0.0011 / -0.0012



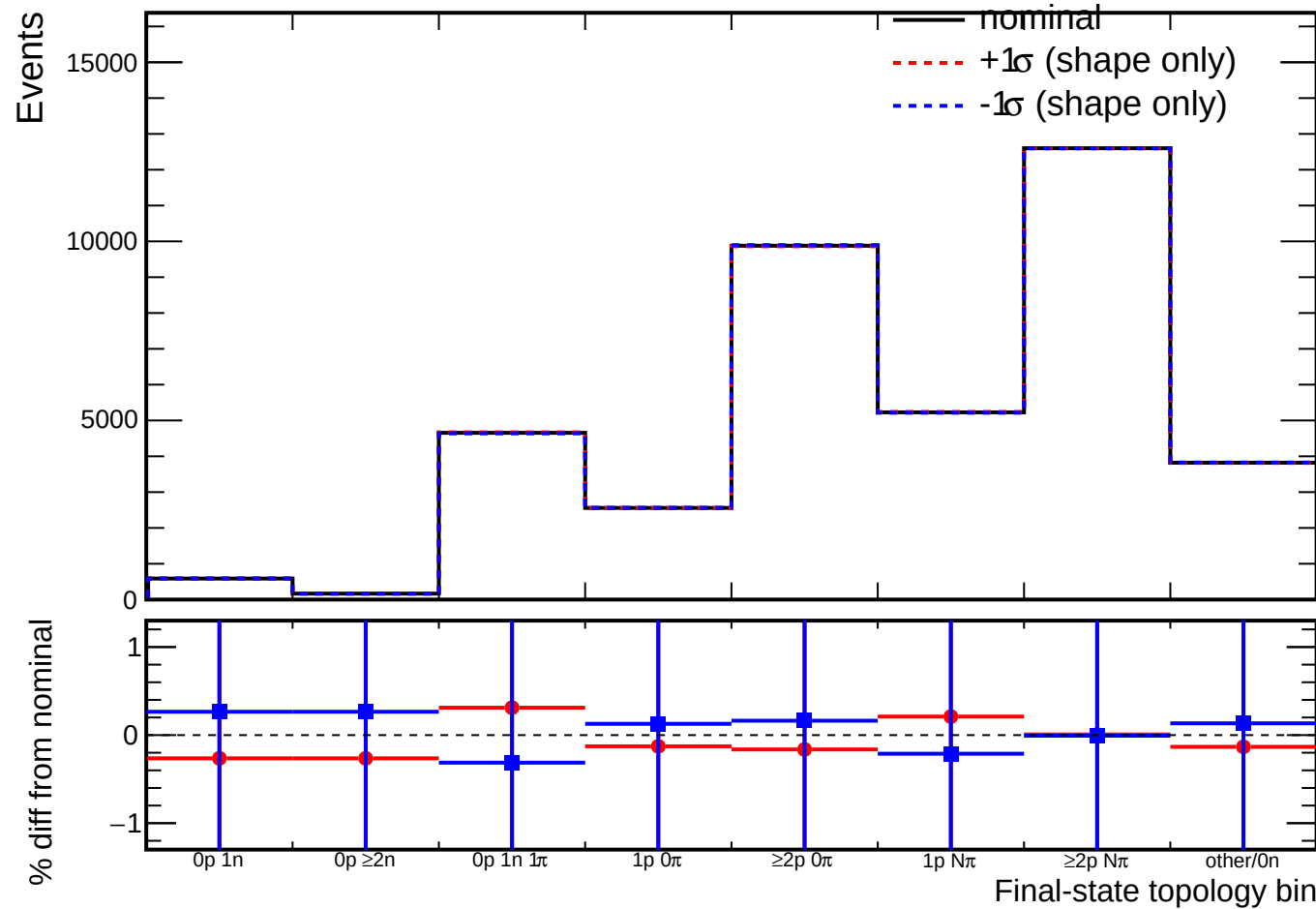
FHC [DIS]: BhtBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.23$ $\chi^2_{\text{shape},-1\sigma}=0.25$ $\Delta\langle x\rangle_{+1\sigma}=-0.0021$ $\Delta\langle x\rangle_{-1\sigma}=+0.0021$
 $\chi^2_{\text{raw},+1\sigma}=0.47$ $\chi^2_{\text{raw},-1\sigma}=0.51$ rate: -0.25% / +0.26% $\Delta\sigma^2$: -0.0027 / +0.0028



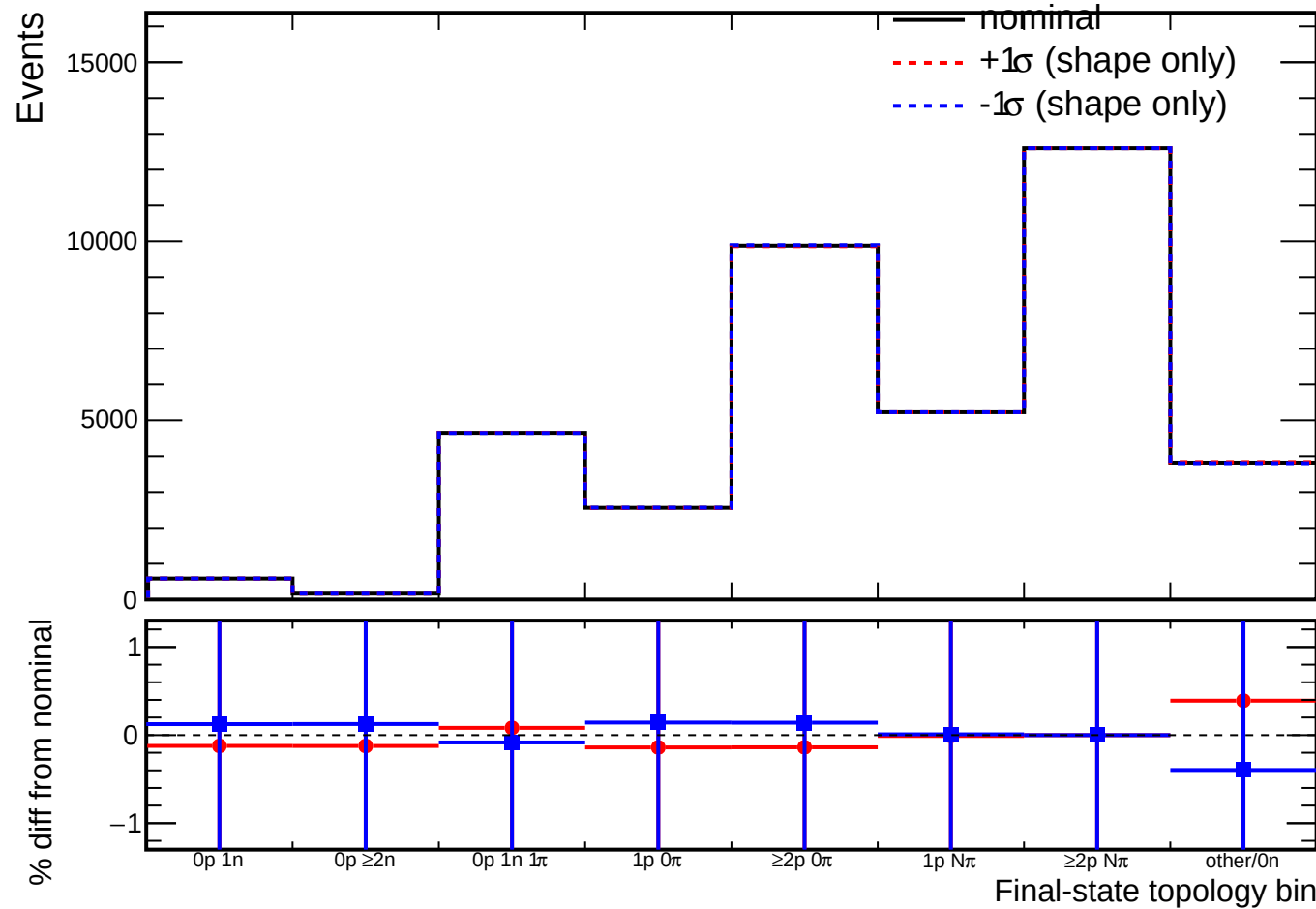
FHC [DIS]: DISvpCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.15$ $\chi^2_{\text{shape},-1\sigma}=0.15$ $\Delta\langle x \rangle_{+1\sigma}=+0.0012$ $\Delta\langle x \rangle_{-1\sigma}=-0.0012$
 $\chi^2_{\text{raw},+1\sigma}=0.50$ $\chi^2_{\text{raw},-1\sigma}=0.50$ rate: +0.30% / -0.30% $\Delta\sigma^2$: -0.0046 / +0.0046



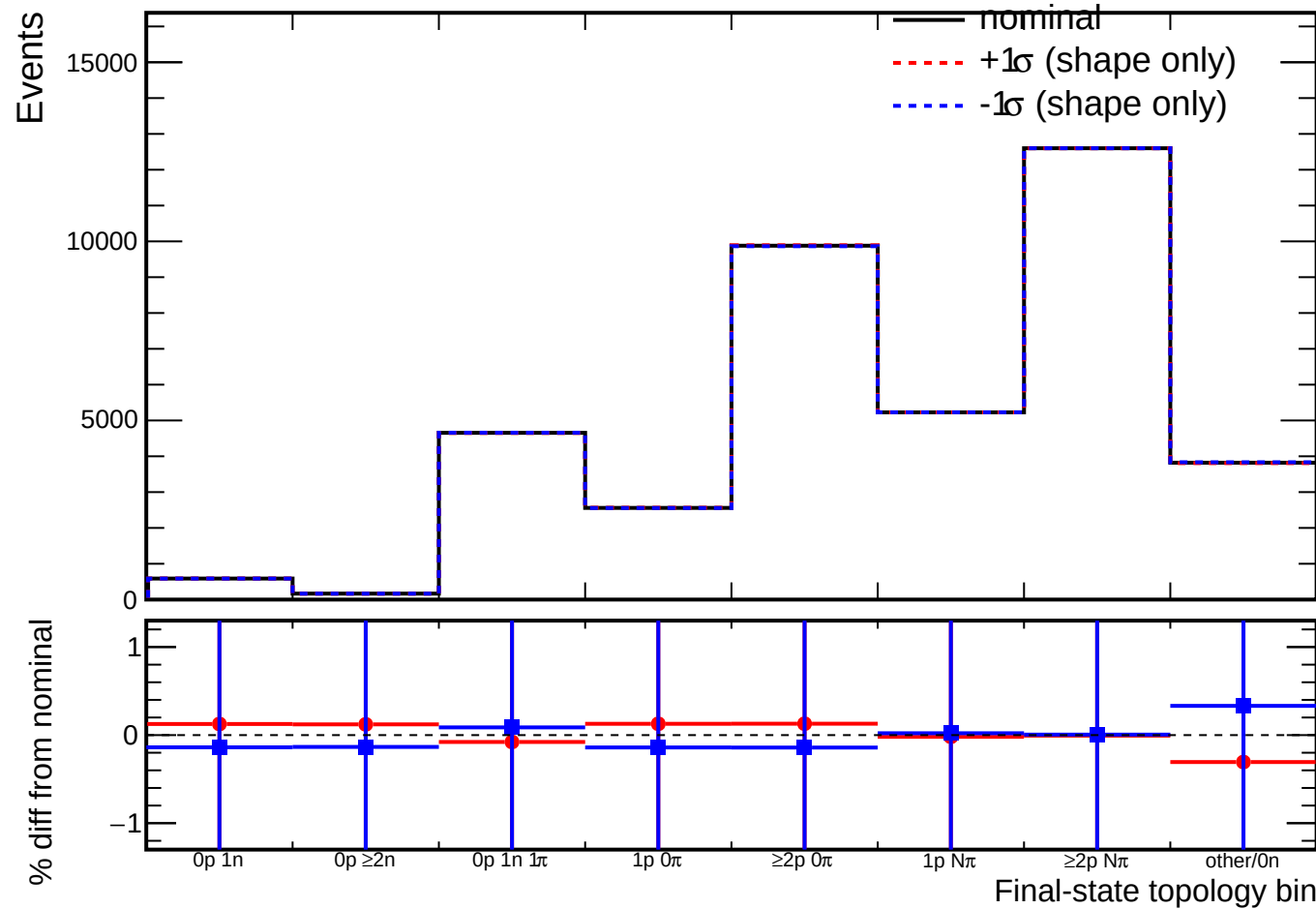
FHC [DIS]: DISvnnC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.11$ $\chi^2_{\text{shape},-1\sigma}=0.11$ $\Delta\langle x\rangle_{+1\sigma}=-0.0005$ $\Delta\langle x\rangle_{-1\sigma}=+0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.39$ $\chi^2_{\text{raw},-1\sigma}=0.39$ rate: +0.26% / -0.26% $\Delta\sigma^2$: +0.0006 / -0.0006



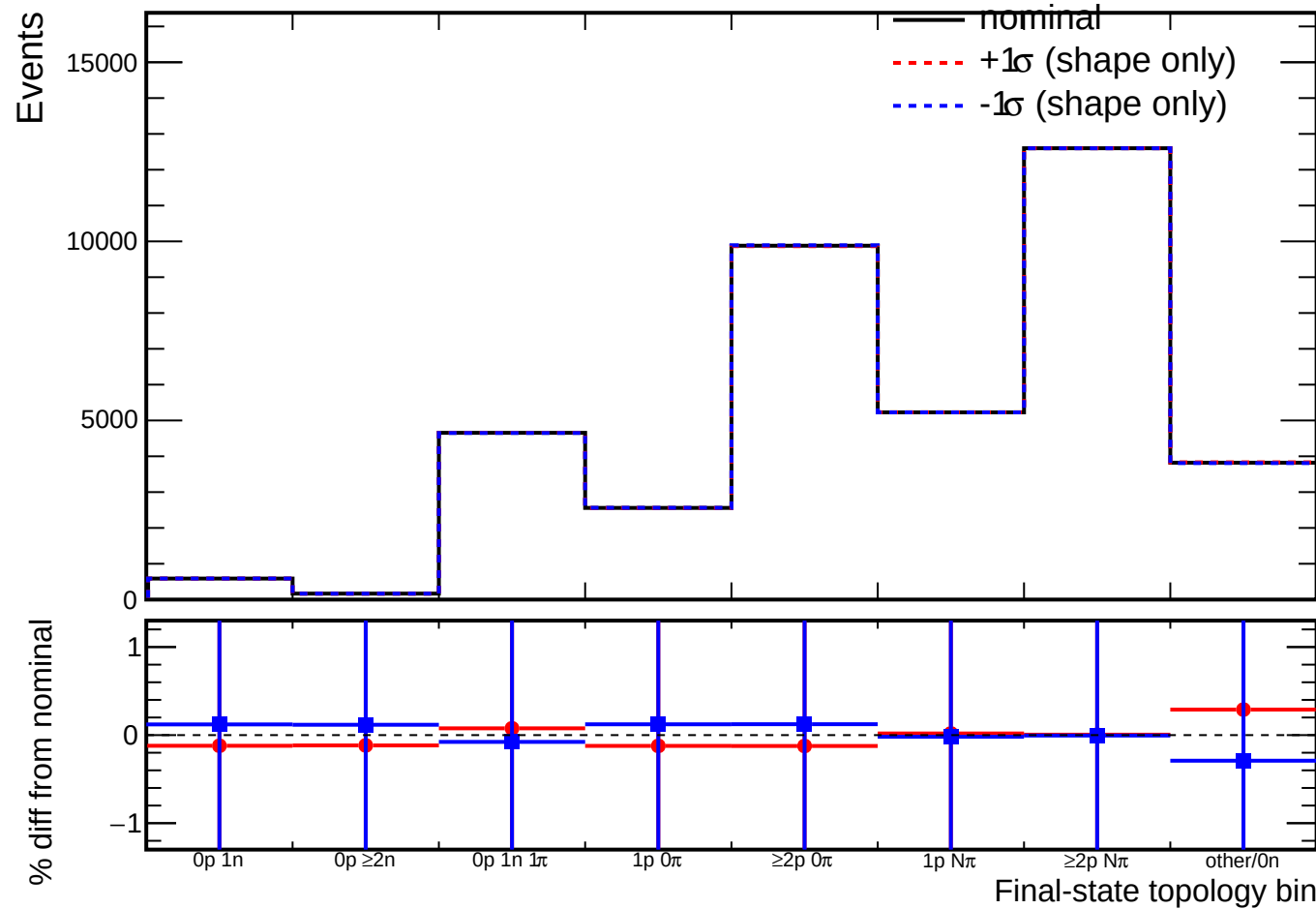
FHC [DIS]: AhtBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.09$ $\chi^2_{\text{shape},-1\sigma}=0.09$ $\Delta\langle x\rangle_{+1\sigma}=+0.0011$ $\Delta\langle x\rangle_{-1\sigma}=-0.0011$
 $\chi^2_{\text{raw},+1\sigma}=0.17$ $\chi^2_{\text{raw},-1\sigma}=0.18$ rate: +0.15% / -0.15% $\Delta\sigma^2$: +0.0018 / -0.0018



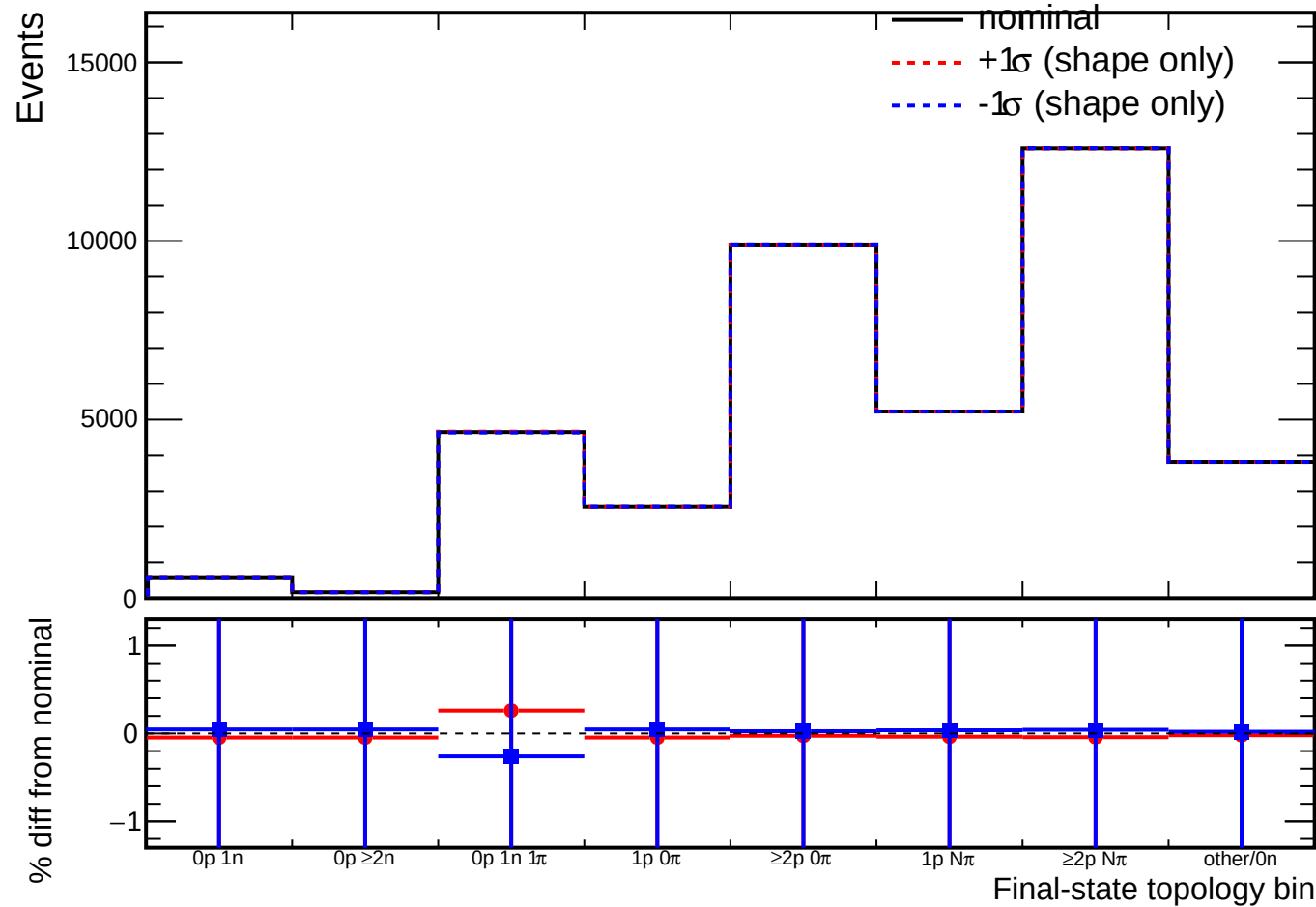
FHC [DIS]: CV1uBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.06$ $\chi^2_{\text{shape},-1\sigma}=0.07$ $\Delta\langle x\rangle_{+1\sigma}=-0.0009$ $\Delta\langle x\rangle_{-1\sigma}=+0.0010$
 $\chi^2_{\text{raw},+1\sigma}=0.14$ $\chi^2_{\text{raw},-1\sigma}=0.16$ rate: -0.14% / +0.15% $\Delta\sigma^2$: -0.0014 / +0.0015



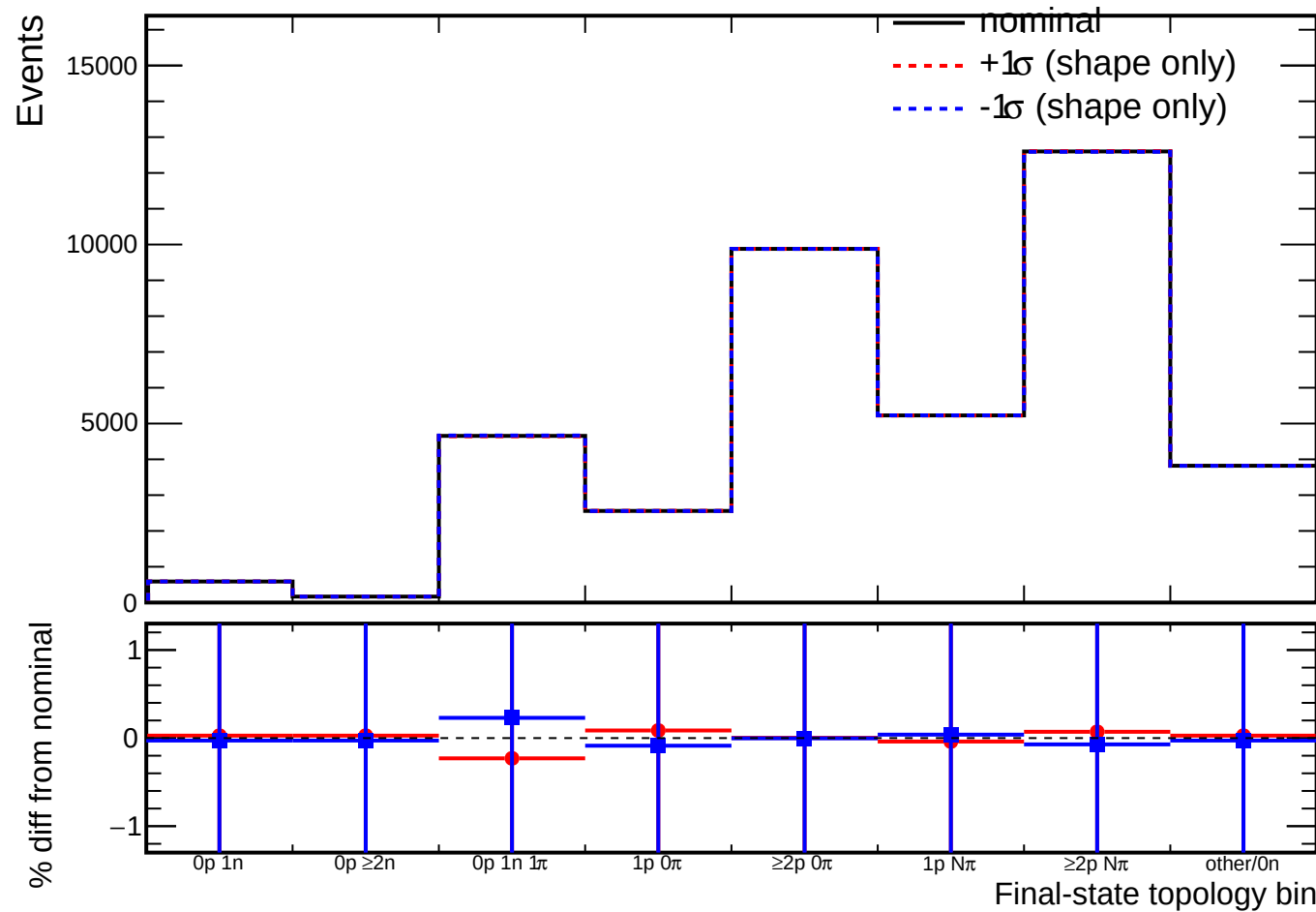
FHC [DIS]: CV2uBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.06$ $\chi^2_{\text{shape},-1\sigma}=0.06$ $\Delta\langle x \rangle_{+1\sigma}=+0.0009$ $\Delta\langle x \rangle_{-1\sigma}=-0.0009$
 $\chi^2_{\text{raw},+1\sigma}=0.12$ $\chi^2_{\text{raw},-1\sigma}=0.12$ rate: +0.13% / -0.13% $\Delta\sigma^2$: +0.0013 / -0.0014



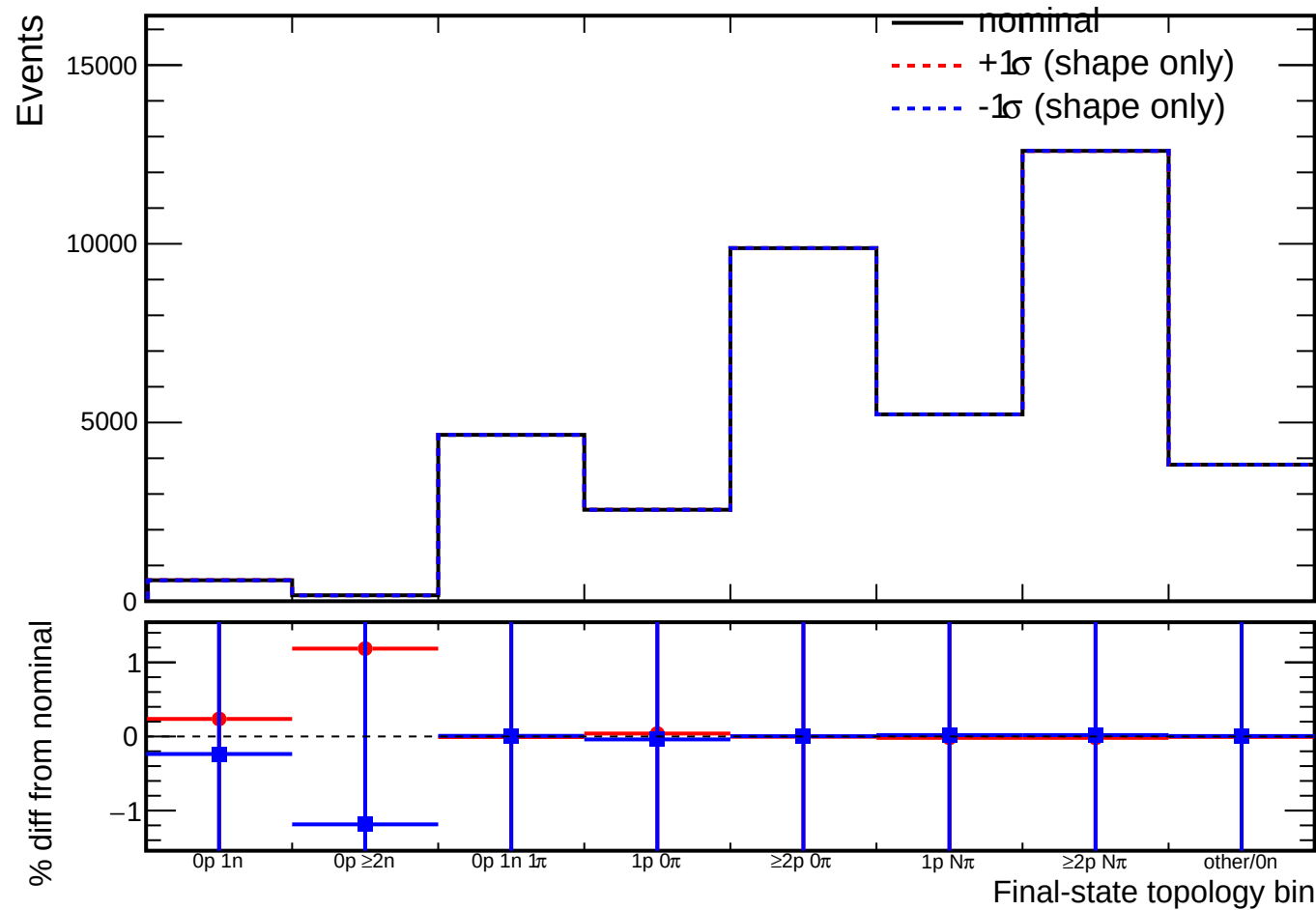
FHC [DIS]: DISvbarpNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.04$ $\chi^2_{\text{shape},-1\sigma}=0.04$ $\Delta\langle x\rangle_{+1\sigma}=-0.0009$ $\Delta\langle x\rangle_{-1\sigma}=+0.0009$
 $\chi^2_{\text{raw},+1\sigma}=0.04$ $\chi^2_{\text{raw},-1\sigma}=0.04$ rate: +0.05% / -0.05% $\Delta\sigma^2$: +0.0016 / -0.0016



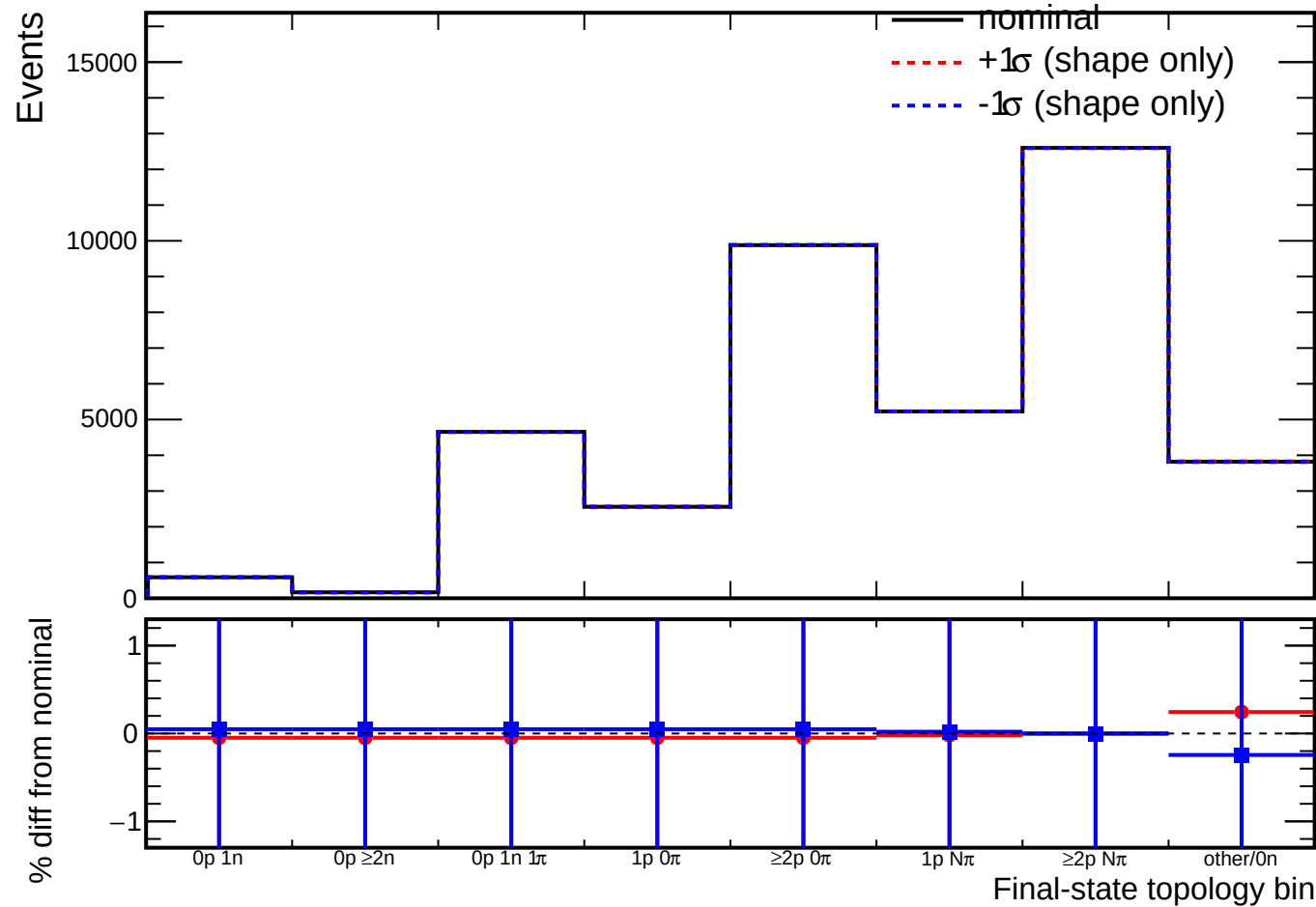
FHC [DIS]: DISNuBarHadronQ0Syst (shape only) $\chi^2_{\text{shape},+1\sigma}=0.03$ $\chi^2_{\text{shape},-1\sigma}=0.03$ $\Delta\langle x \rangle_{+1\sigma}=+0.0010$ $\Delta\langle x \rangle_{-1\sigma}=-0.0010$
 $\chi^2_{\text{raw},+1\sigma}=0.04$ $\chi^2_{\text{raw},-1\sigma}=0.04$ rate: -0.03% / +0.03% $\Delta\sigma^2$: -0.0012 / +0.0012



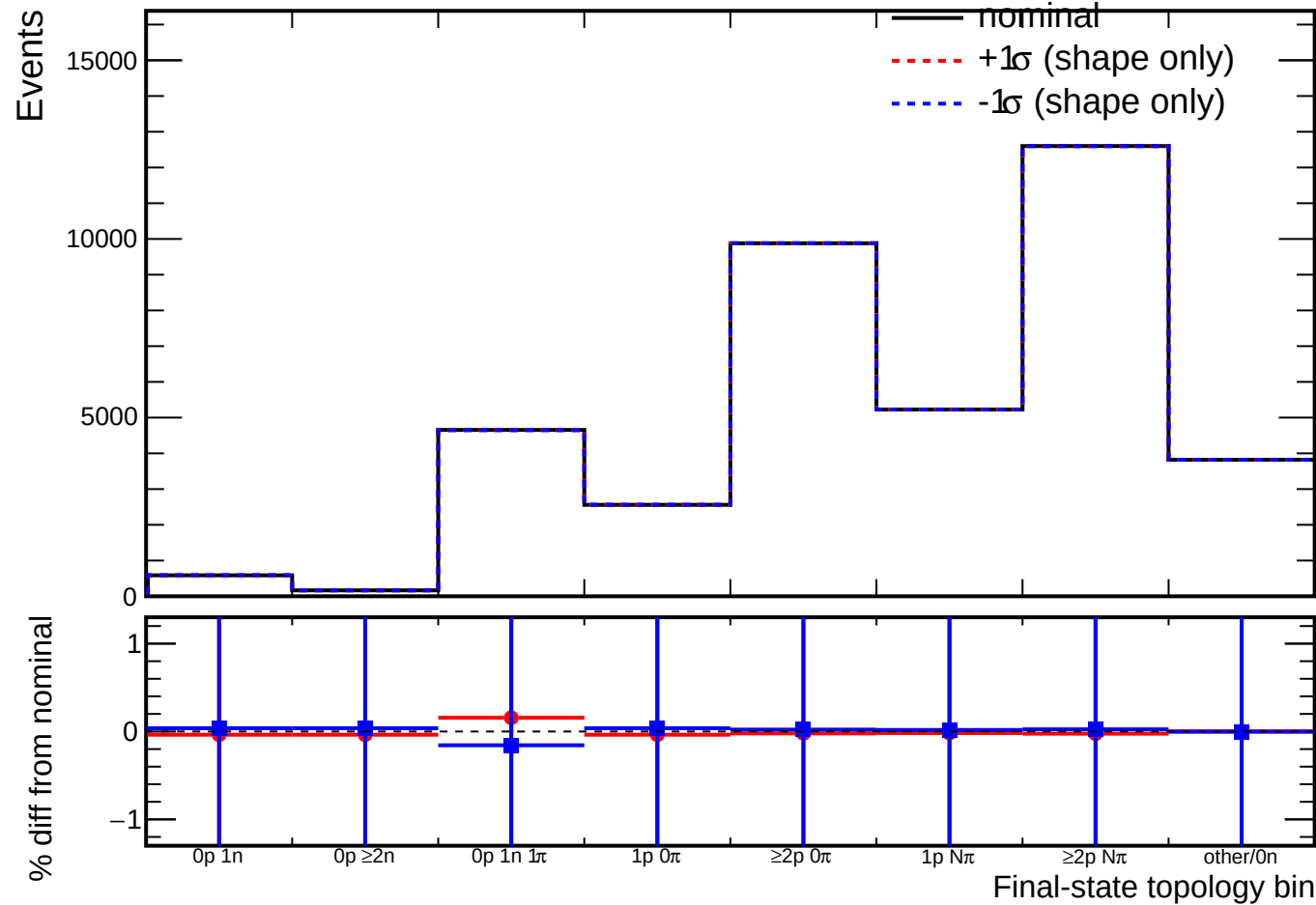
FHC [DIS]: DISvncC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.03$ $\chi^2_{\text{shape},-1\sigma}=0.03$ $\Delta\langle x\rangle_{+1\sigma}=-0.0005$ $\Delta\langle x\rangle_{-1\sigma}=+0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.03$ $\chi^2_{\text{raw},-1\sigma}=0.03$ rate: +0.02% / -0.02% $\Delta\sigma^2$: +0.0013 / -0.0013



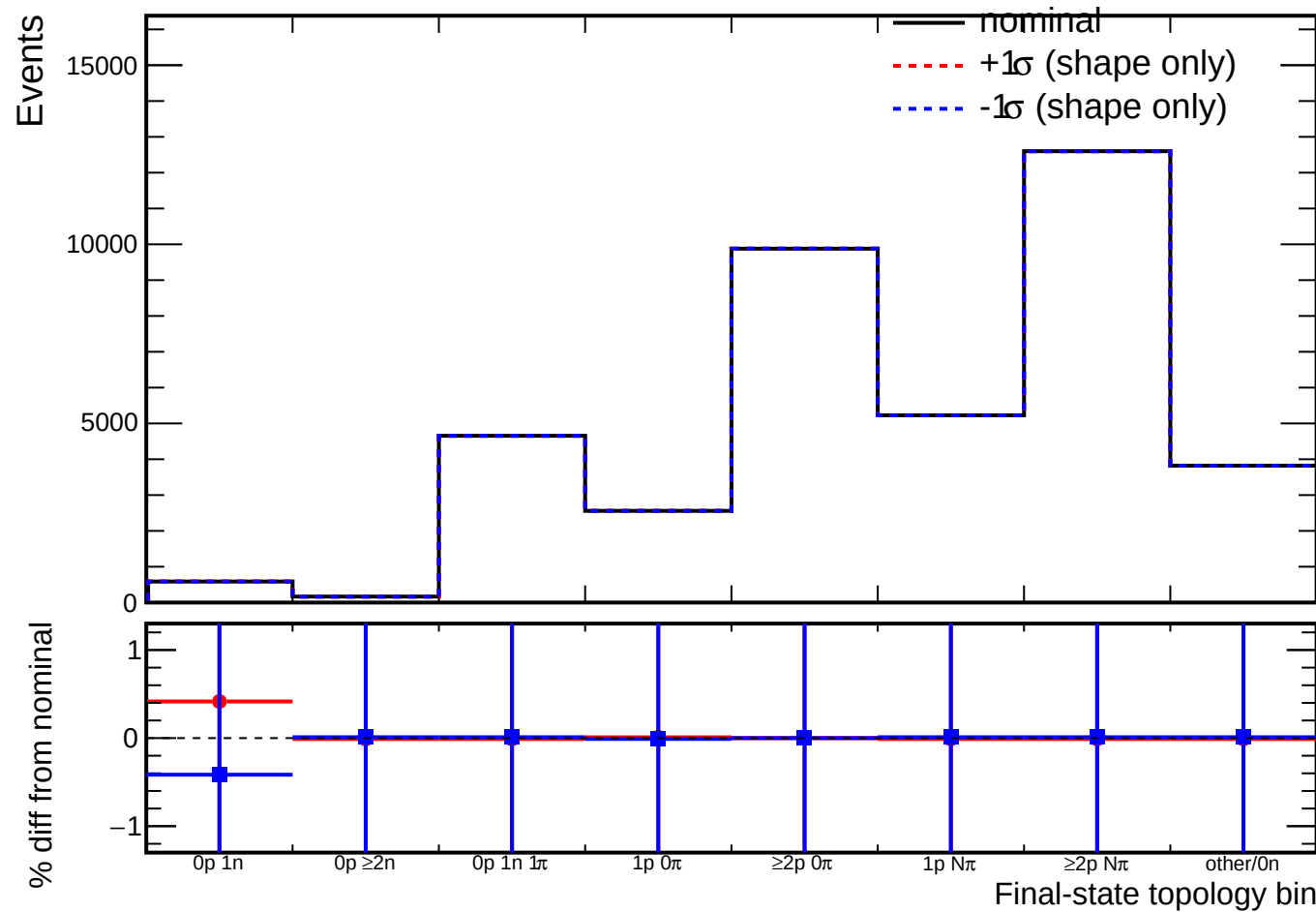
FHC [DIS]: DISvbarpCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.03$ $\chi^2_{\text{shape},-1\sigma}=0.03$ $\Delta\langle x\rangle_{+1\sigma}=+0.0009$ $\Delta\langle x\rangle_{-1\sigma}=-0.0009$
 $\chi^2_{\text{raw},+1\sigma}=0.04$ $\chi^2_{\text{raw},-1\sigma}=0.04$ rate: +0.05% / -0.05% $\Delta\sigma^2$: +0.0005 / -0.0005



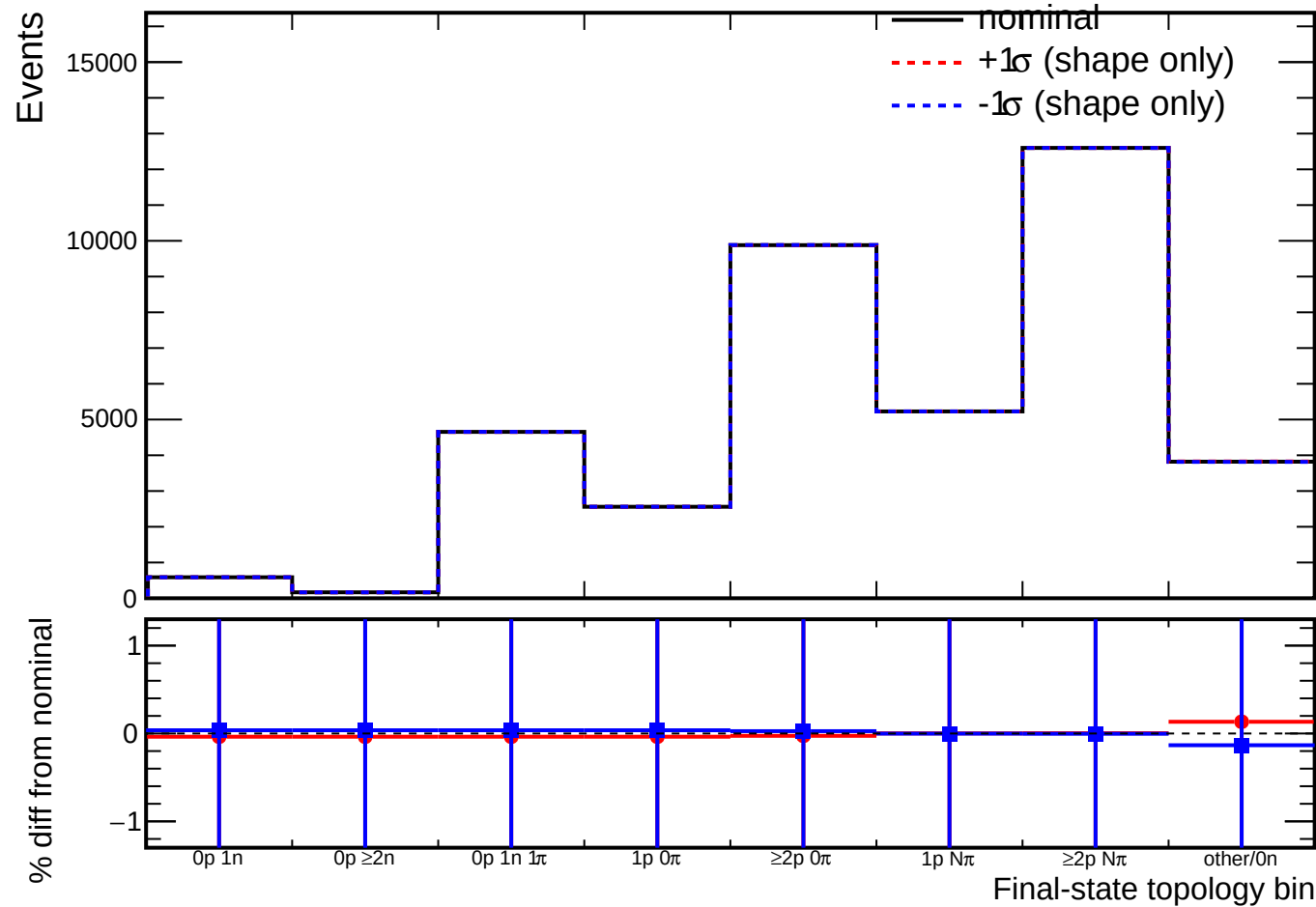
FHC [DIS]: DISvbnCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=-0.0005$ $\Delta\langle x\rangle_{-1\sigma}=+0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.02$ $\chi^2_{\text{raw},-1\sigma}=0.02$ rate: +0.04% / -0.04% $\Delta\sigma^2$: +0.0010 / -0.0010



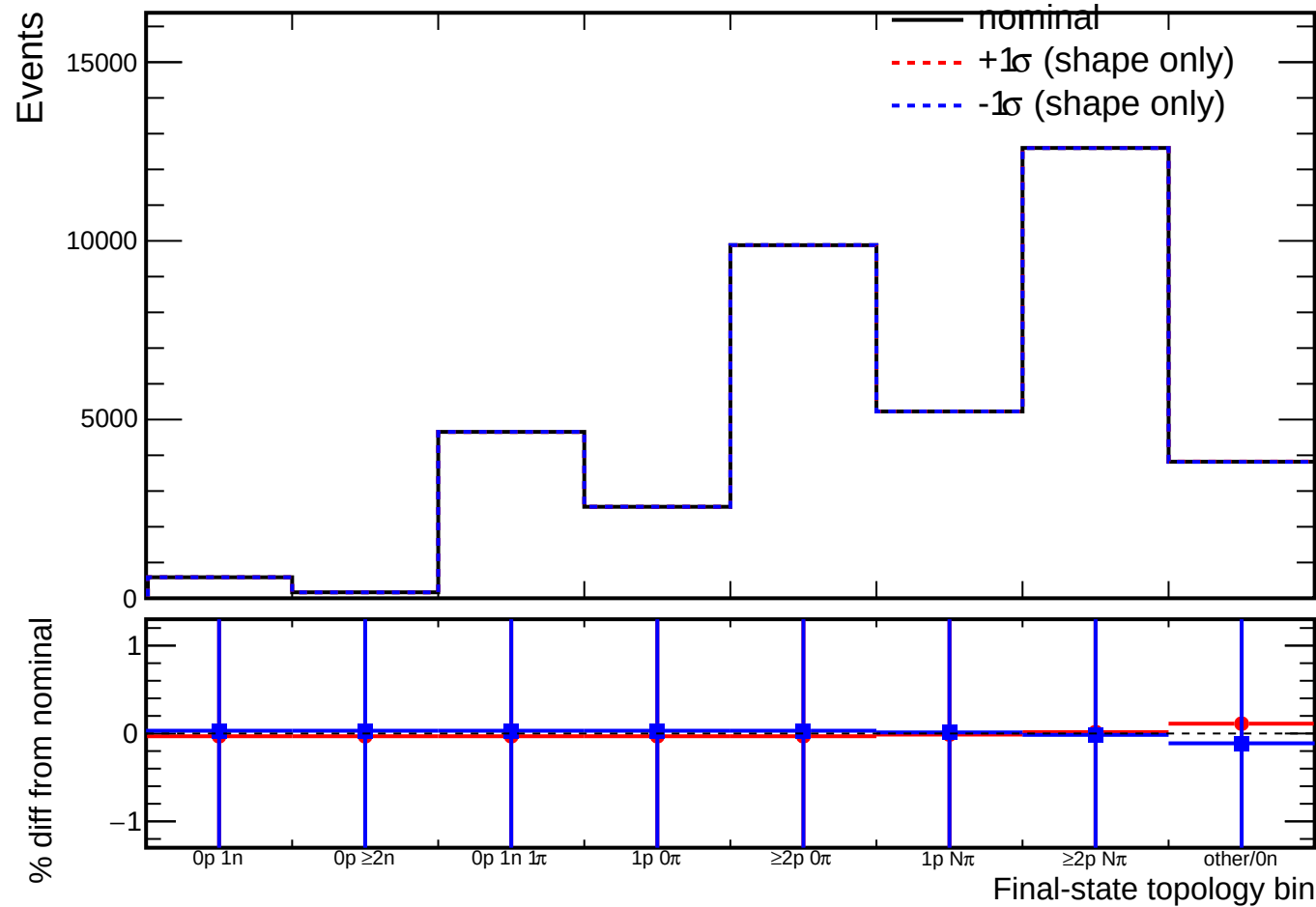
FHC [DIS]: DISvNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x \rangle_{+1\sigma}=-0.0003$ $\Delta\langle x \rangle_{-1\sigma}=+0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.01% / -0.01% $\Delta\sigma^2$: +0.0012 / -0.0012



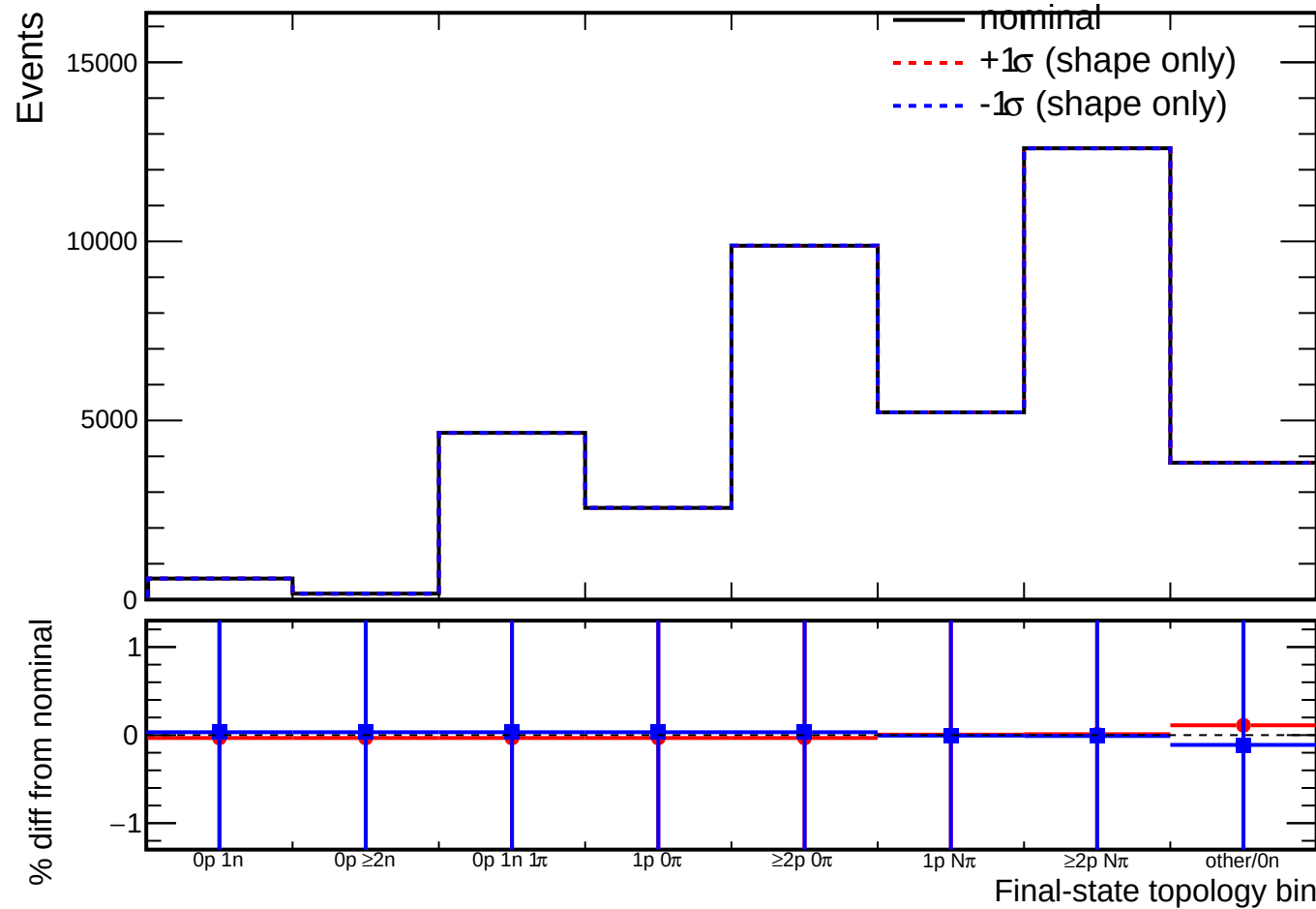
FHC [DIS]: DISvbnCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x \rangle_{+1\sigma}=+0.0005$ $\Delta\langle x \rangle_{-1\sigma}=-0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.04% / -0.04% $\Delta\sigma^2$: +0.0002 / -0.0002



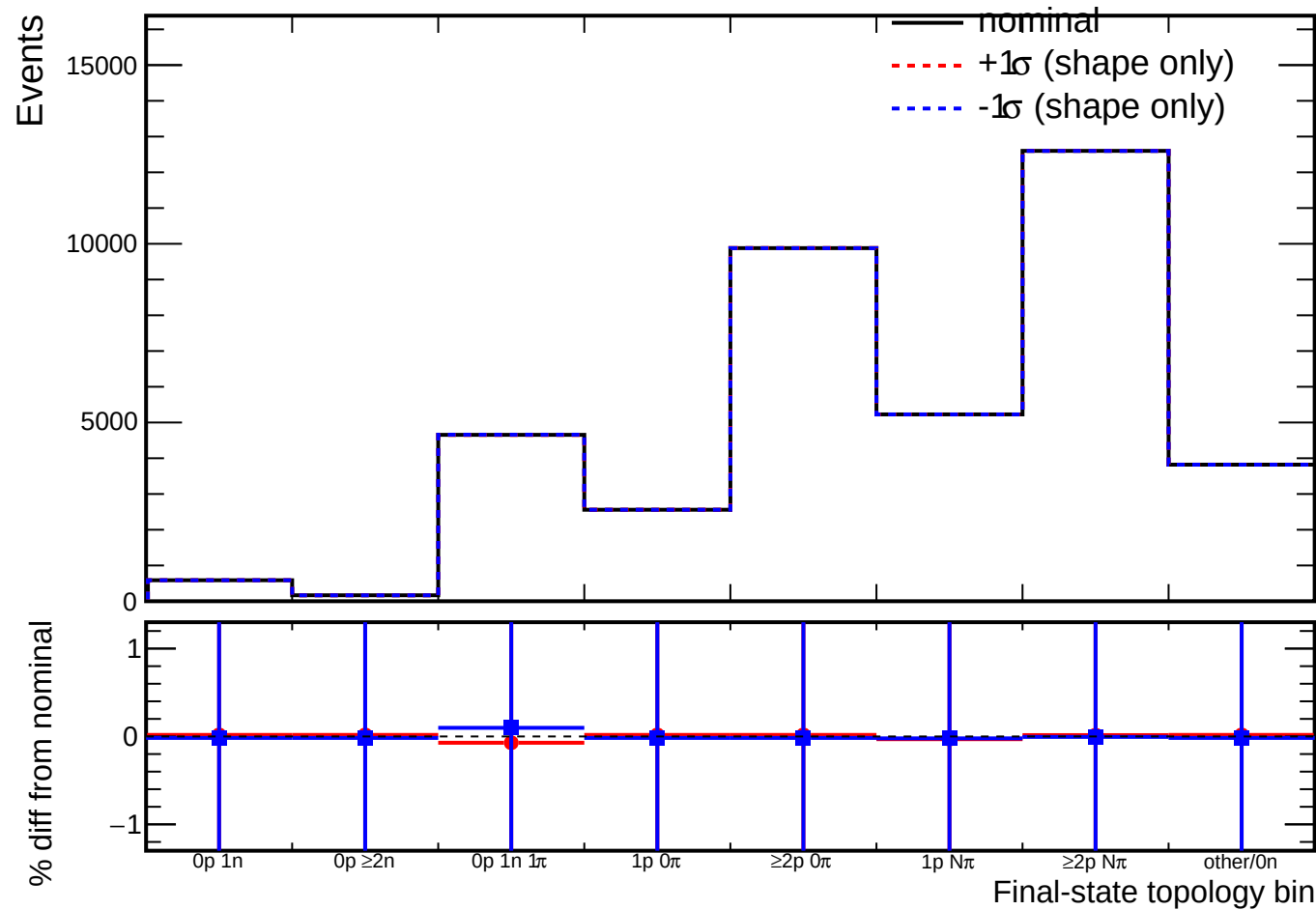
FHC [DIS]: DISvbarpCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x \rangle_{+1\sigma}=+0.0005$ $\Delta\langle x \rangle_{-1\sigma}=-0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.03% / -0.03% $\Delta\sigma^2$: +0.0002 / -0.0002



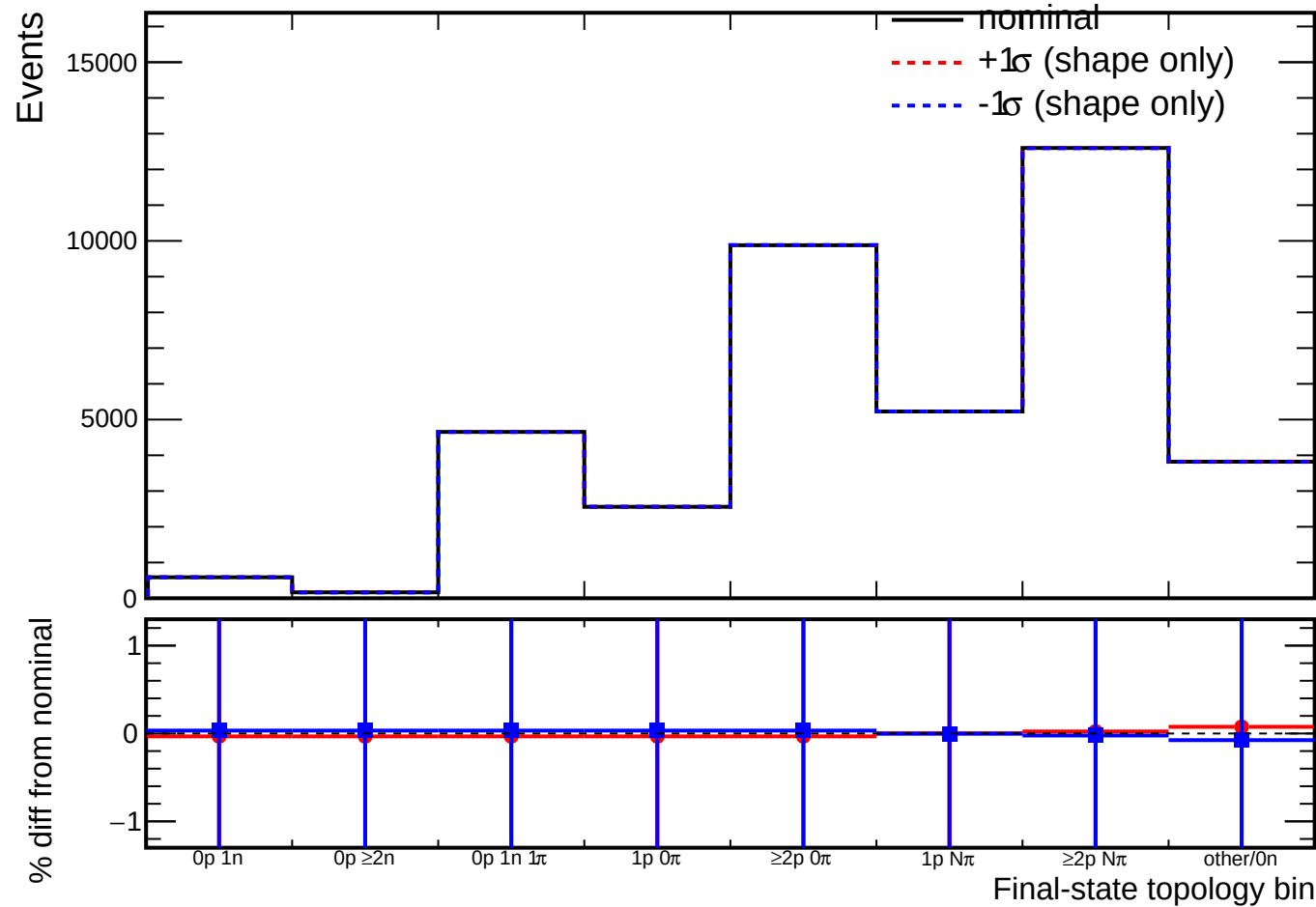
FHC [DIS]: DISvbnNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x \rangle_{+1\sigma}=+0.0005$ $\Delta\langle x \rangle_{-1\sigma}=-0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.03% / -0.03% $\Delta\sigma^2$: +0.0001 / -0.0001



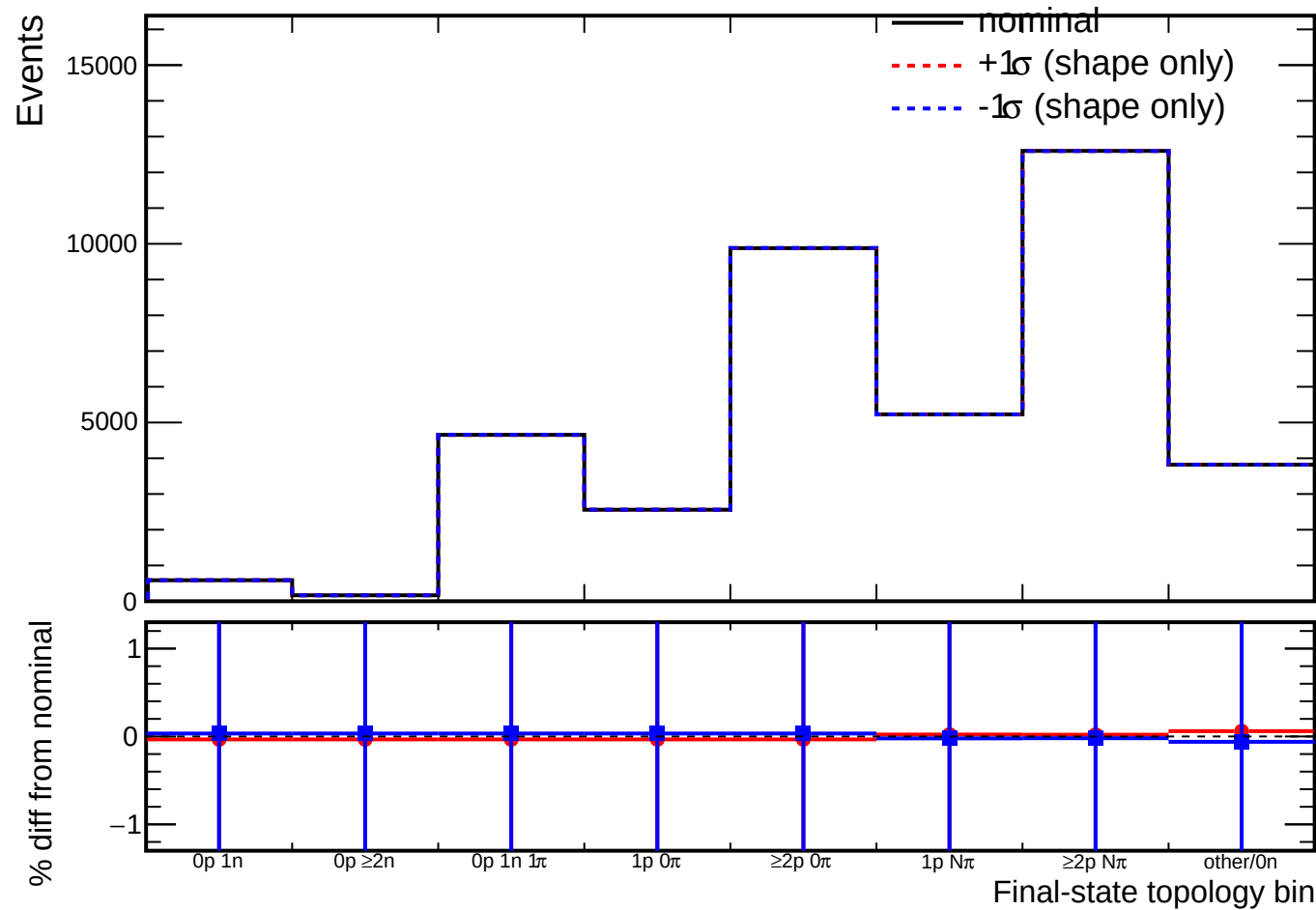
FHC [DIS]: AGKYxF1pi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=+0.0003$ $\Delta\langle x\rangle_{-1\sigma}=-0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: -0.02% / +0.02% $\Delta\sigma^2$: -0.0003 / +0.0006



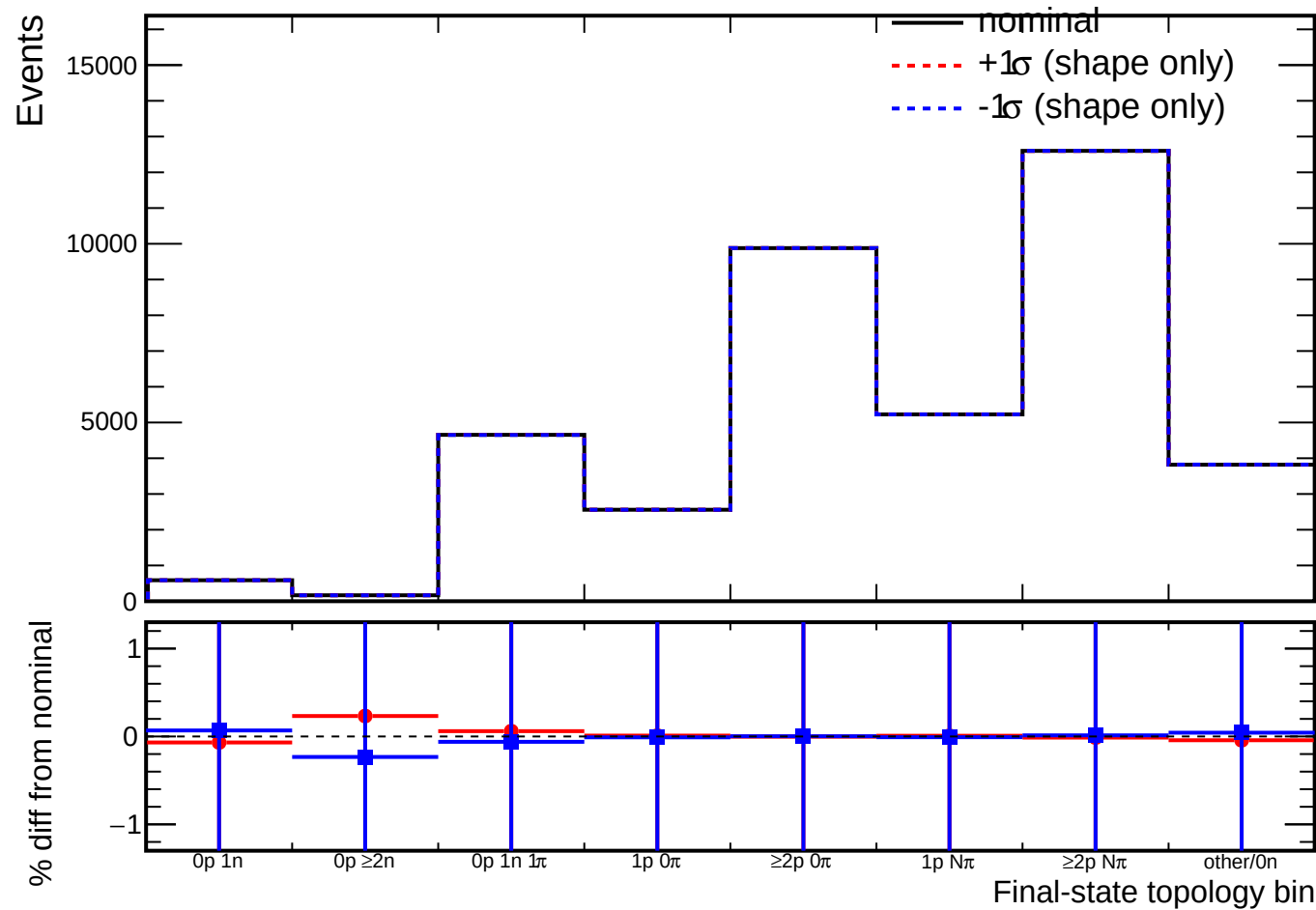
FHC [DIS]: DISvbnrCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0005$ $\Delta\langle x\rangle_{-1\sigma}=-0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.03% / -0.03% $\Delta\sigma^2$: +0.0000 / -0.0000



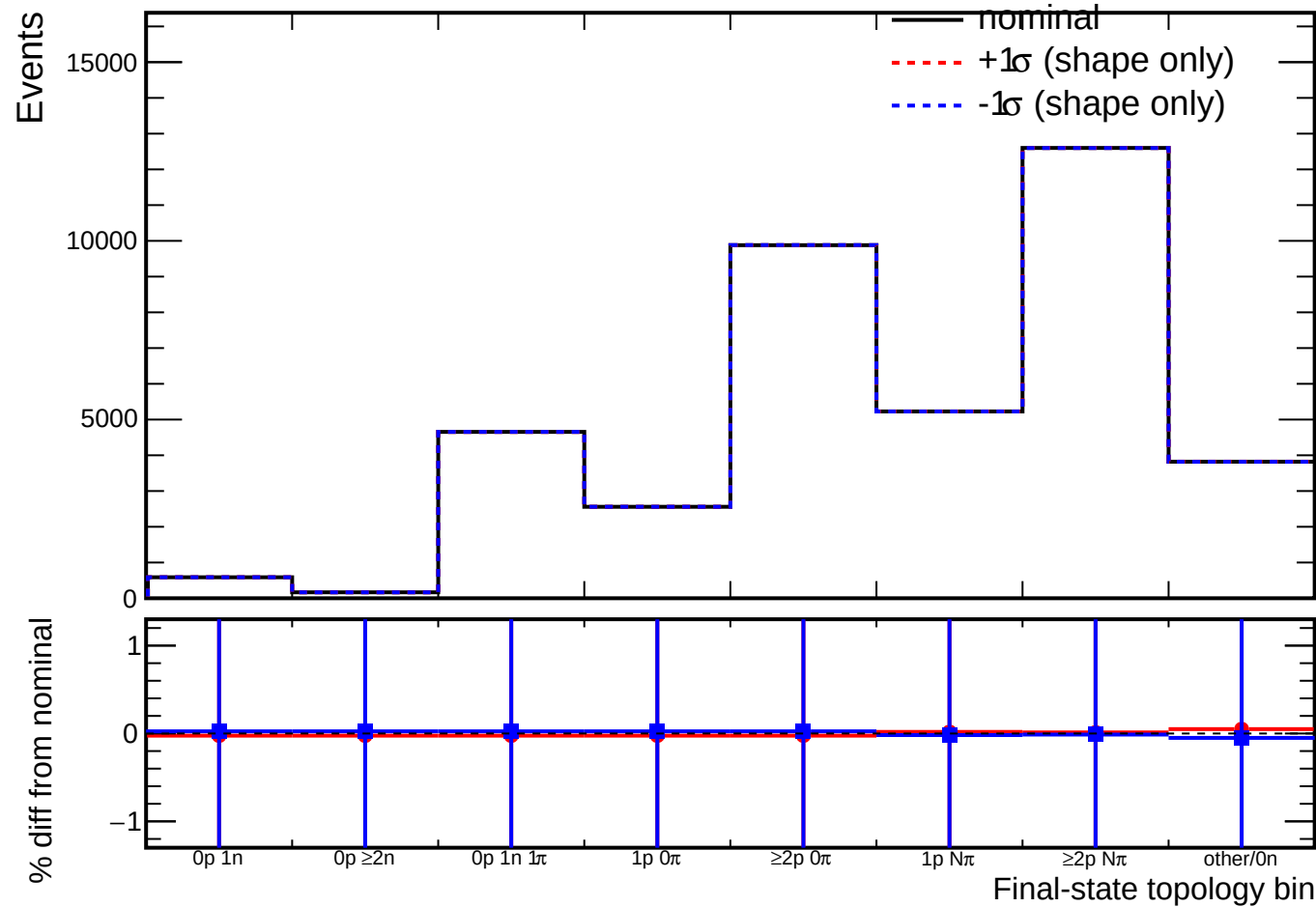
FHC [DIS]: DISvbnNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0005$ $\Delta\langle x\rangle_{-1\sigma}=-0.0005$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.03% / -0.03% $\Delta\sigma^2$: -0.0001 / +0.0001



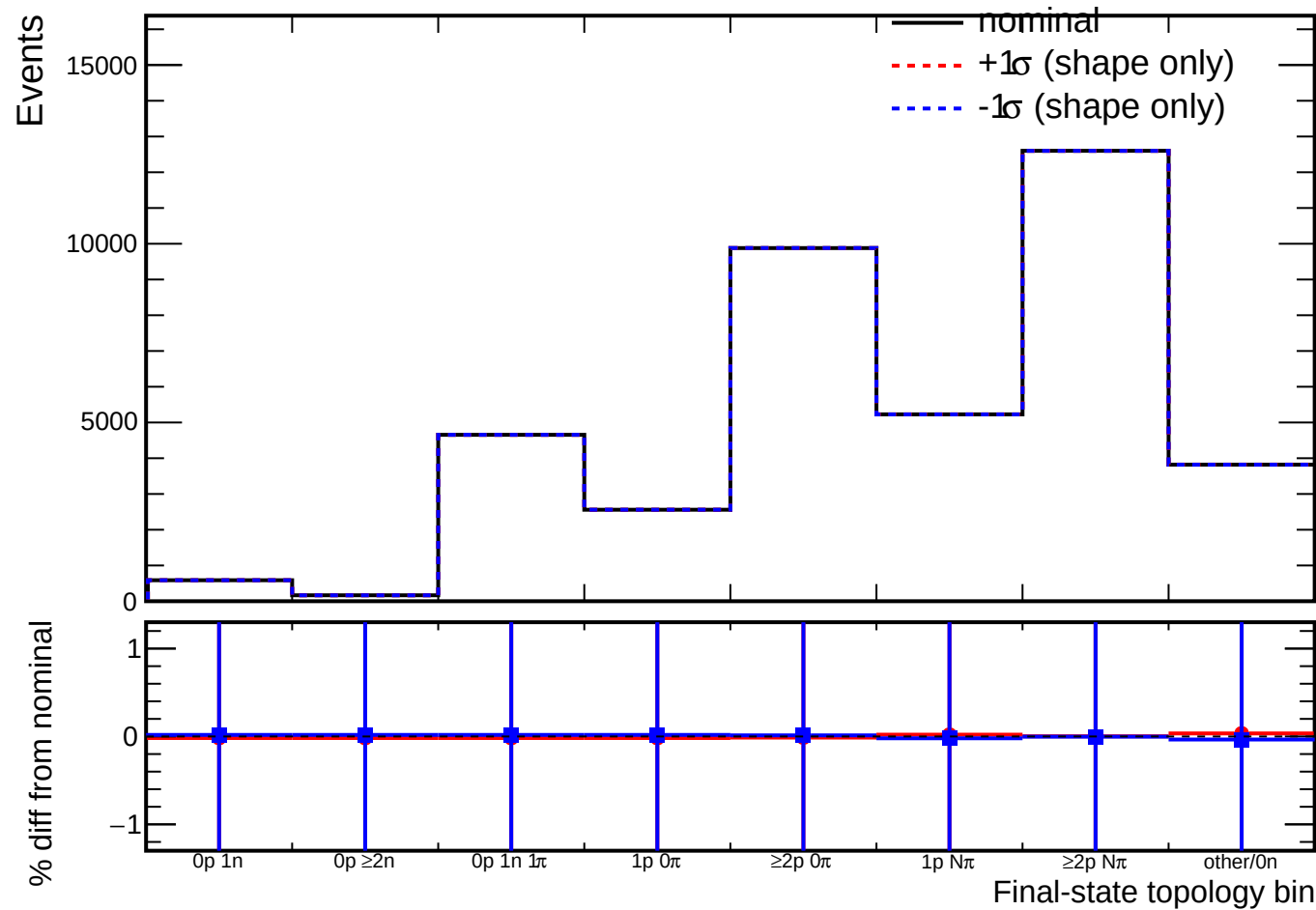
FHC [DIS]: DISvbarpCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0003$ $\Delta\langle x\rangle_{-1\sigma}=+0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.02$ $\chi^2_{\text{raw},-1\sigma}=0.02$ rate: +0.07% / -0.07% $\Delta\sigma^2$: +0.0002 / -0.0002



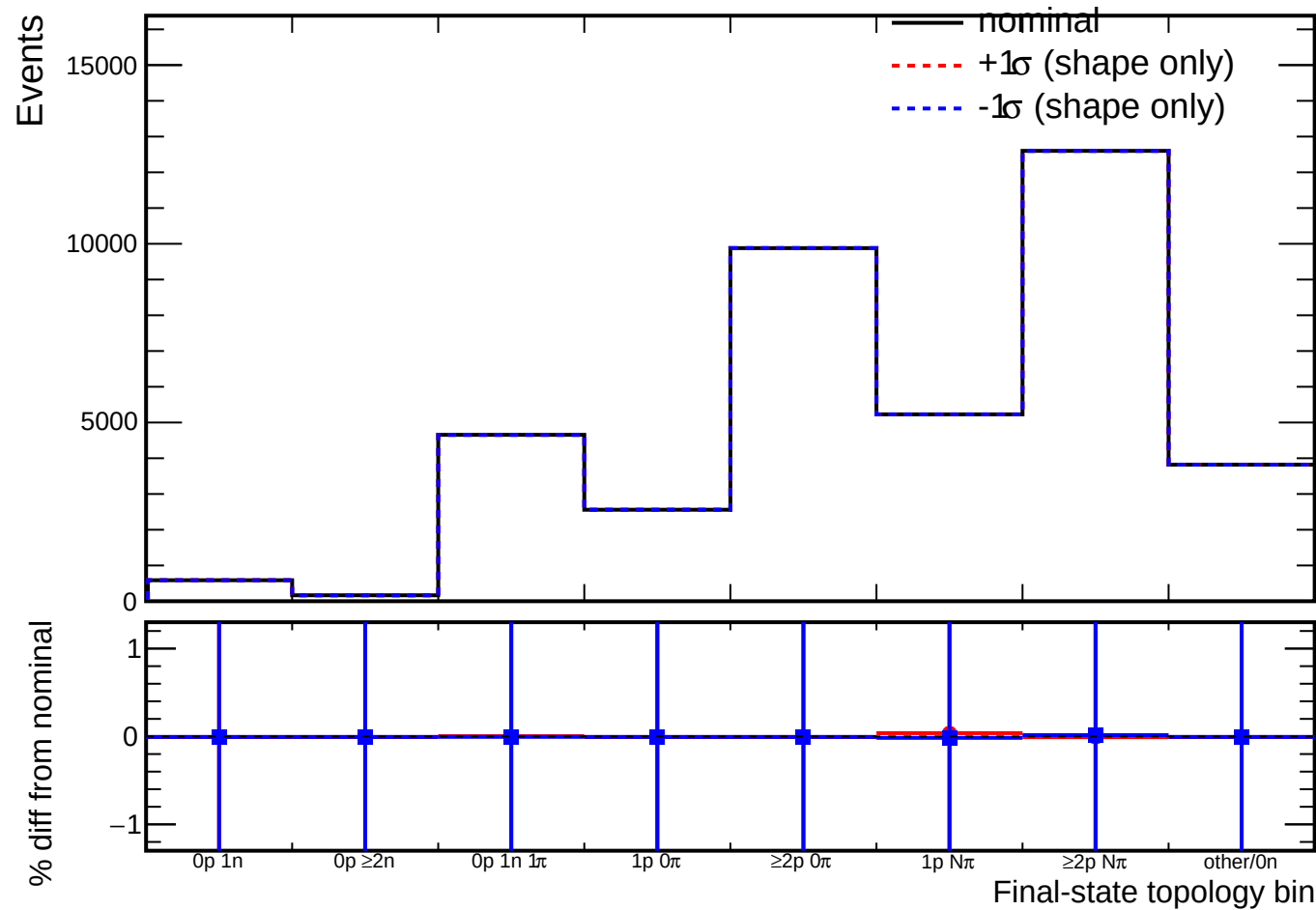
FHC [DIS]: DISvbarpNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0003$ $\Delta\langle x\rangle_{-1\sigma}=-0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.02% / -0.02% $\Delta\sigma^2$: -0.0001 / +0.0001



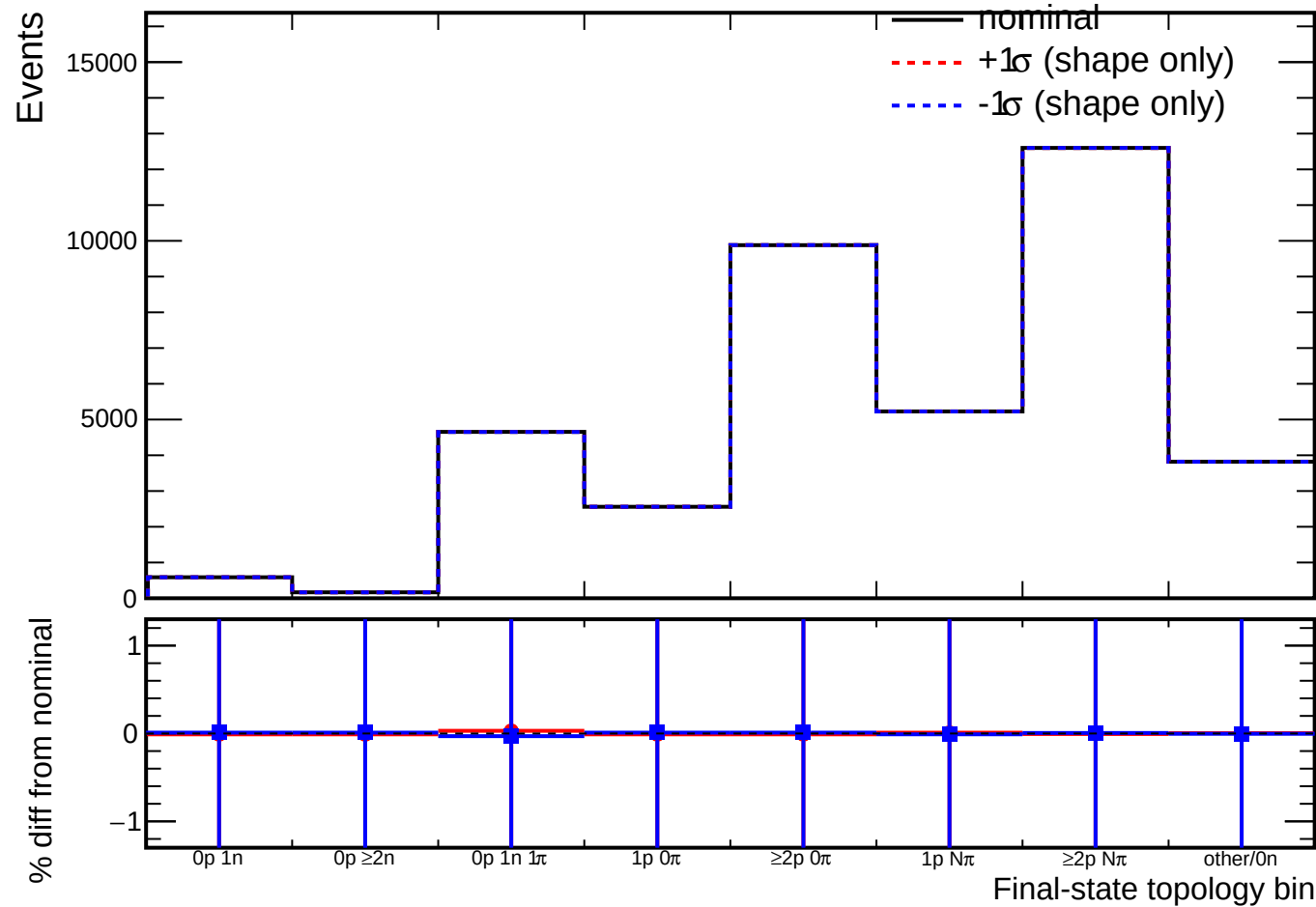
FHC [DIS]: DISvbarpNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0002$ $\Delta\langle x\rangle_{-1\sigma}=-0.0002$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.02% / -0.02% $\Delta\sigma^2$: -0.0001 / +0.0001



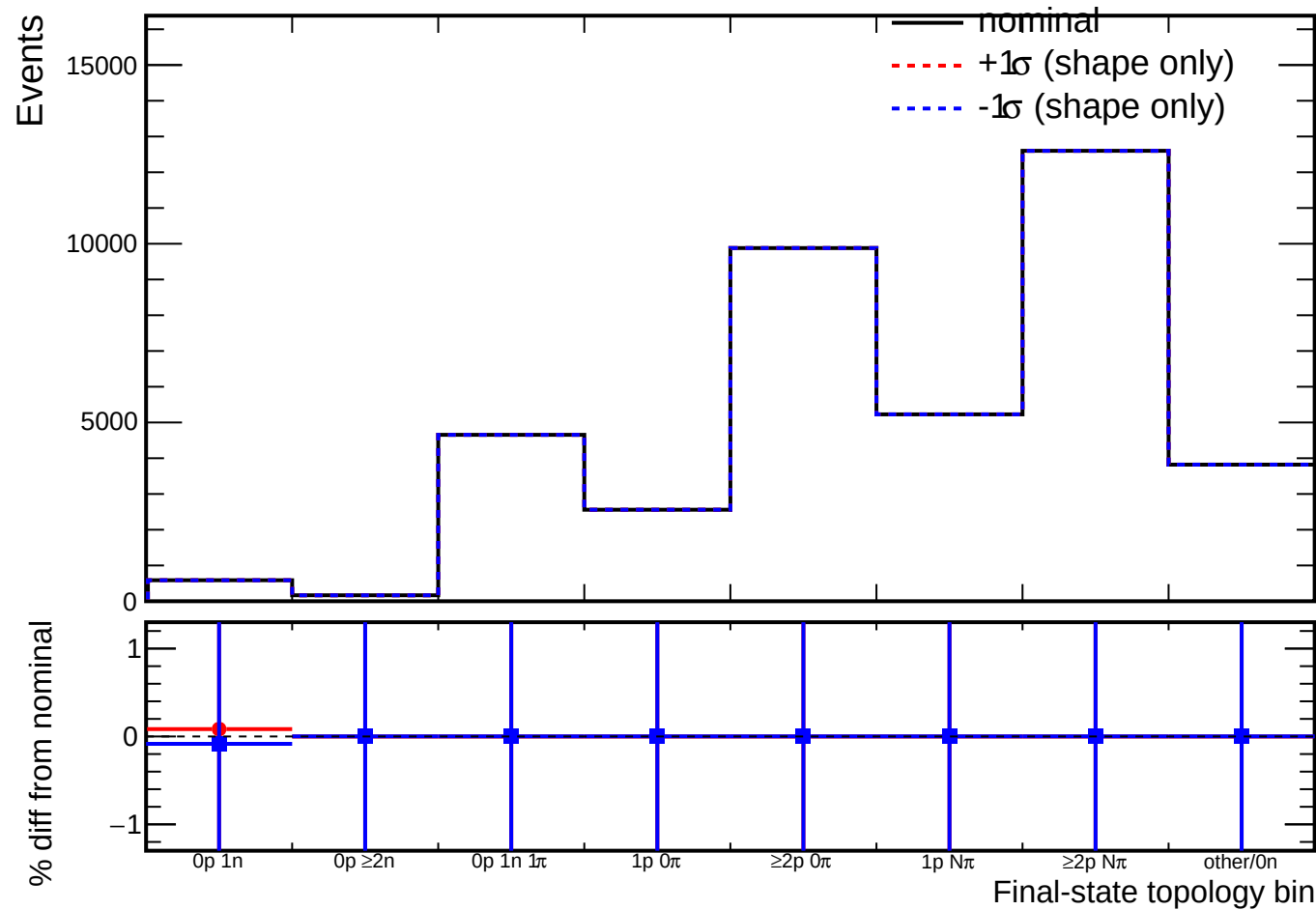
FHC [DIS]: AGKYpT1pi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x \rangle_{+1\sigma}=-0.0000$ $\Delta\langle x \rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / +0.01% $\Delta\sigma^2$: -0.0001 / -0.0000



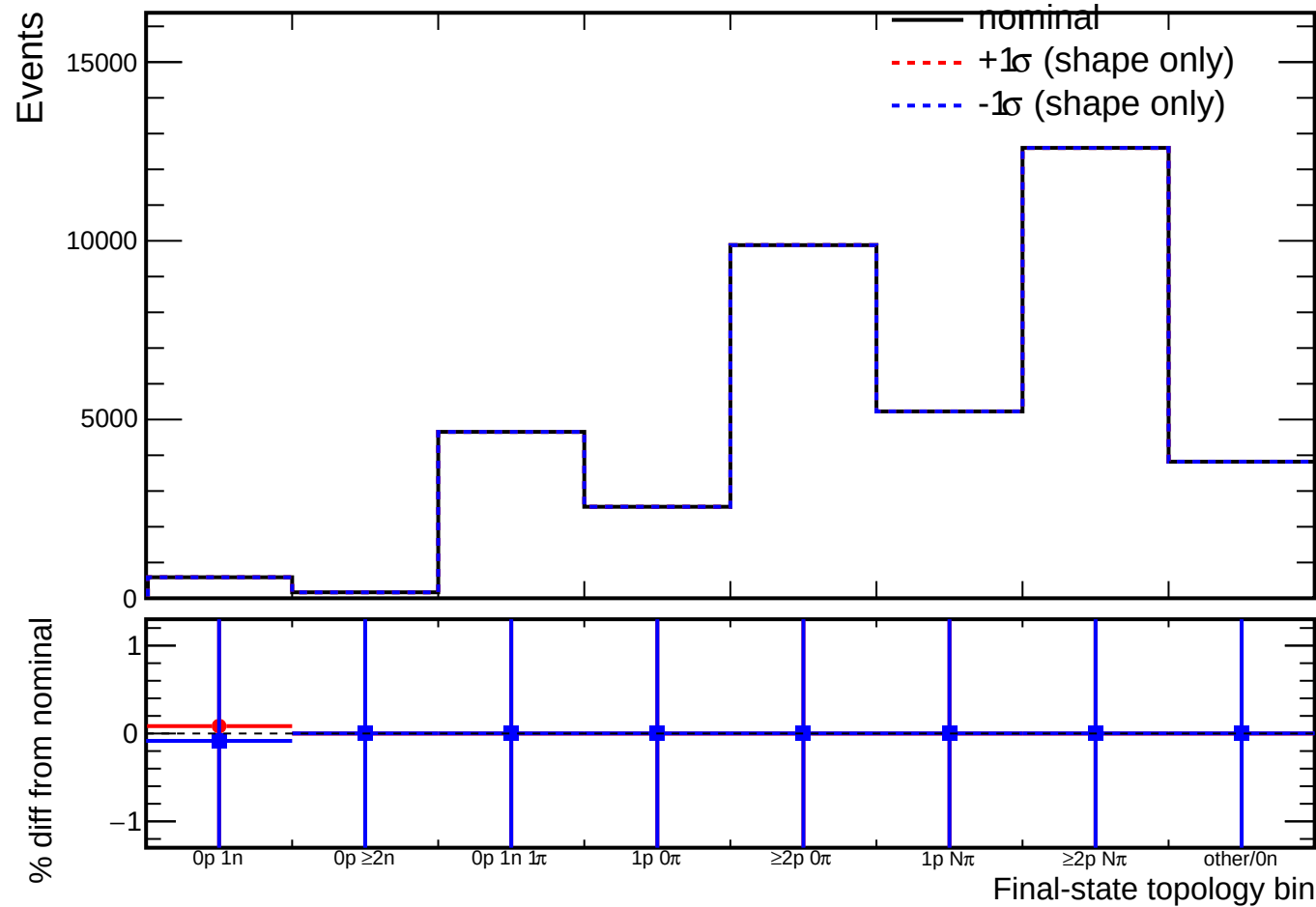
FHC [DIS]: DISvbnNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0001$ $\Delta\langle x\rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / -0.01% $\Delta\sigma^2$: +0.0002 / -0.0002



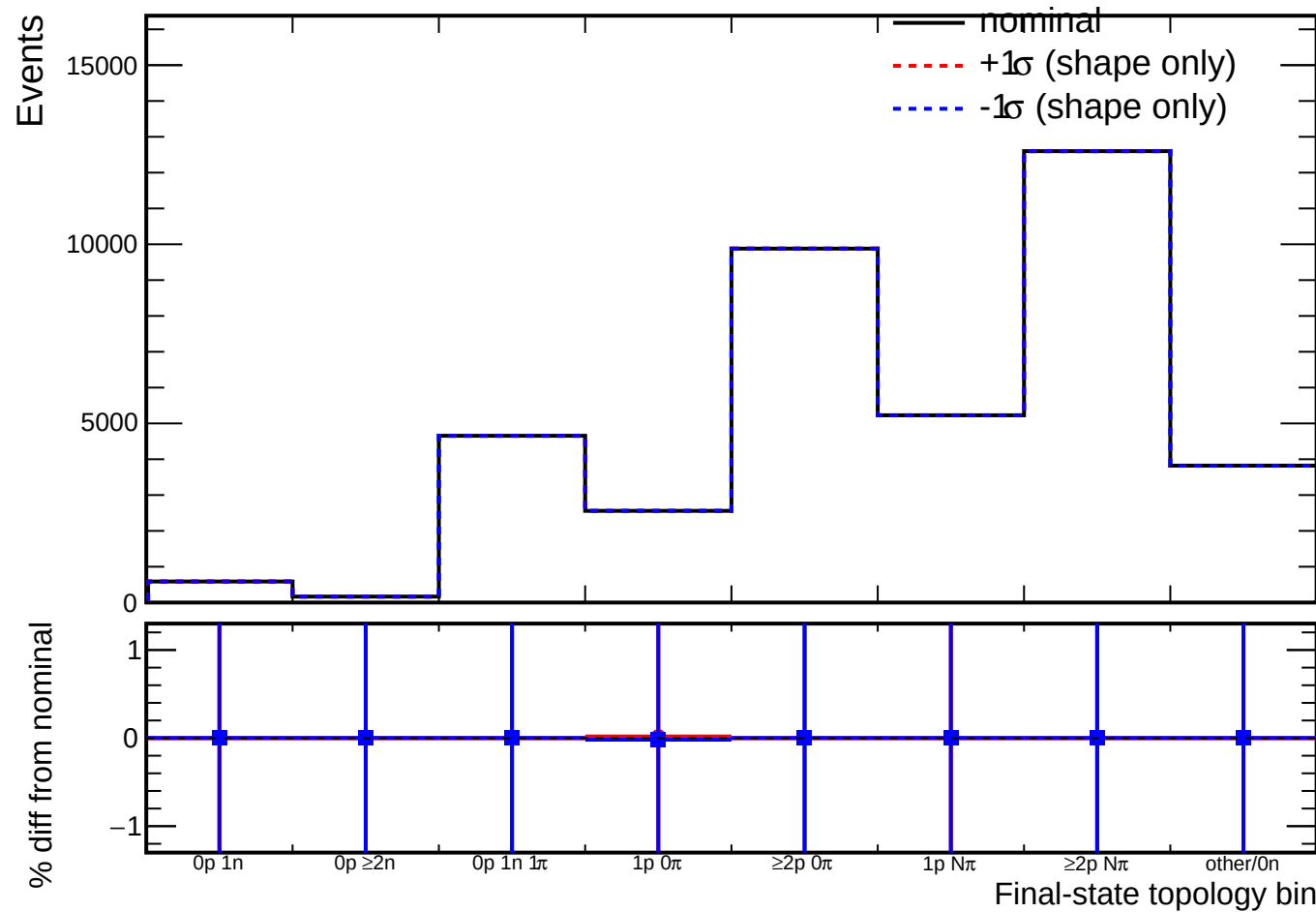
FHC [DIS]: DISvpCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0001$ $\Delta\langle x\rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / -0.00% $\Delta\sigma^2$: +0.0002 / -0.0002



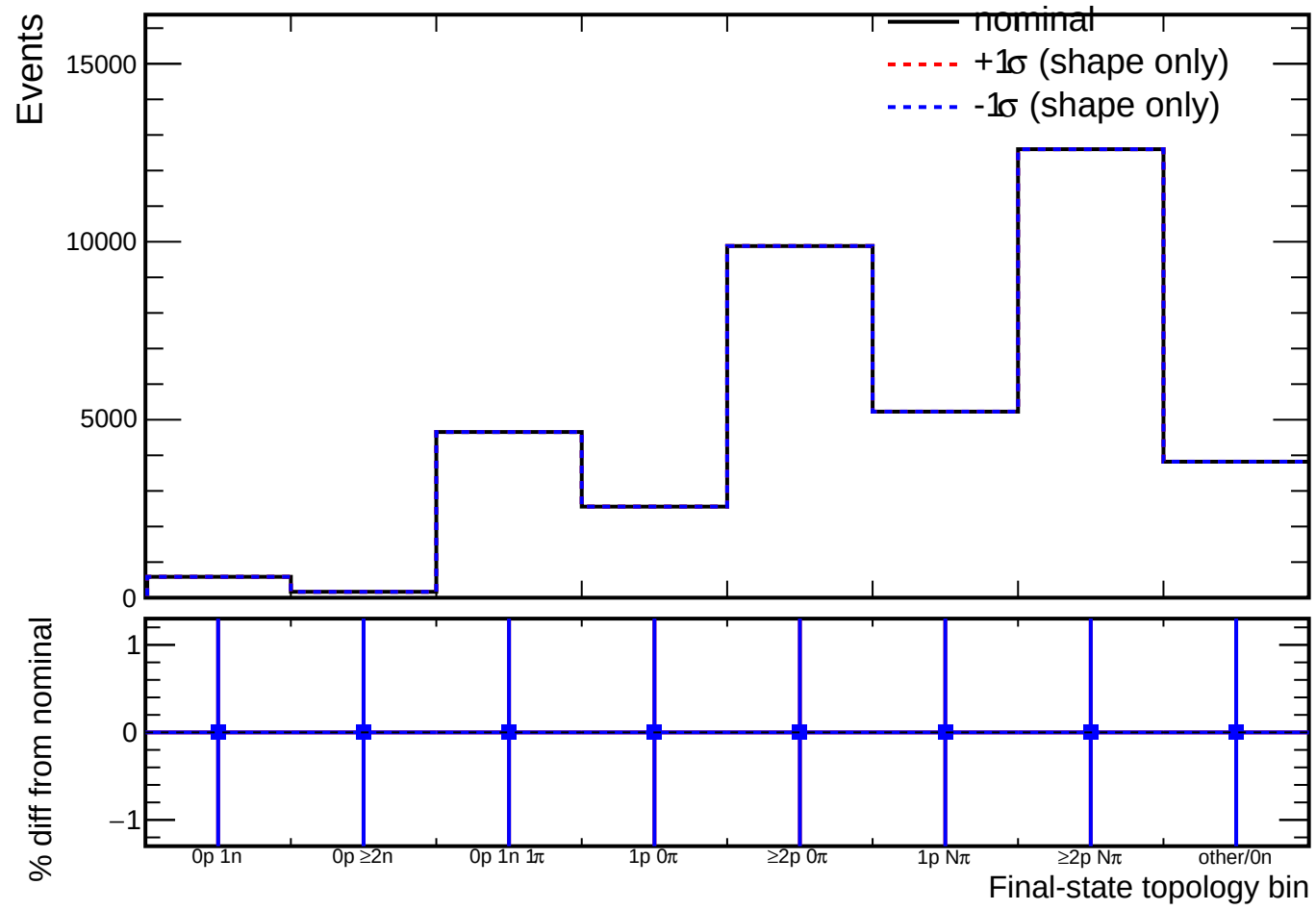
FHC [DIS]: DISvbnNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0001$ $\Delta\langle x\rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / -0.00% $\Delta\sigma^2$: +0.0002 / -0.0002



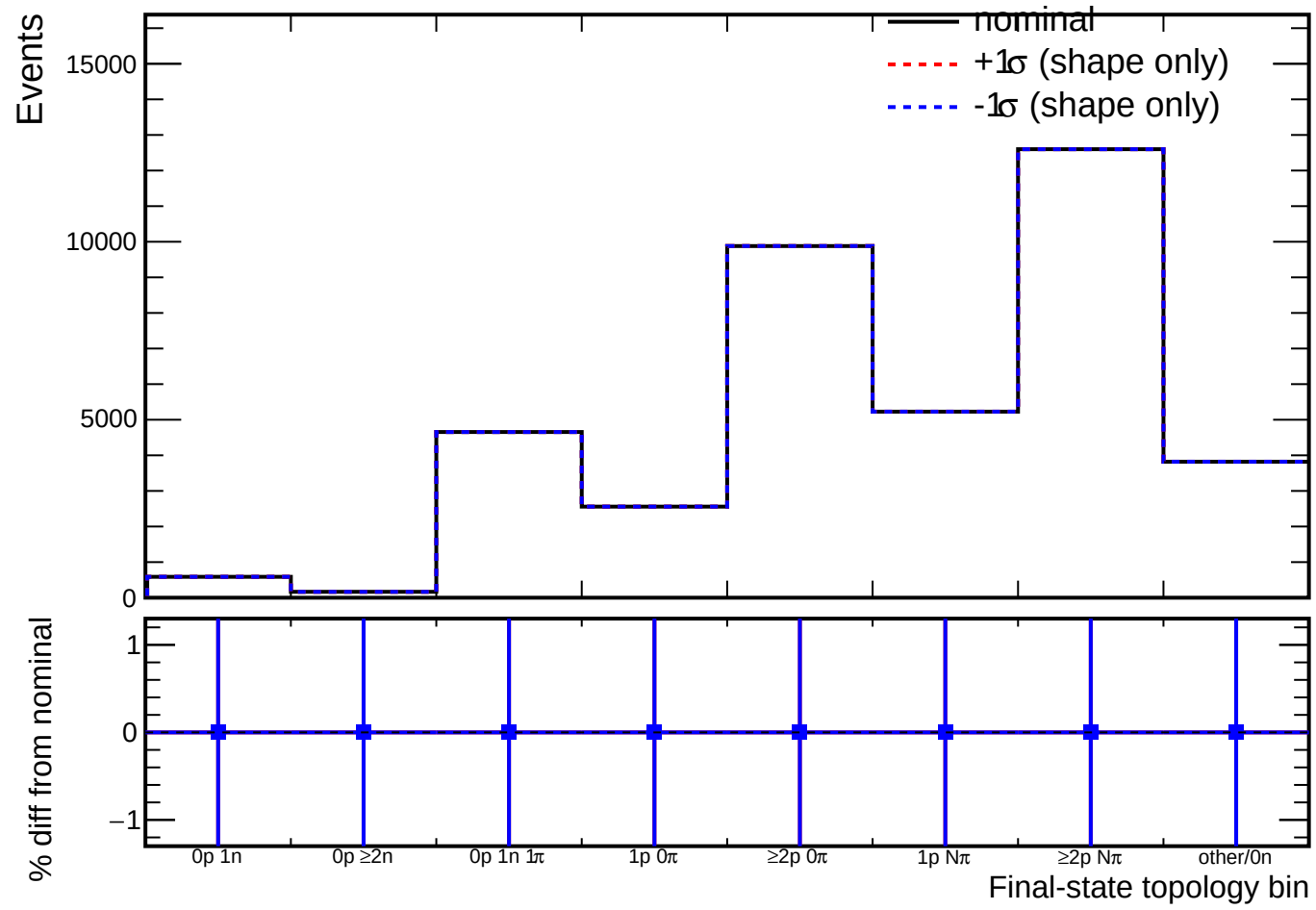
FHC [DIS]: DISvpNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0000$ $\Delta\langle x\rangle_{-1\sigma}=+0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / -0.00% $\Delta\sigma^2$: +0.0000 / -0.0000



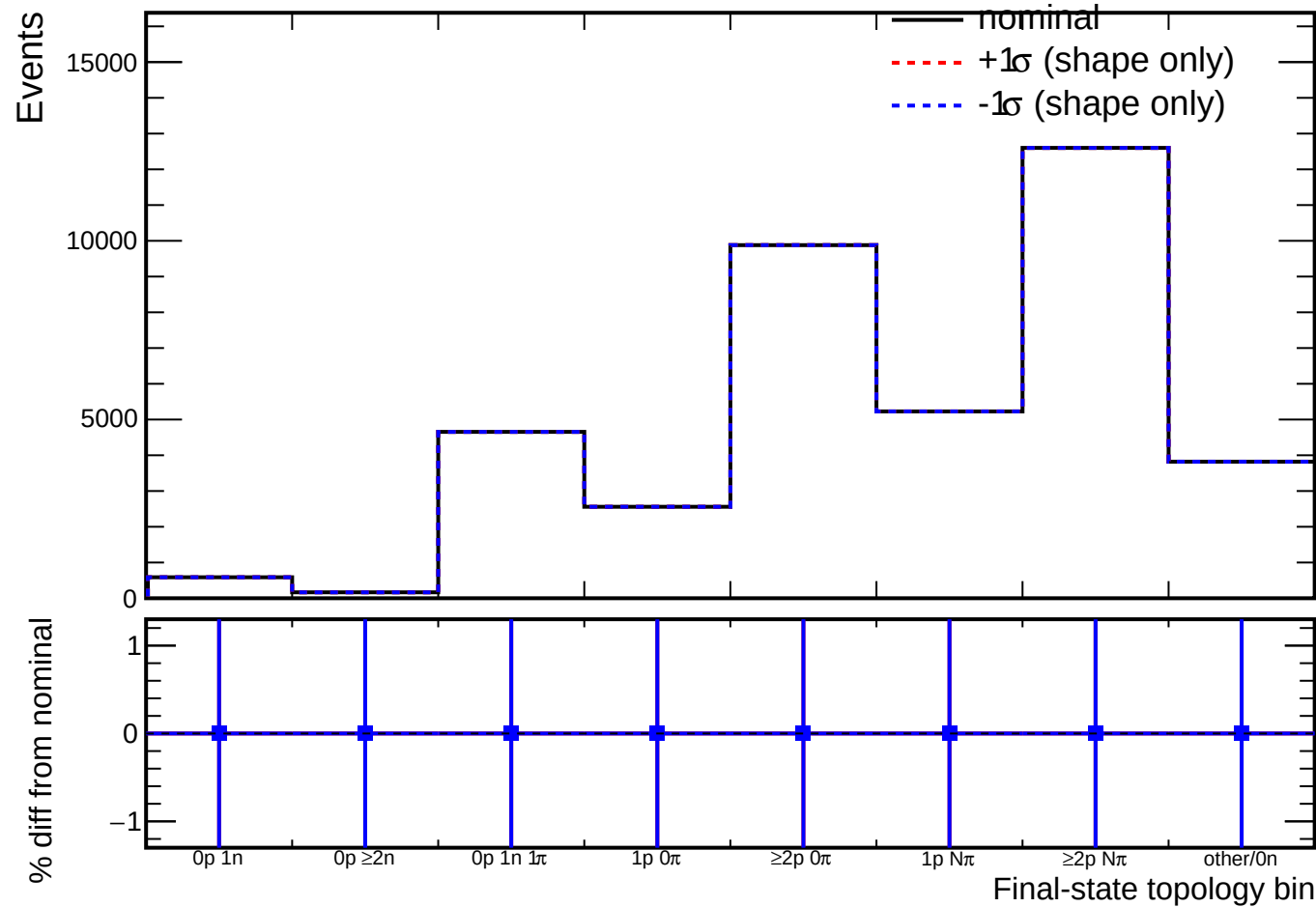
FHC [DIS]: DISvbarpNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



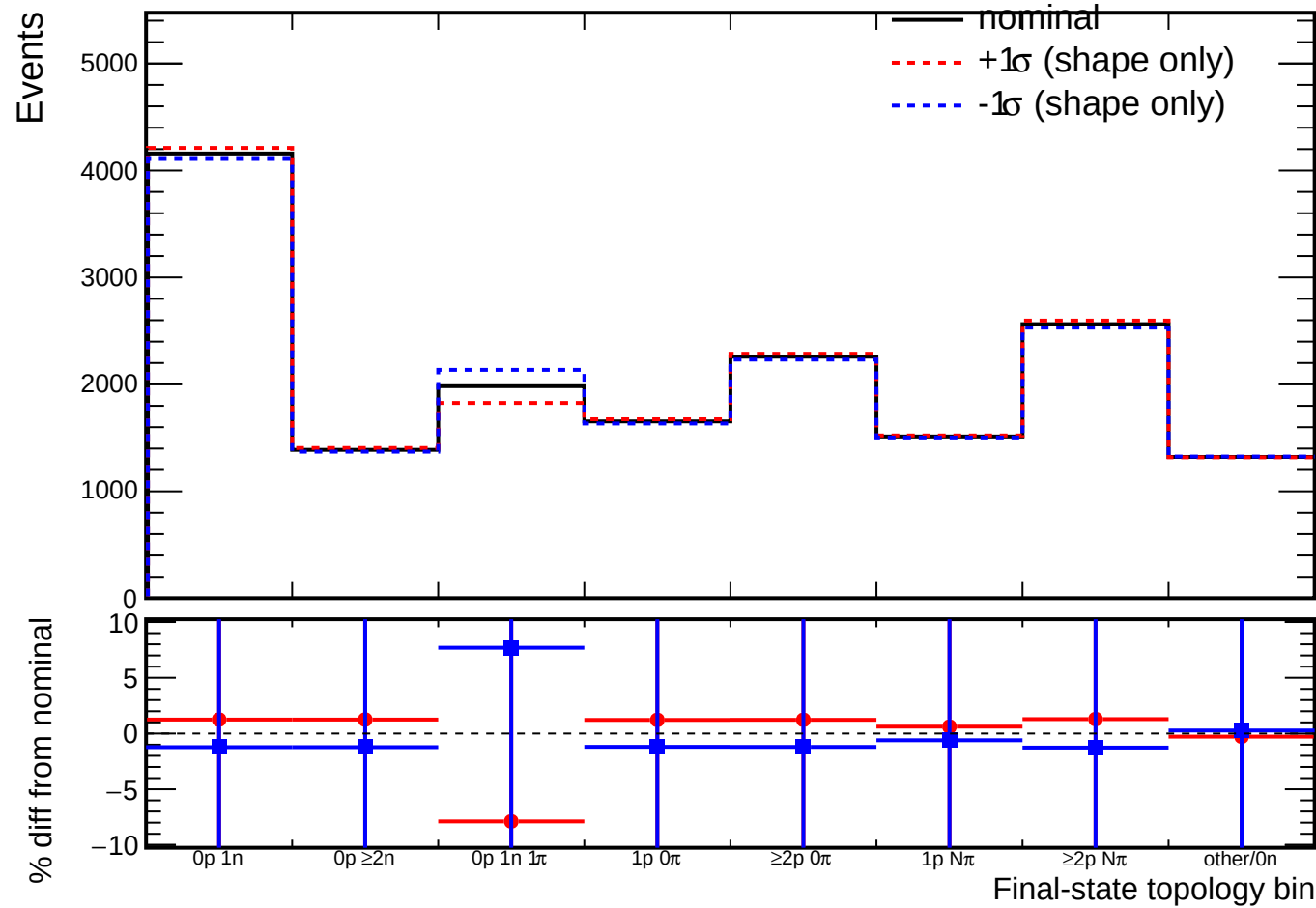
FHC [DIS]: DISvbarpCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



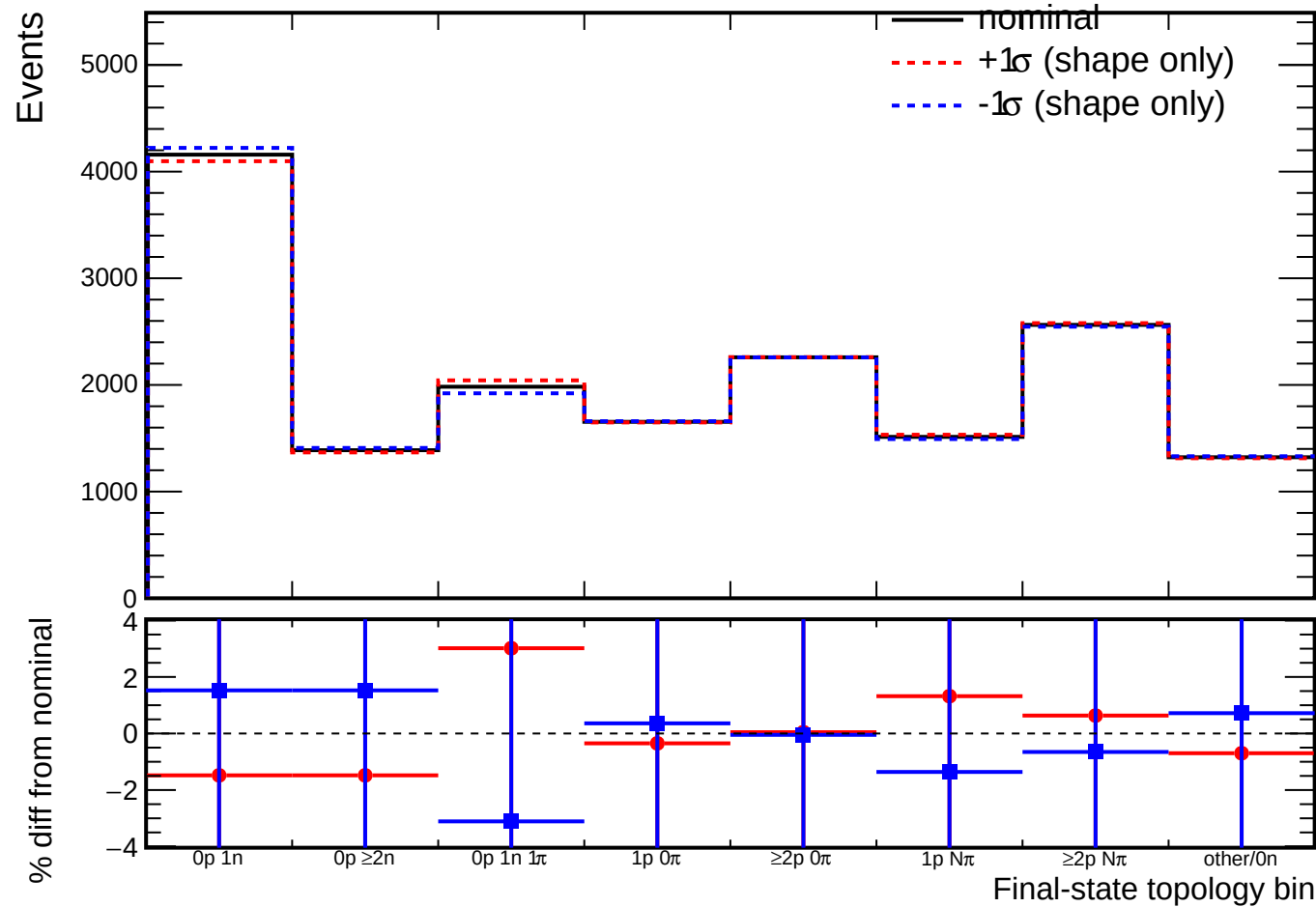
FHC [DIS]: DISvbnCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



RHC [DIS]: DISNuBarHAdronQ0Syst (shape only) $\chi^2_{\text{shape}, +1\sigma} = 14.29$ $\chi^2_{\text{shape}, -1\sigma} = 13.60$ $\Delta\langle x \rangle_{+1\sigma} = +0.0057$ $\Delta\langle x \rangle_{-1\sigma} = -0.0056$
 $\chi^2_{\text{raw}, +1\sigma} = 16.52$ $\chi^2_{\text{raw}, -1\sigma} = 16.52$ rate: -1.24% / +1.24% $\Delta\sigma^2$: +0.0401 / -0.0391

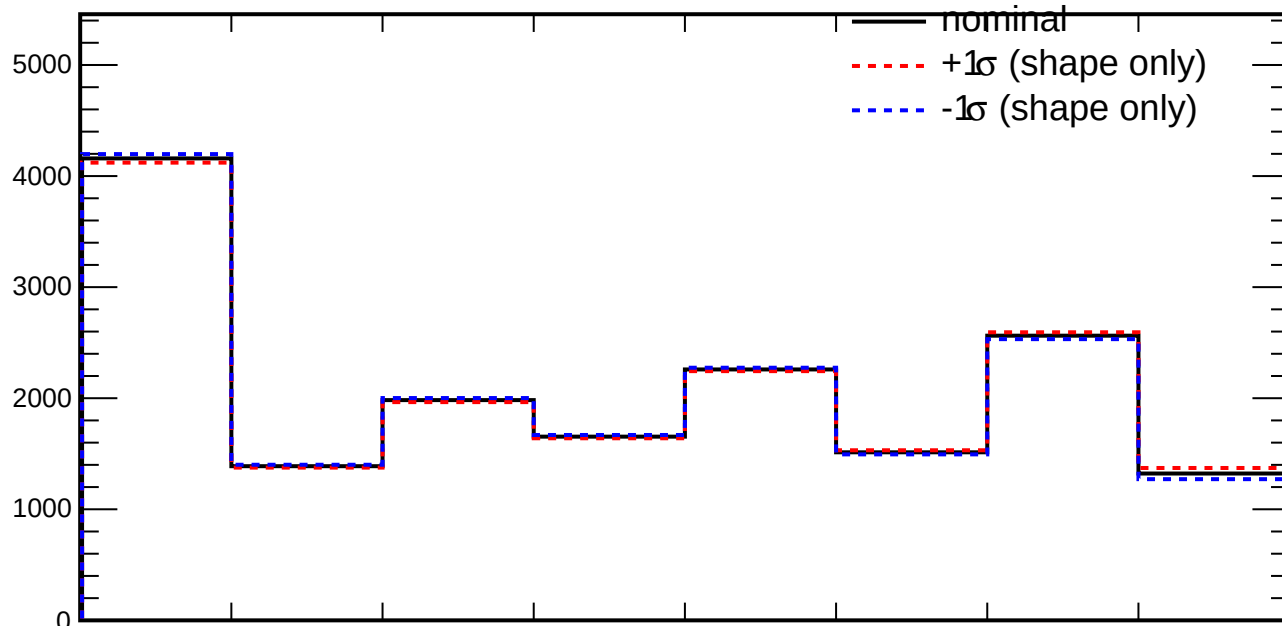


RHC [DIS]: DISvbarpCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=3.47$ $\chi^2_{\text{shape},-1\sigma}=3.69$ $\Delta\langle x\rangle_{+1\sigma}=+0.0130$ $\Delta\langle x\rangle_{-1\sigma}=-0.0134$
 $\chi^2_{\text{raw},+1\sigma}=7.38$ $\chi^2_{\text{raw},-1\sigma}=7.38$ rate: +1.50% / -1.50% $\Delta\sigma^2$: -0.0312 / +0.0318

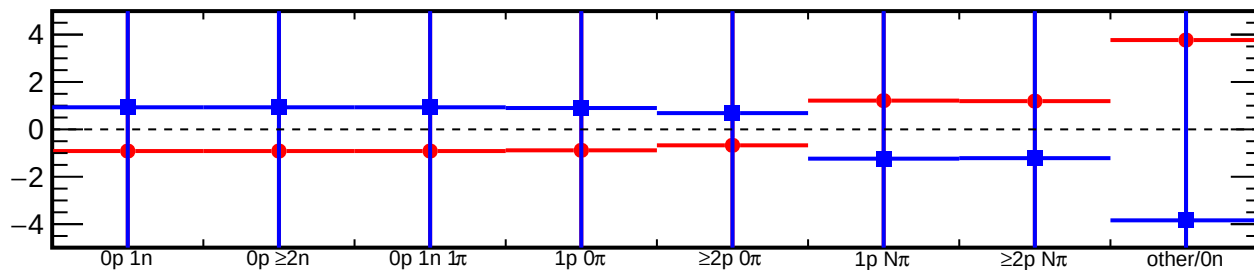


RHC [DIS]: DISvbnCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=3.33$ $\chi^2_{\text{shape},-1\sigma}=3.45$ $\Delta\langle x\rangle_{+1\sigma}=+0.0279$ $\Delta\langle x\rangle_{-1\sigma}=-0.0284$
 $\chi^2_{\text{raw},+1\sigma}=4.83$ $\chi^2_{\text{raw},-1\sigma}=4.83$ rate: +0.92% / -0.92% $\Delta\sigma^2$: +0.0385 / -0.0408

Events

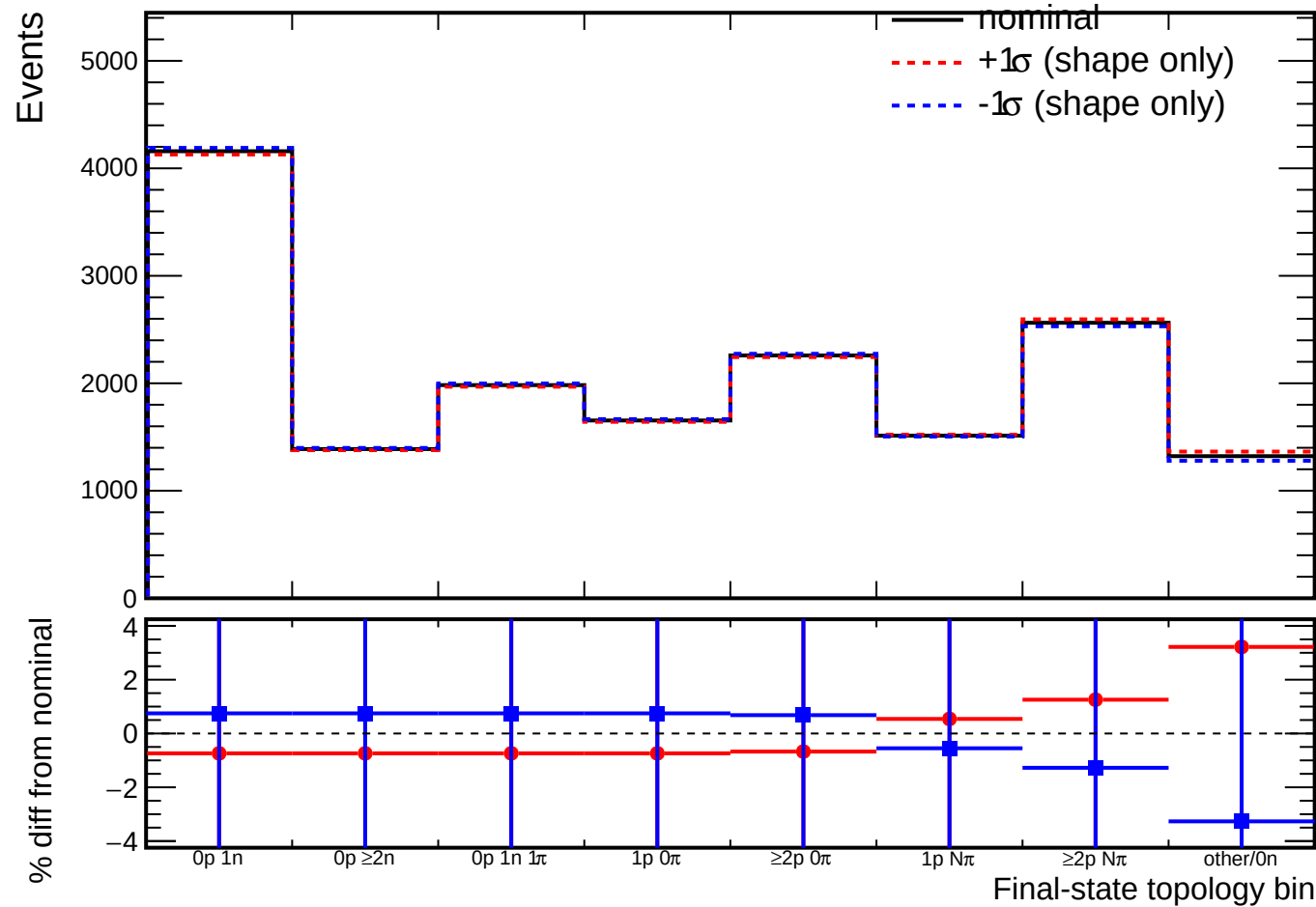


% diff from nominal

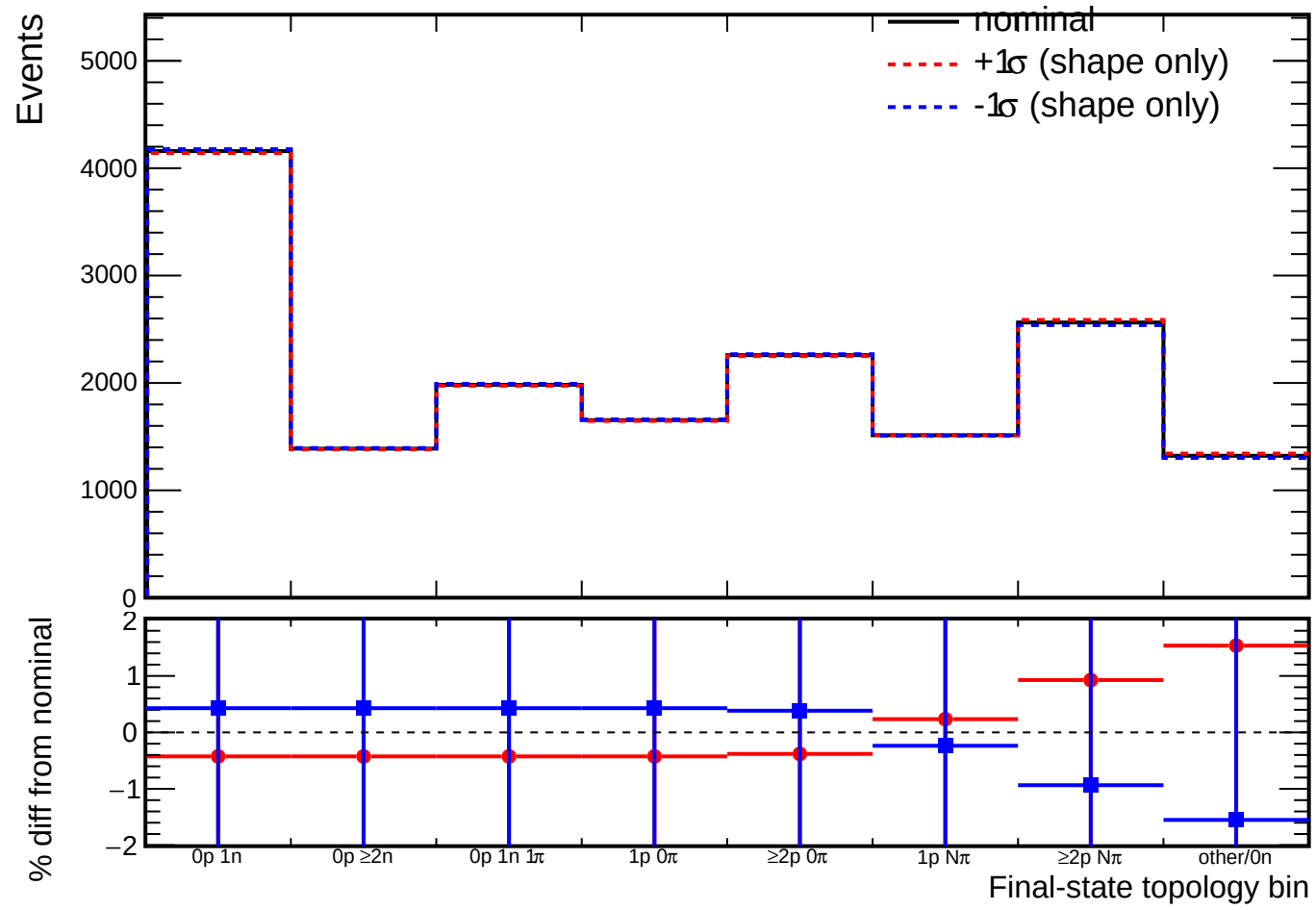


Final-state topology bin

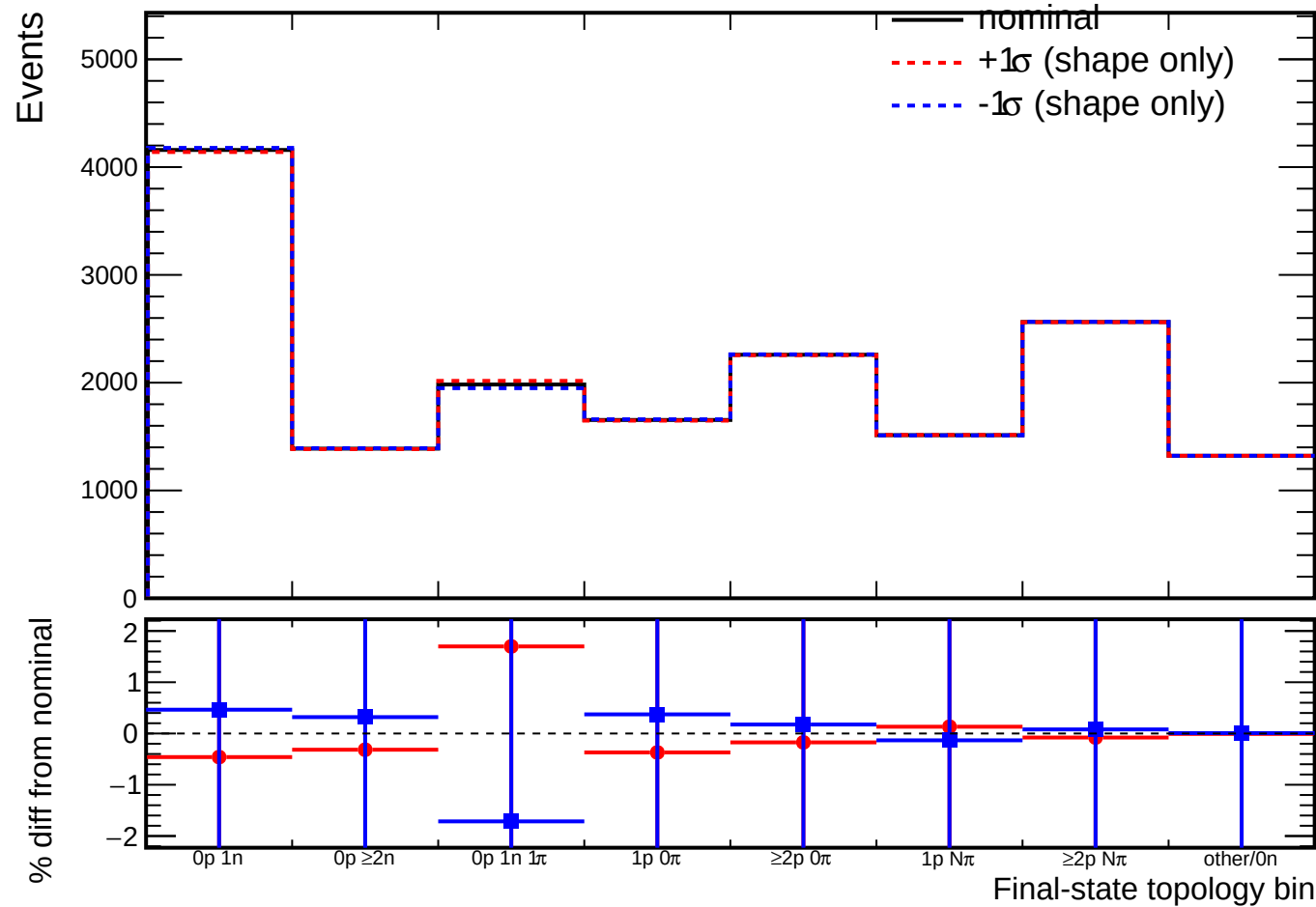
RHC [DIS]: DISvbarpCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=2.42$ $\chi^2_{\text{shape},-1\sigma}=2.49$ $\Delta\langle x\rangle_{+1\sigma}=+0.0235$ $\Delta\langle x\rangle_{-1\sigma}=-0.0238$
 $\chi^2_{\text{raw},+1\sigma}=3.39$ $\chi^2_{\text{raw},-1\sigma}=3.39$ rate: +0.74% / -0.74% $\Delta\sigma^2$: +0.0356 / -0.0372



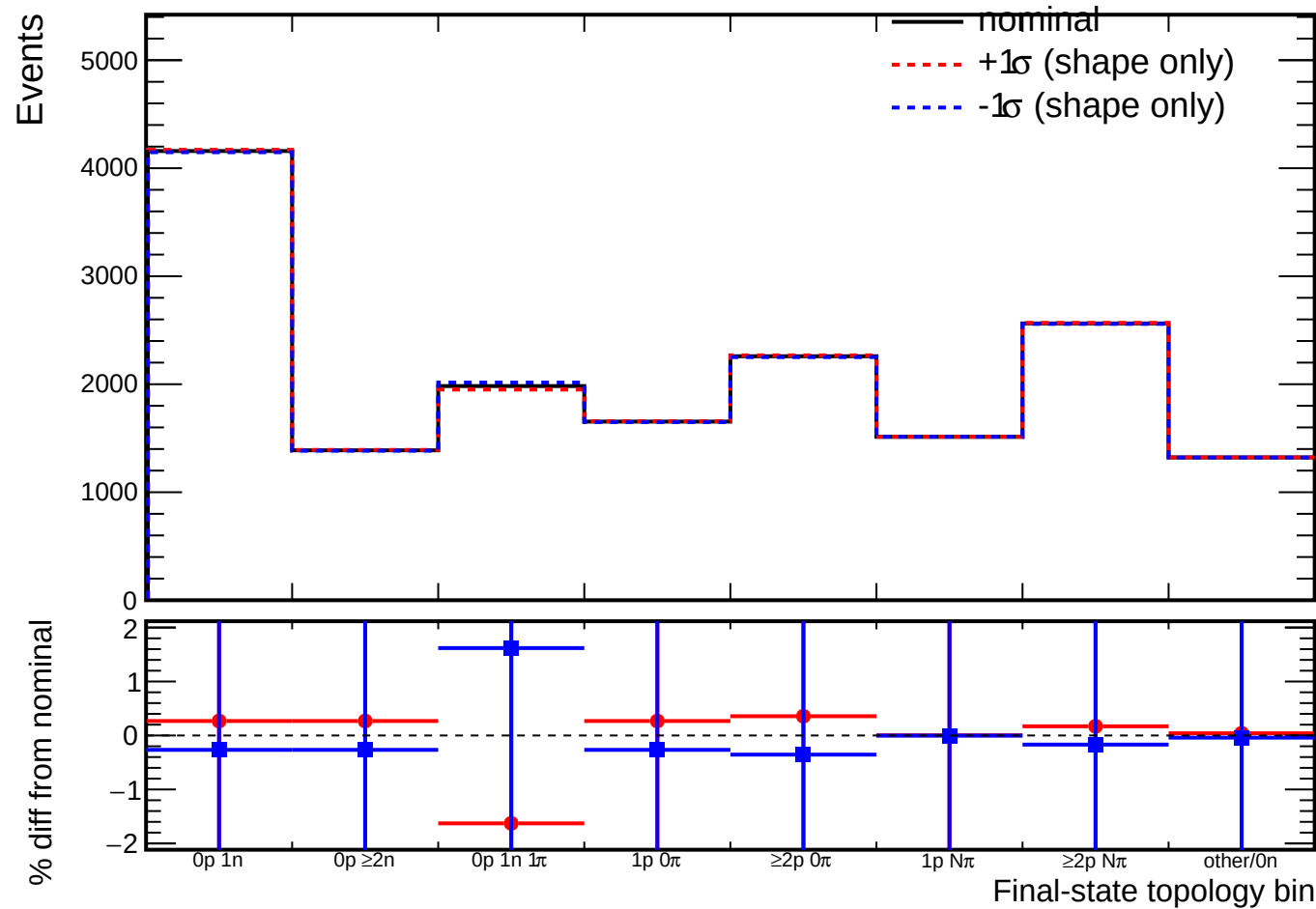
RHC [DIS]: DISvbnCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.74$ $\chi^2_{\text{shape},-1\sigma}=0.75$ $\Delta\langle x\rangle_{+1\sigma}=+0.0133$ $\Delta\langle x\rangle_{-1\sigma}=-0.0134$
 $\chi^2_{\text{raw},+1\sigma}=1.05$ $\chi^2_{\text{raw},-1\sigma}=1.05$ rate: +0.43% / -0.43% $\Delta\sigma^2$: +0.0192 / -0.0197



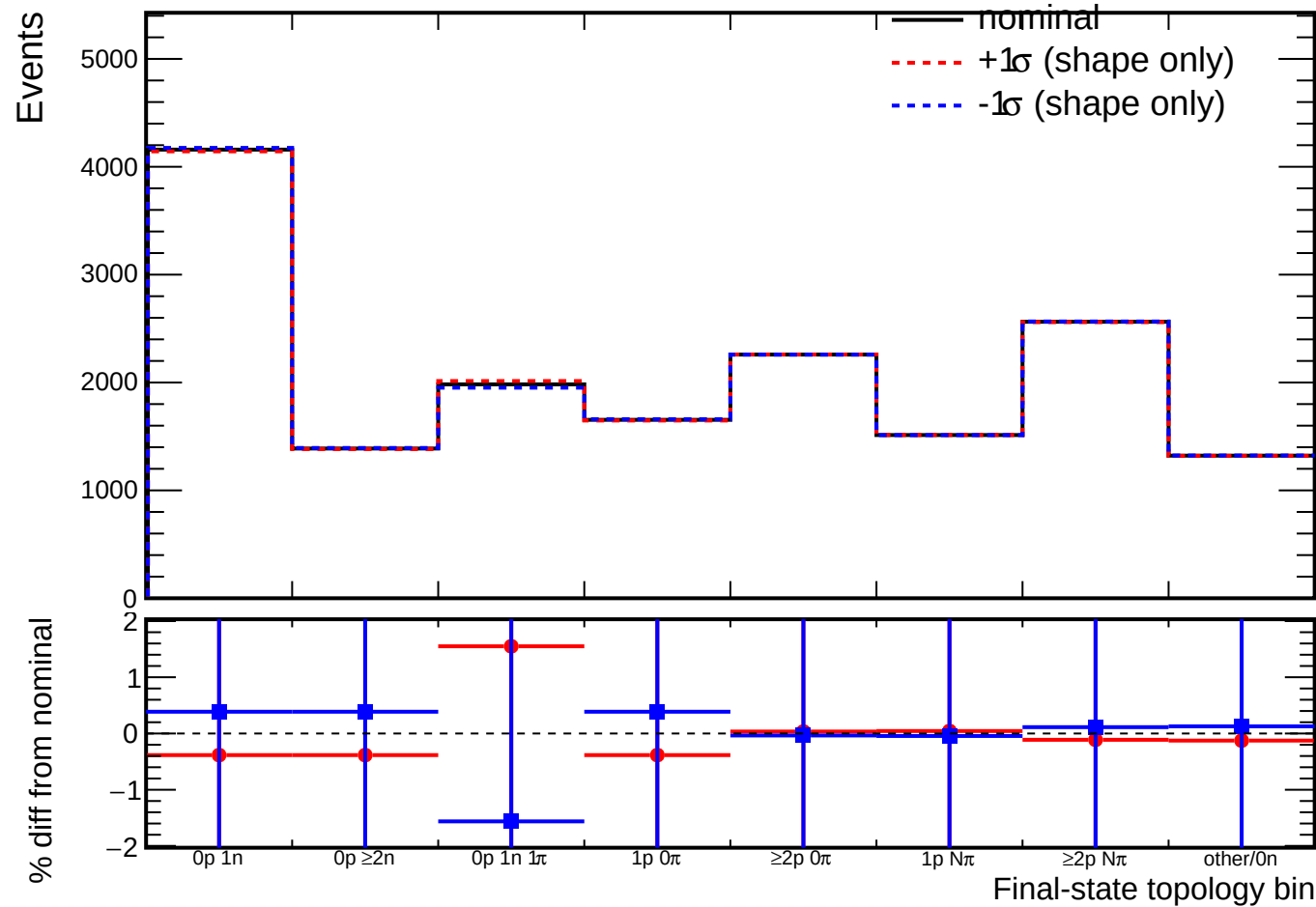
RHC [DIS]: DiSvbarnCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.71$ $\chi^2_{\text{shape},-1\sigma}=0.72$ $\Delta\langle x\rangle_{+1\sigma}=+0.0015$ $\Delta\langle x\rangle_{-1\sigma}=-0.0016$
 $\chi^2_{\text{raw},+1\sigma}=1.07$ $\chi^2_{\text{raw},-1\sigma}=1.07$ rate: +0.46% / -0.46% $\Delta\sigma^2$: -0.0104 / +0.0105



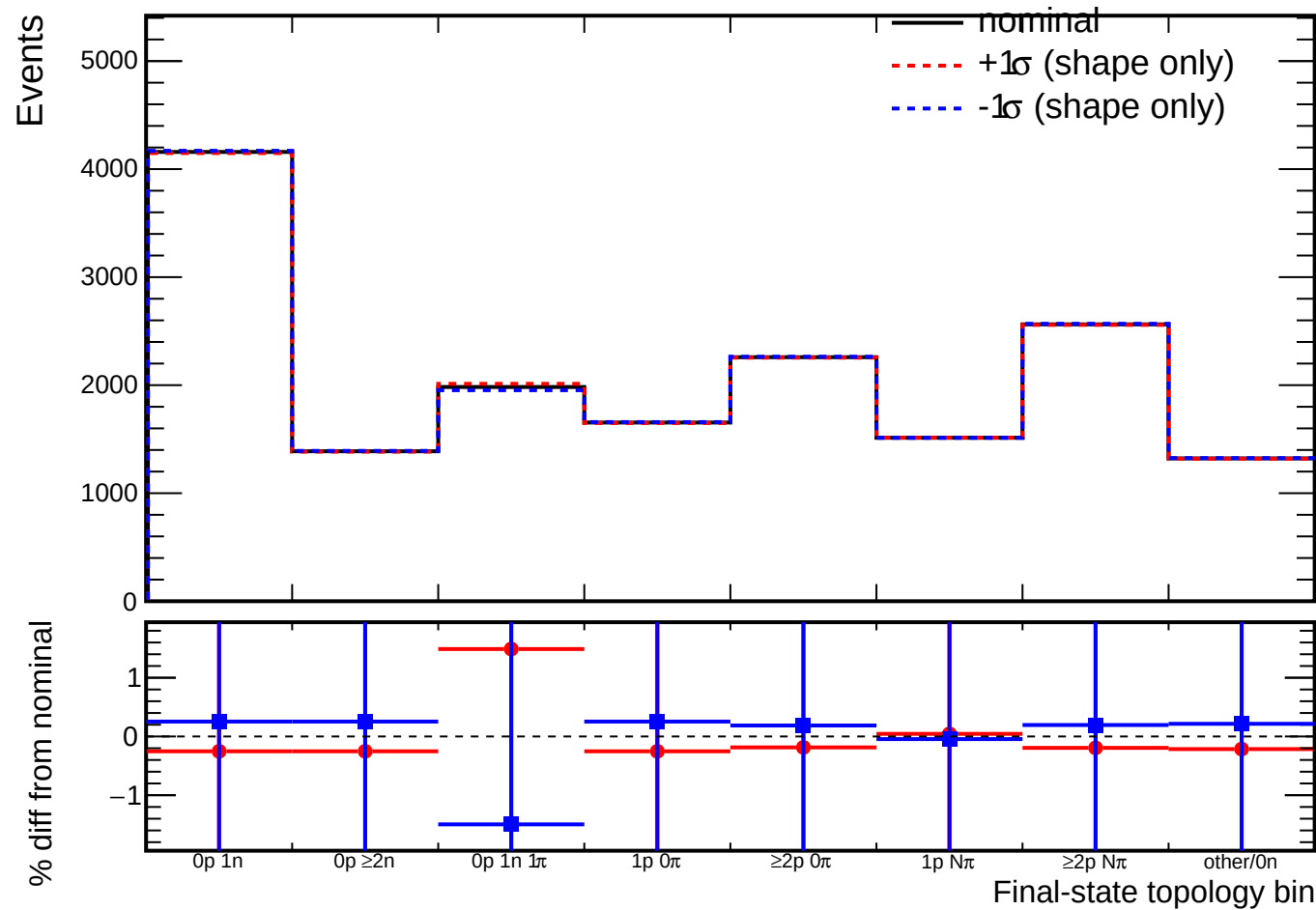
RHC [DIS]: DISNuHadronQ1Syst (shape only) $\chi^2_{\text{shape},+1\sigma}=0.61$ $\chi^2_{\text{shape},-1\sigma}=0.61$ $\Delta\langle x\rangle_{+1\sigma}=+0.0009$ $\Delta\langle x\rangle_{-1\sigma}=-0.0009$
 $\chi^2_{\text{raw},+1\sigma}=0.73$ $\chi^2_{\text{raw},-1\sigma}=0.73$ rate: -0.27% / +0.27% $\Delta\sigma^2$: +0.0081 / -0.0081



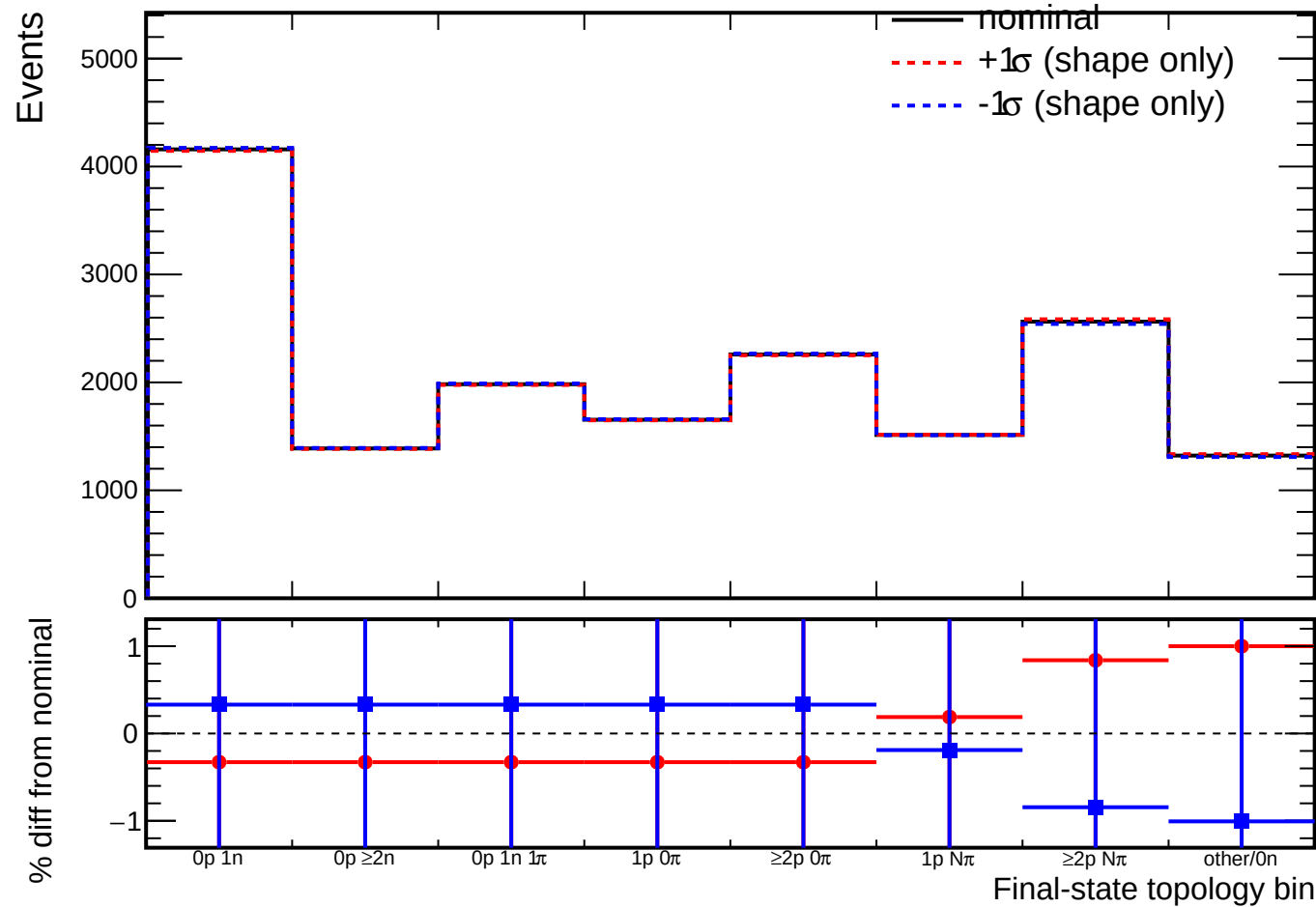
RHC [DIS]: DISvncC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.59$ $\chi^2_{\text{shape},-1\sigma}=0.60$ $\Delta\langle x \rangle_{+1\sigma}=+0.0009$ $\Delta\langle x \rangle_{-1\sigma}=-0.0009$
 $\chi^2_{\text{raw},+1\sigma}=0.84$ $\chi^2_{\text{raw},-1\sigma}=0.84$ rate: +0.39% / -0.39% $\Delta\sigma^2$: -0.0110 / +0.0111



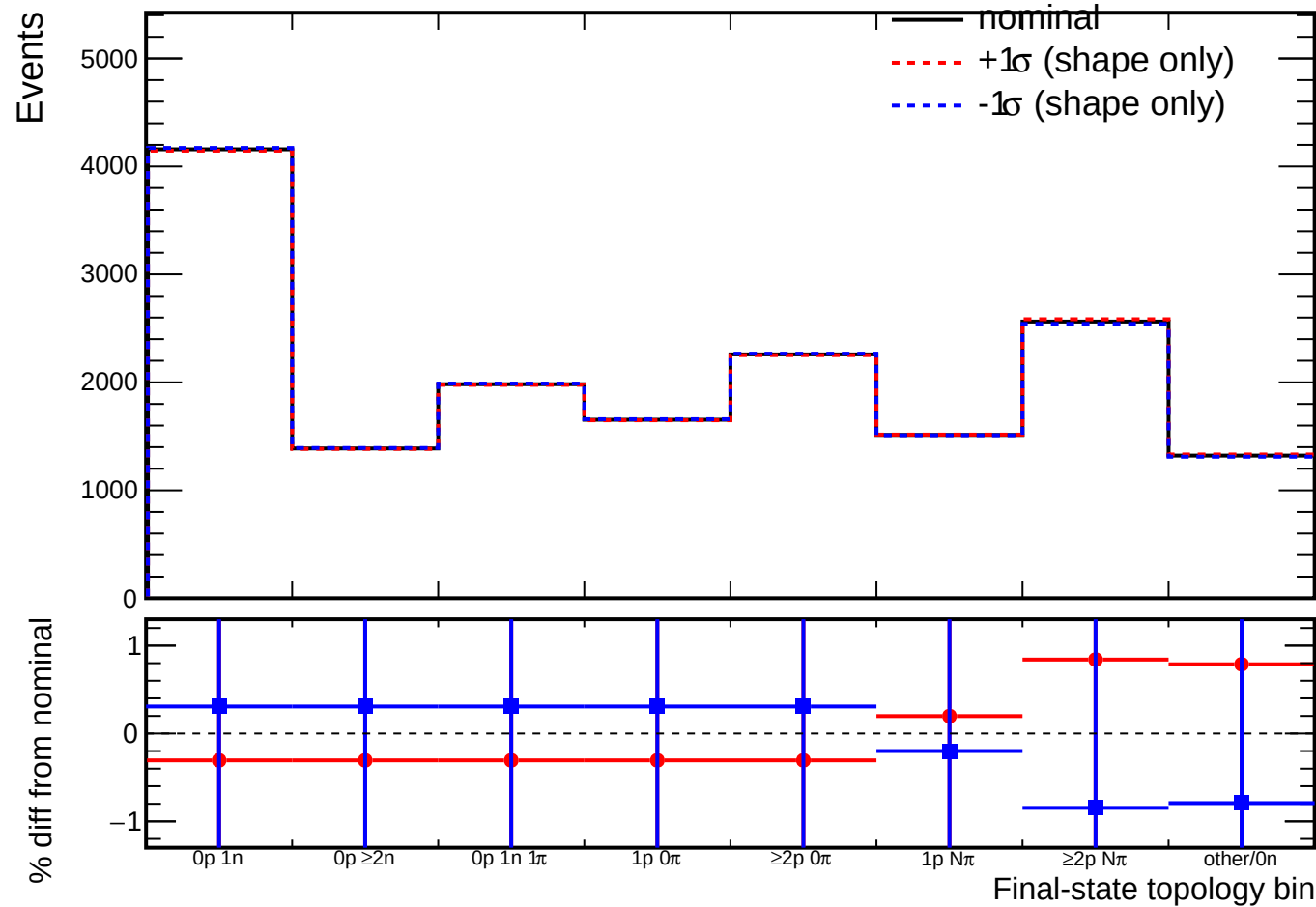
RHC [DIS]: DISvbarpNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.51$ $\chi^2_{\text{shape},-1\sigma}=0.51$ $\Delta\langle x\rangle_{+1\sigma}=-0.0012$ $\Delta\langle x\rangle_{-1\sigma}=+0.0012$
 $\chi^2_{\text{raw},+1\sigma}=0.62$ $\chi^2_{\text{raw},-1\sigma}=0.62$ rate: +0.25% / -0.25% $\Delta\sigma^2$: -0.0100 / +0.0100



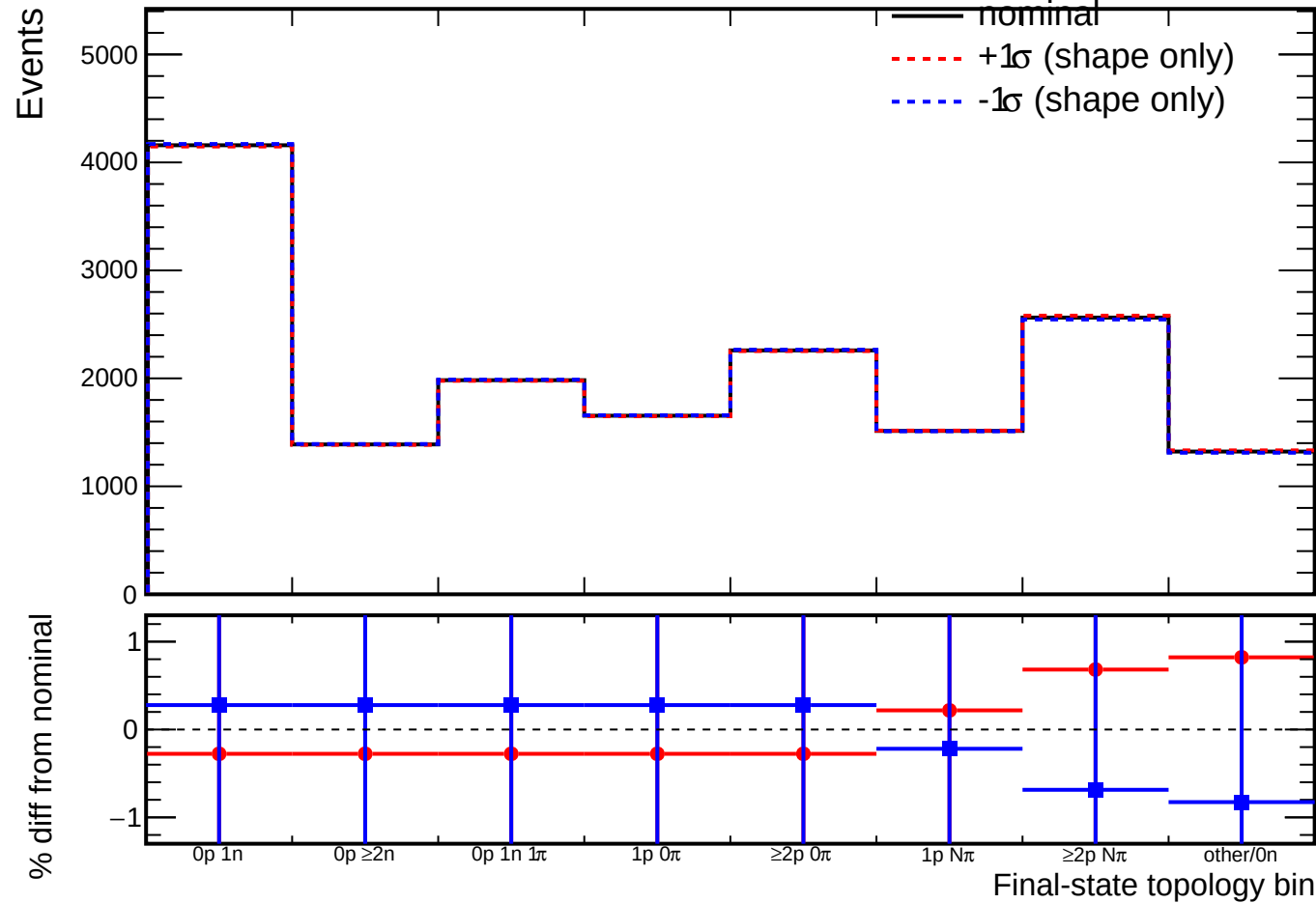
RHC [DIS]: DISvbarpCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.44$ $\chi^2_{\text{shape},-1\sigma}=0.45$ $\Delta\langle x\rangle_{+1\sigma}=+0.0102$ $\Delta\langle x\rangle_{-1\sigma}=-0.0103$
 $\chi^2_{\text{raw},+1\sigma}=0.63$ $\chi^2_{\text{raw},-1\sigma}=0.63$ rate: +0.33% / -0.33% $\Delta\sigma^2$: +0.0142 / -0.0145



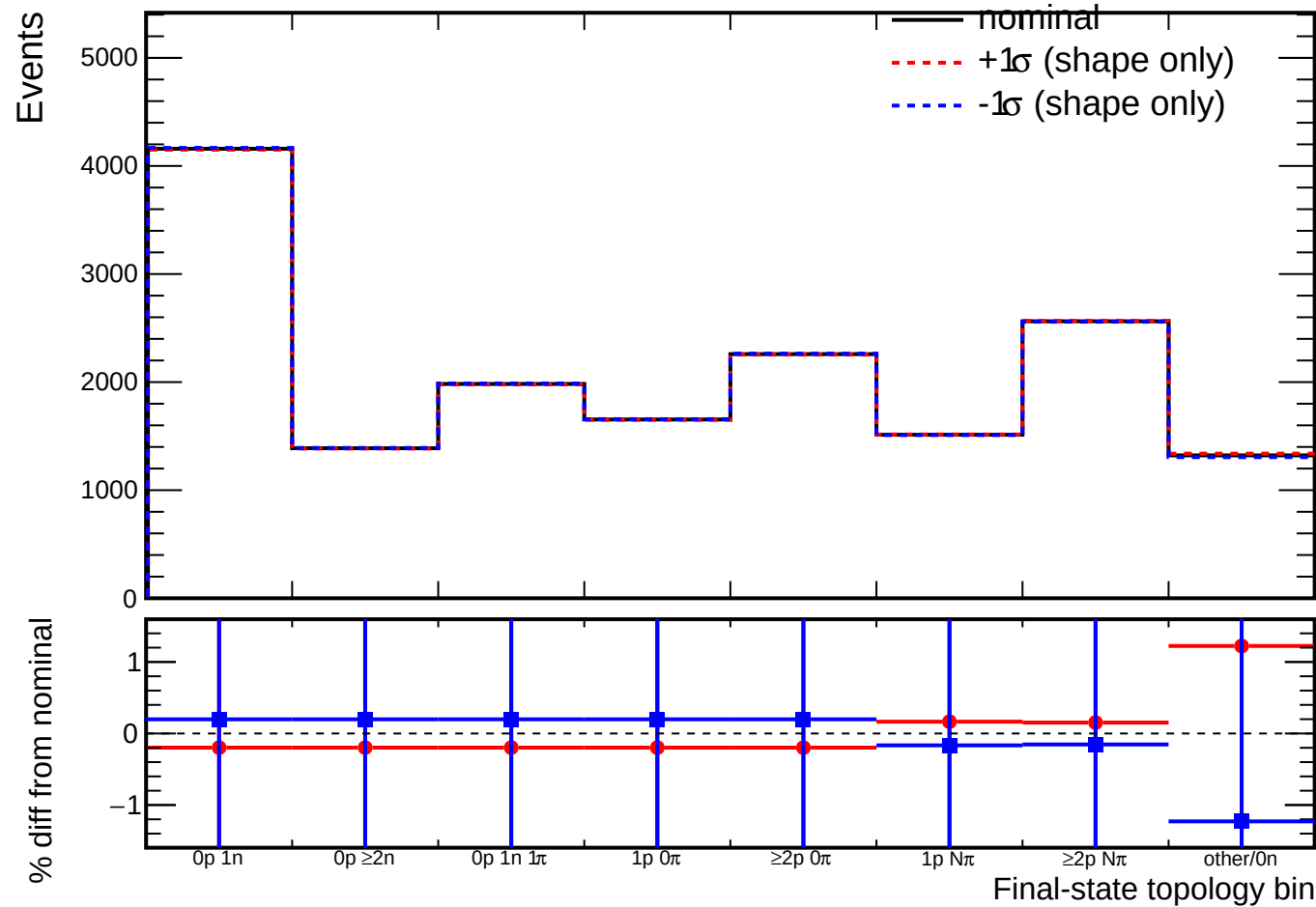
RHC [DIS]: DISvCC3pi_2020 (shape only) $\chi^2_{\text{shape}, +1\sigma} = 0.38$ $\chi^2_{\text{shape}, -1\sigma} = 0.38$ $\Delta\langle x \rangle_{+1\sigma} = +0.0094$ $\Delta\langle x \rangle_{-1\sigma} = -0.0094$
 $\chi^2_{\text{raw}, +1\sigma} = 0.54$ $\chi^2_{\text{raw}, -1\sigma} = 0.54$ rate: +0.31% / -0.31% $\Delta\sigma^2$: +0.0123 / -0.0126



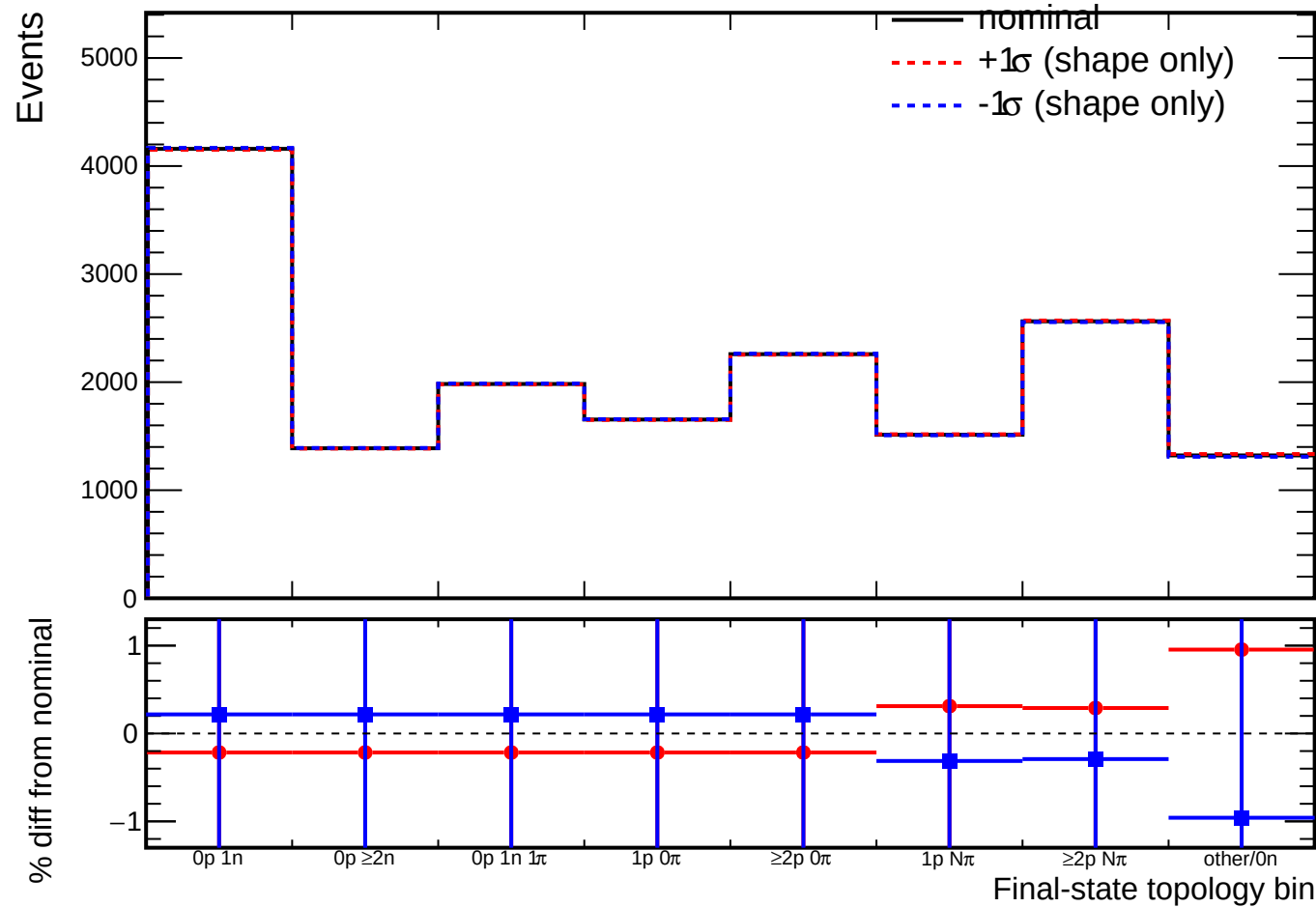
RHC [DIS]: DISvpCC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.30$ $\chi^2_{\text{shape},-1\sigma}=0.31$ $\Delta\langle x \rangle_{+1\sigma}=+0.0085$ $\Delta\langle x \rangle_{-1\sigma}=-0.0086$
 $\chi^2_{\text{raw},+1\sigma}=0.43$ $\chi^2_{\text{raw},-1\sigma}=0.43$ rate: +0.28% / -0.28% $\Delta\sigma^2$: +0.0116 / -0.0118



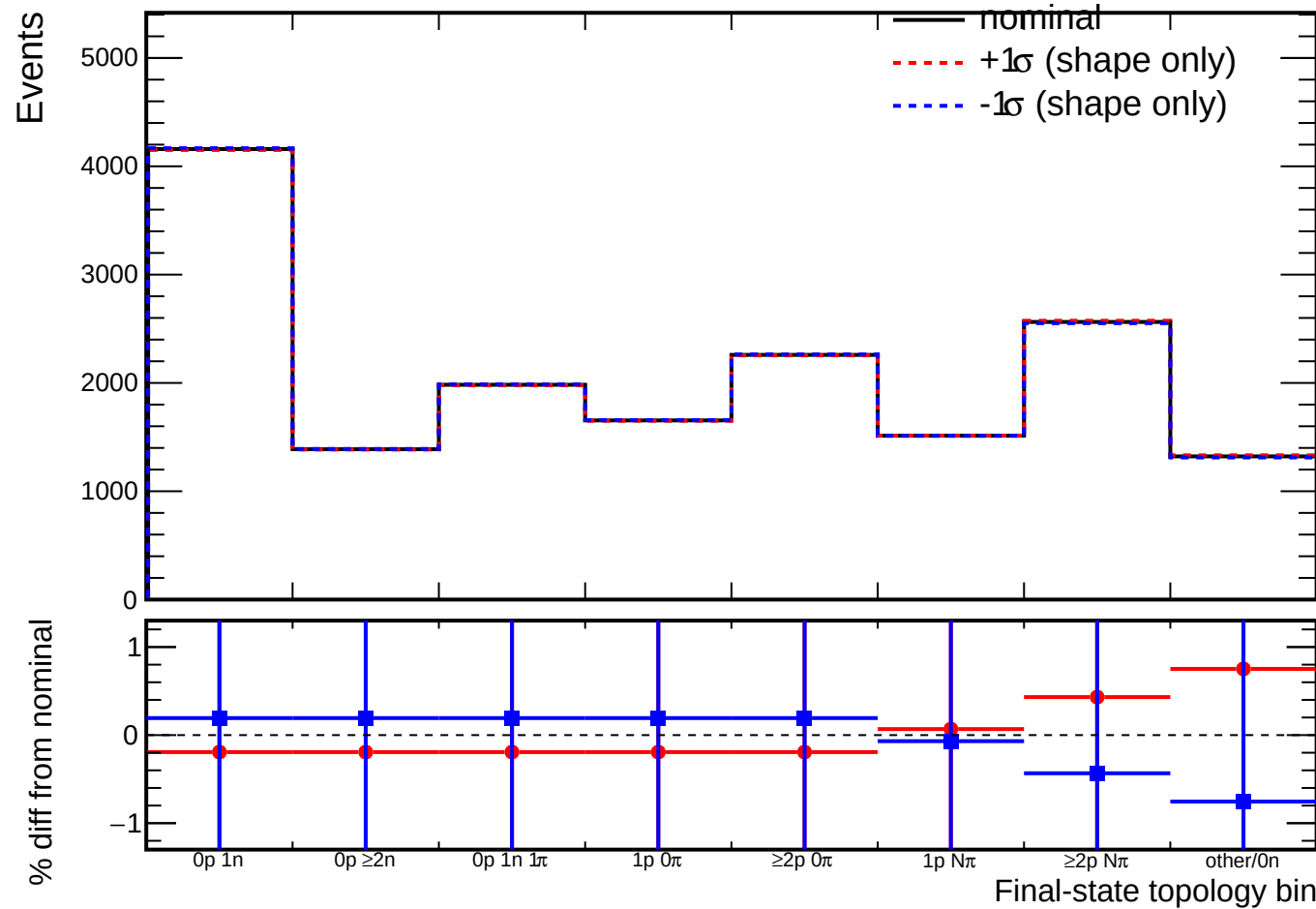
RHC [DIS]: DISvbnrNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.25$ $\chi^2_{\text{shape},-1\sigma}=0.25$ $\Delta\langle x\rangle_{+1\sigma}=+0.0066$ $\Delta\langle x\rangle_{-1\sigma}=-0.0066$
 $\chi^2_{\text{raw},+1\sigma}=0.32$ $\chi^2_{\text{raw},-1\sigma}=0.32$ rate: +0.20% / -0.20% $\Delta\sigma^2$: +0.0117 / -0.0118



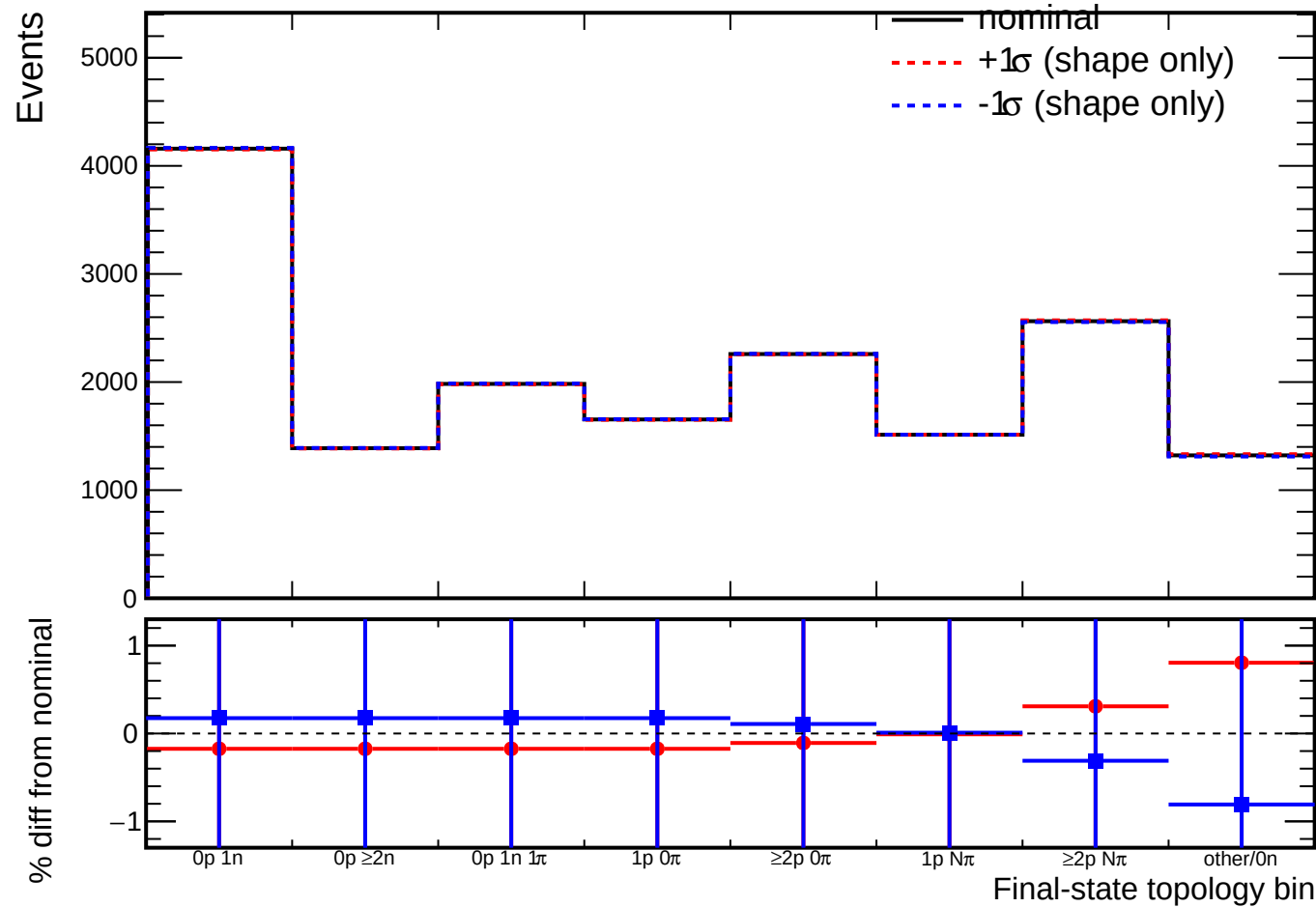
RHC [DIS]: DISvnCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.21$ $\chi^2_{\text{shape},-1\sigma}=0.21$ $\Delta\langle x \rangle_{+1\sigma}=+0.0068$ $\Delta\langle x \rangle_{-1\sigma}=-0.0068$
 $\chi^2_{\text{raw},+1\sigma}=0.29$ $\chi^2_{\text{raw},-1\sigma}=0.29$ rate: +0.22% / -0.22% $\Delta\sigma^2$: +0.0101 / -0.0103



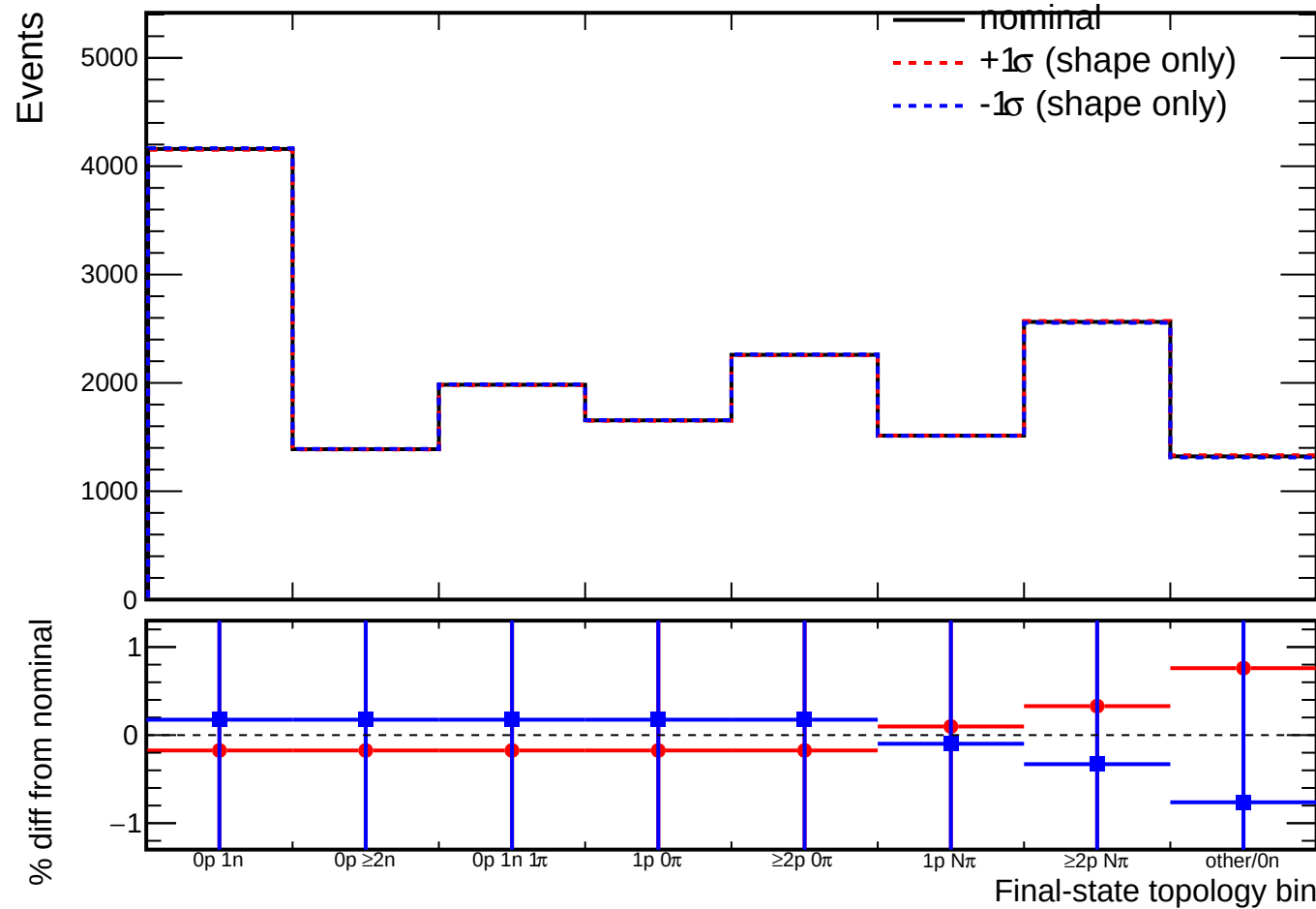
RHC [DIS]: DISvbarpNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.17$ $\chi^2_{\text{shape},-1\sigma}=0.17$ $\Delta\langle x\rangle_{+1\sigma}=+0.0062$ $\Delta\langle x\rangle_{-1\sigma}=-0.0062$
 $\chi^2_{\text{raw},+1\sigma}=0.23$ $\chi^2_{\text{raw},-1\sigma}=0.23$ rate: +0.19% / -0.19% $\Delta\sigma^2$: +0.0094 / -0.0095



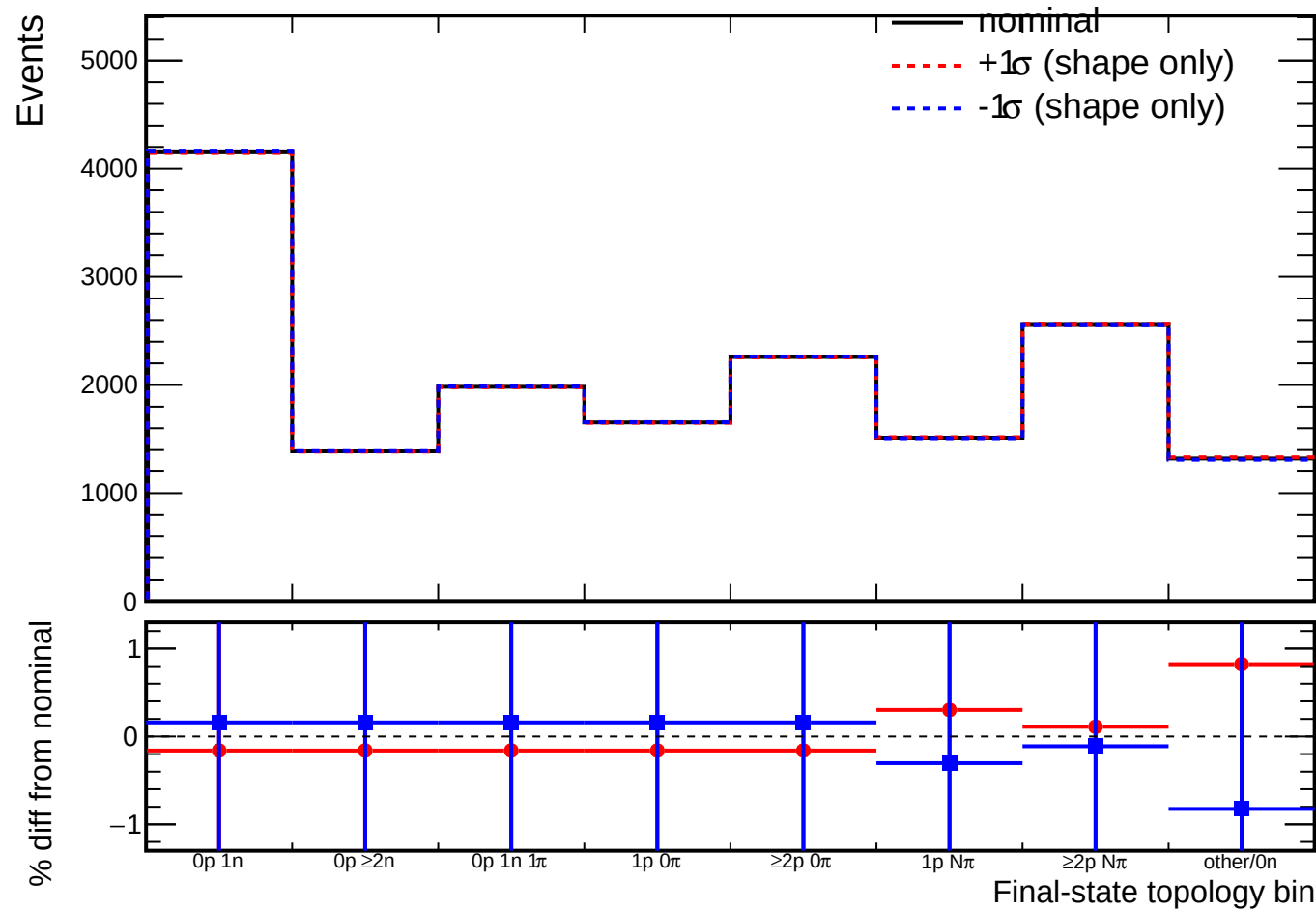
RHC [DIS]: DISvpCC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.14$ $\chi^2_{\text{shape},-1\sigma}=0.14$ $\Delta\langle x\rangle_{+1\sigma}=+0.0056$ $\Delta\langle x\rangle_{-1\sigma}=-0.0056$
 $\chi^2_{\text{raw},+1\sigma}=0.19$ $\chi^2_{\text{raw},-1\sigma}=0.19$ rate: +0.17% / -0.17% $\Delta\sigma^2$: +0.0088 / -0.0089



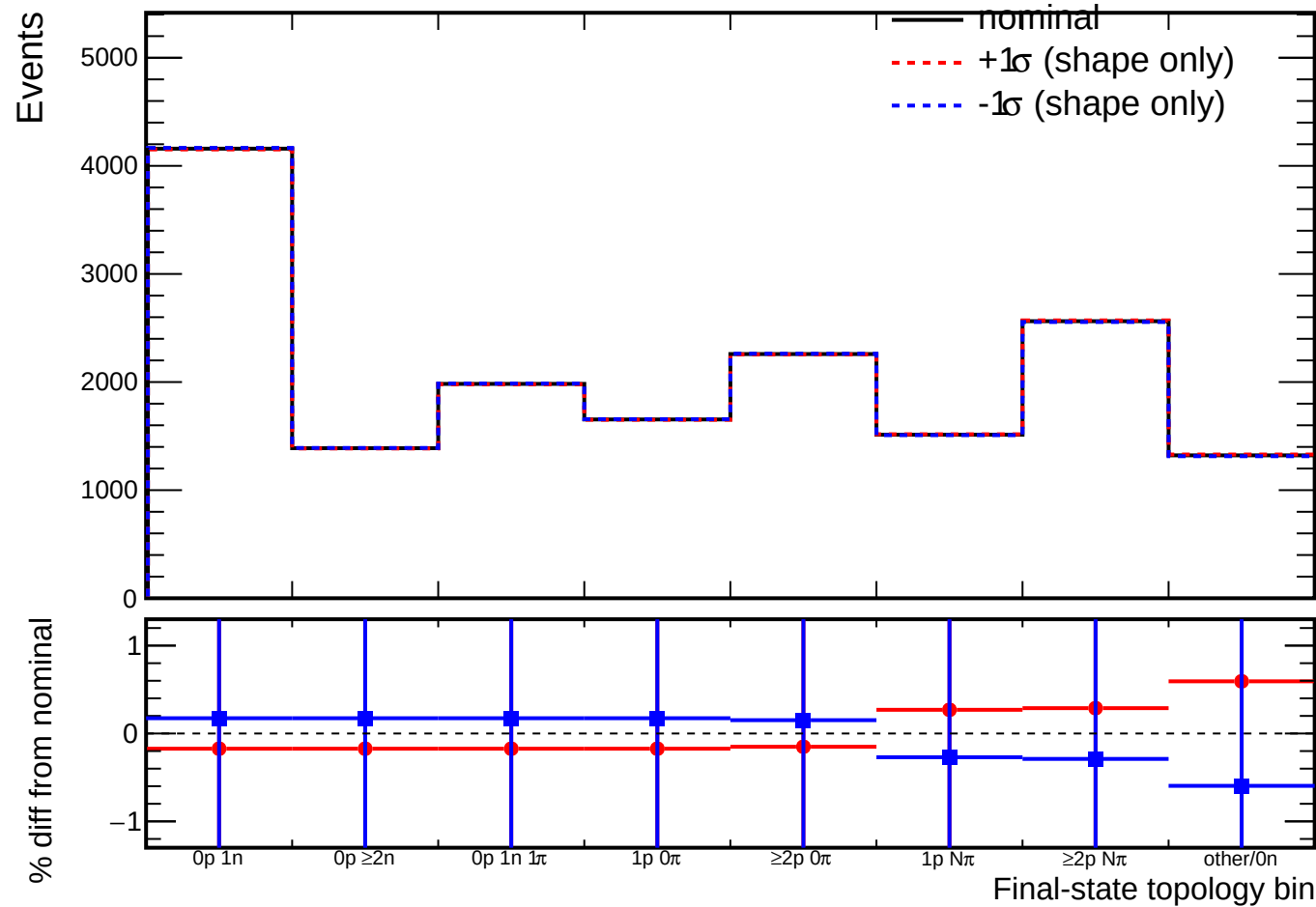
RHC [DIS]: DISvpNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.14$ $\chi^2_{\text{shape},-1\sigma}=0.14$ $\Delta\langle x\rangle_{+1\sigma}=+0.0056$ $\Delta\langle x\rangle_{-1\sigma}=-0.0056$
 $\chi^2_{\text{raw},+1\sigma}=0.19$ $\chi^2_{\text{raw},-1\sigma}=0.19$ rate: +0.17% / -0.17% $\Delta\sigma^2$: +0.0088 / -0.0089



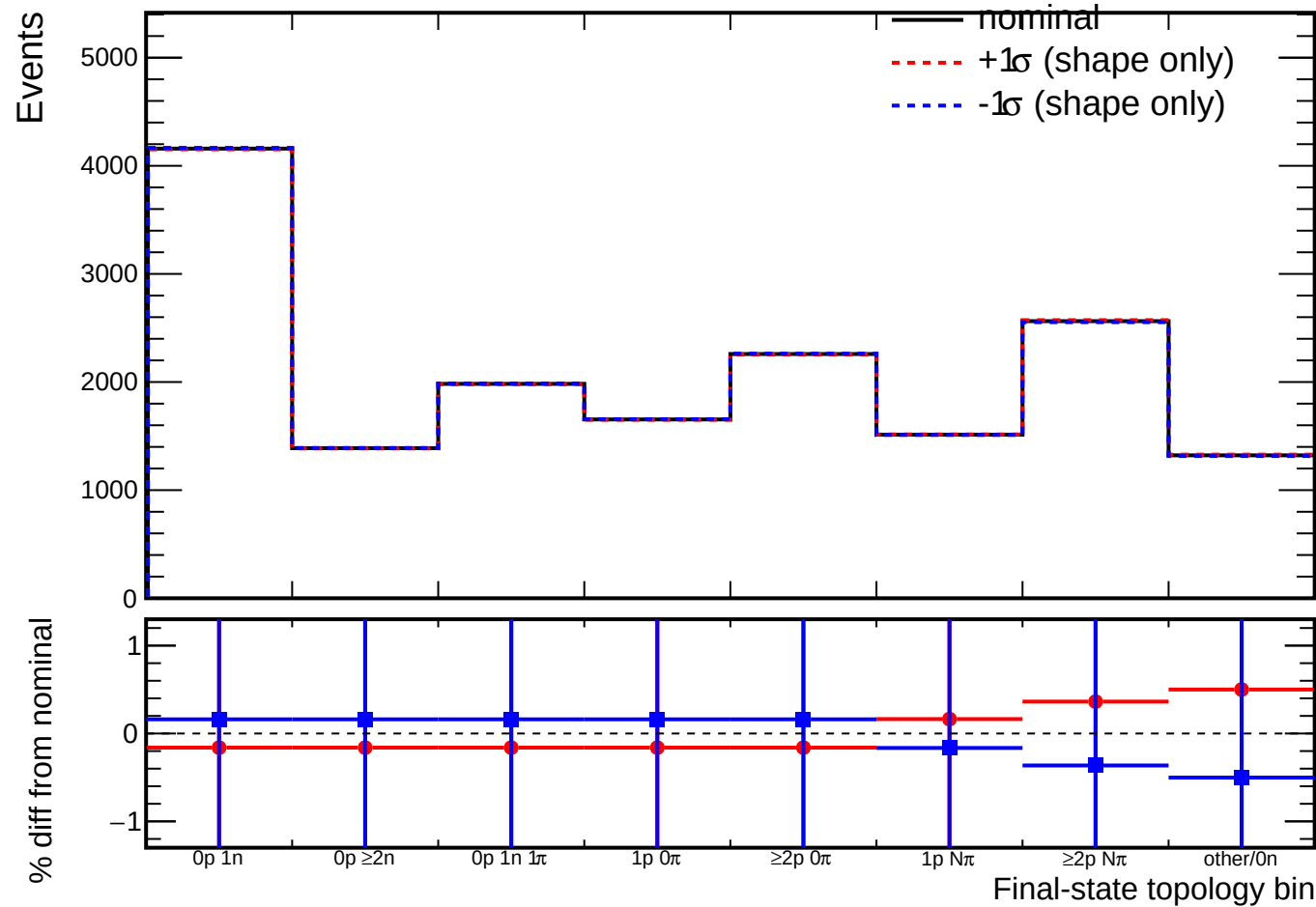
RHC [DIS]: DISvbarpNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.14$ $\chi^2_{\text{shape},-1\sigma}=0.14$ $\Delta\langle x\rangle_{+1\sigma}=+0.0050$ $\Delta\langle x\rangle_{-1\sigma}=-0.0051$
 $\chi^2_{\text{raw},+1\sigma}=0.18$ $\chi^2_{\text{raw},-1\sigma}=0.18$ rate: +0.16% / -0.16% $\Delta\sigma^2$: +0.0078 / -0.0079



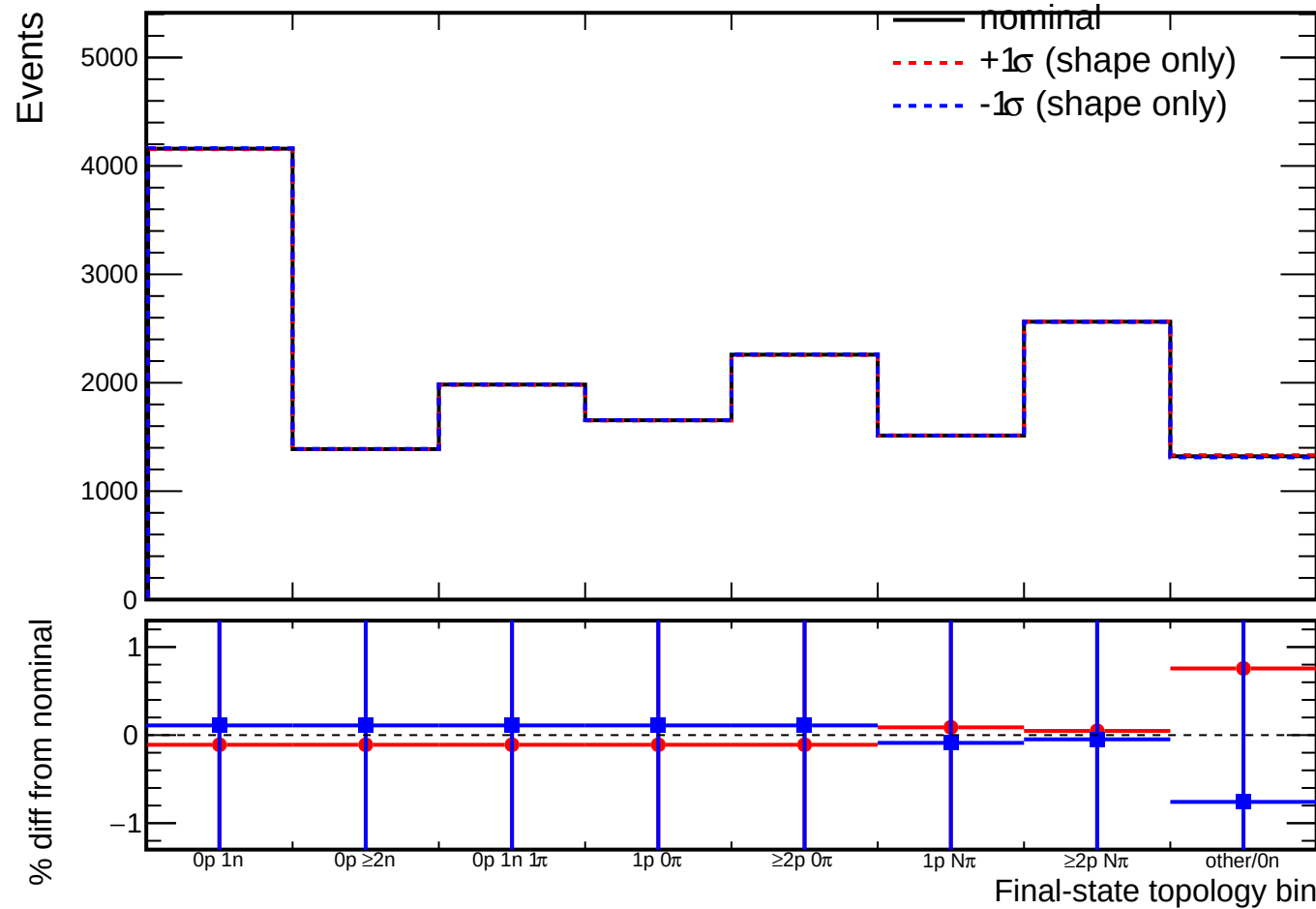
RHC [DIS]: DISvbnNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.11$ $\chi^2_{\text{shape},-1\sigma}=0.11$ $\Delta\langle x\rangle_{+1\sigma}=+0.0052$ $\Delta\langle x\rangle_{-1\sigma}=-0.0053$
 $\chi^2_{\text{raw},+1\sigma}=0.16$ $\chi^2_{\text{raw},-1\sigma}=0.16$ rate: +0.17% / -0.17% $\Delta\sigma^2$: +0.0069 / -0.0070



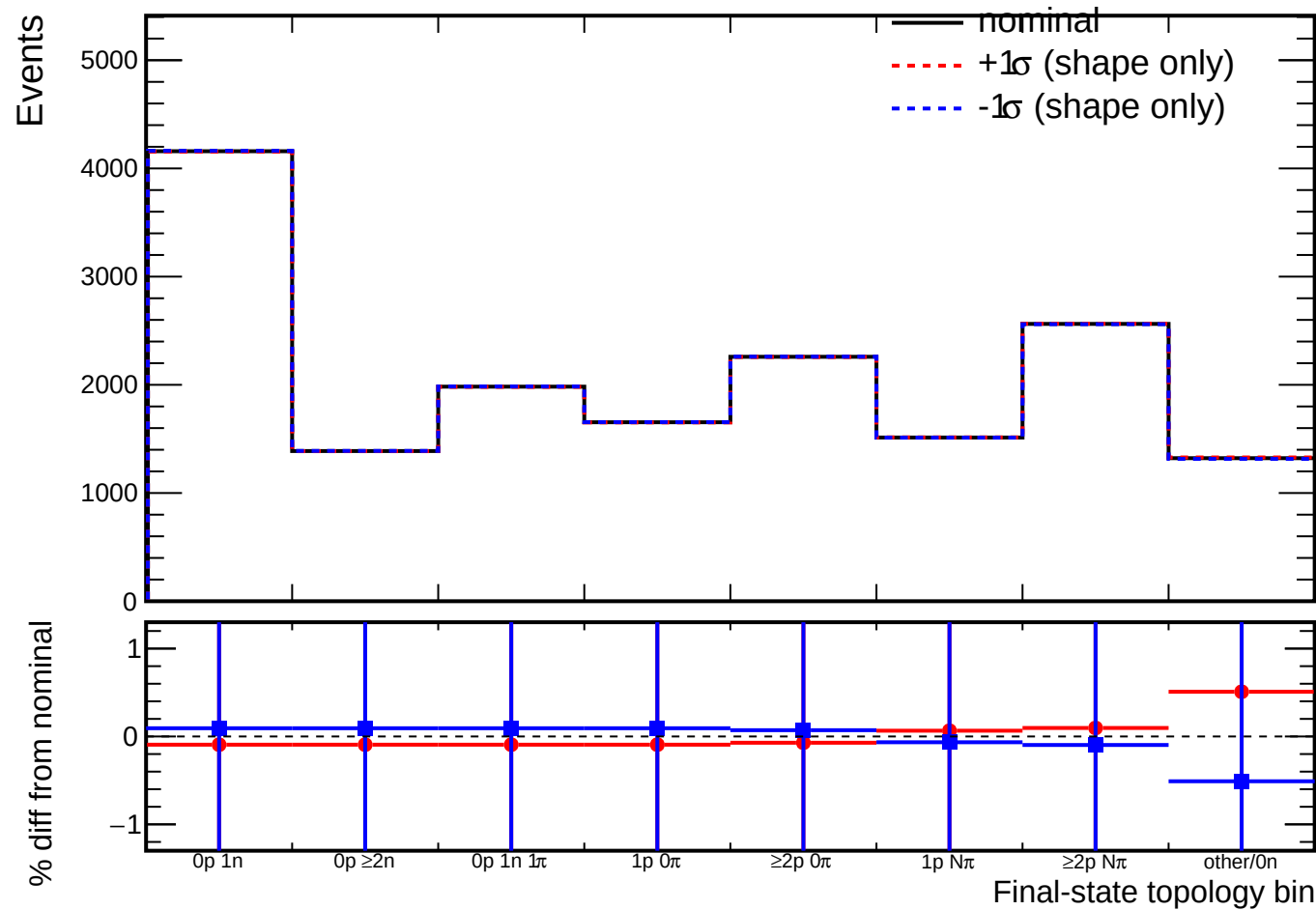
RHC [DIS]: DISvNC3pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.10$ $\chi^2_{\text{shape},-1\sigma}=0.10$ $\Delta\langle x \rangle_{+1\sigma}=+0.0050$ $\Delta\langle x \rangle_{-1\sigma}=-0.0050$
 $\chi^2_{\text{raw},+1\sigma}=0.14$ $\chi^2_{\text{raw},-1\sigma}=0.14$ rate: +0.16% / -0.16% $\Delta\sigma^2$: +0.0067 / -0.0068



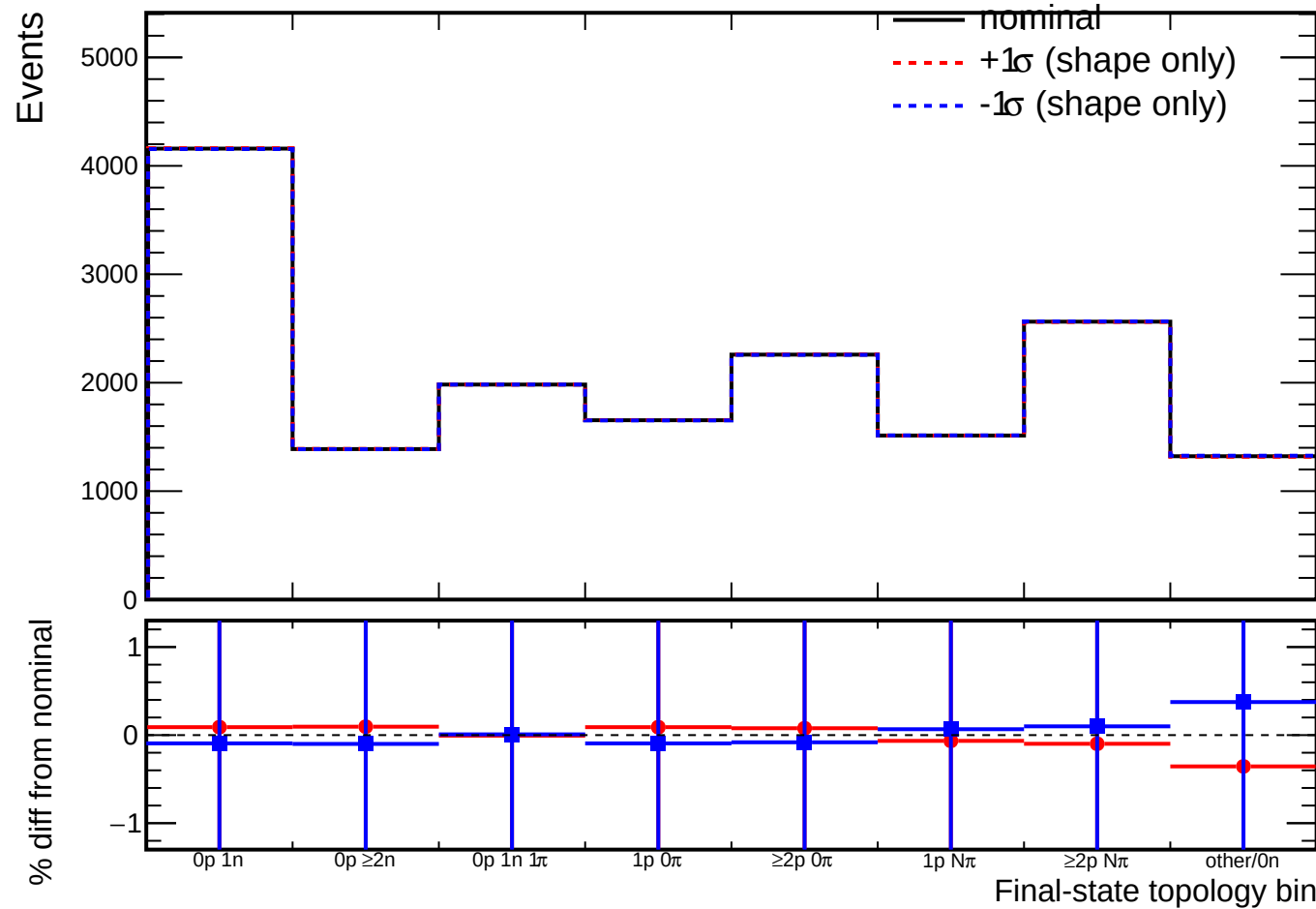
RHC [DIS]: DISvpNC2pi_2020 (shape only) $\chi^2_{\text{shape}, +1\sigma} = 0.09$ $\chi^2_{\text{shape}, -1\sigma} = 0.09$ $\Delta\langle x \rangle_{+1\sigma} = +0.0037$ $\Delta\langle x \rangle_{-1\sigma} = -0.0037$
 $\chi^2_{\text{raw}, +1\sigma} = 0.11$ $\chi^2_{\text{raw}, -1\sigma} = 0.11$ rate: +0.11% / -0.11% $\Delta\sigma^2$: +0.0069 / -0.0070



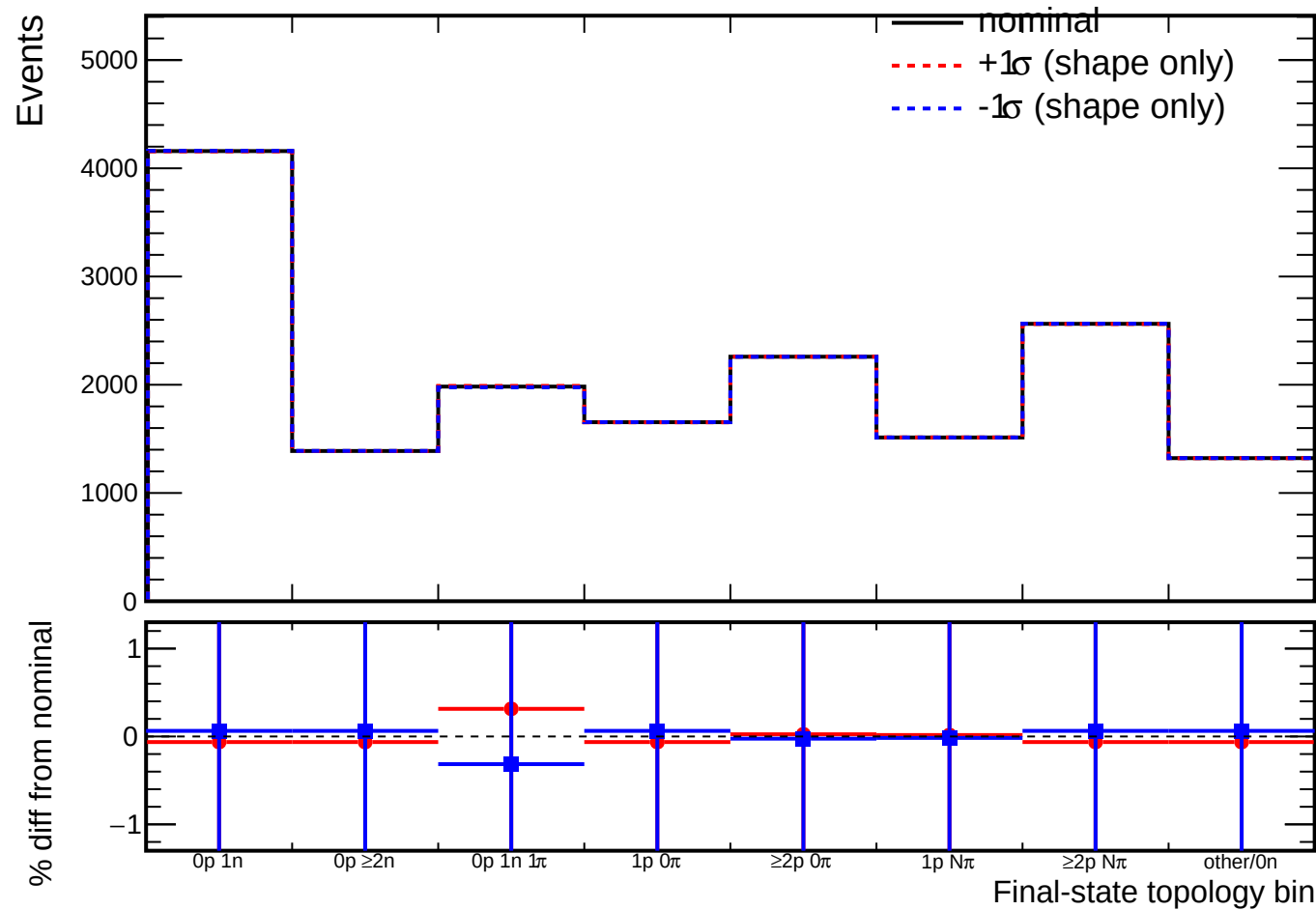
RHC [DIS]: DISvnNC2pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.05$ $\chi^2_{\text{shape},-1\sigma}=0.05$ $\Delta\langle x\rangle_{+1\sigma}=+0.0030$ $\Delta\langle x\rangle_{-1\sigma}=-0.0030$
 $\chi^2_{\text{raw},+1\sigma}=0.06$ $\chi^2_{\text{raw},-1\sigma}=0.06$ rate: +0.09% / -0.09% $\Delta\sigma^2$: +0.0050 / -0.0050



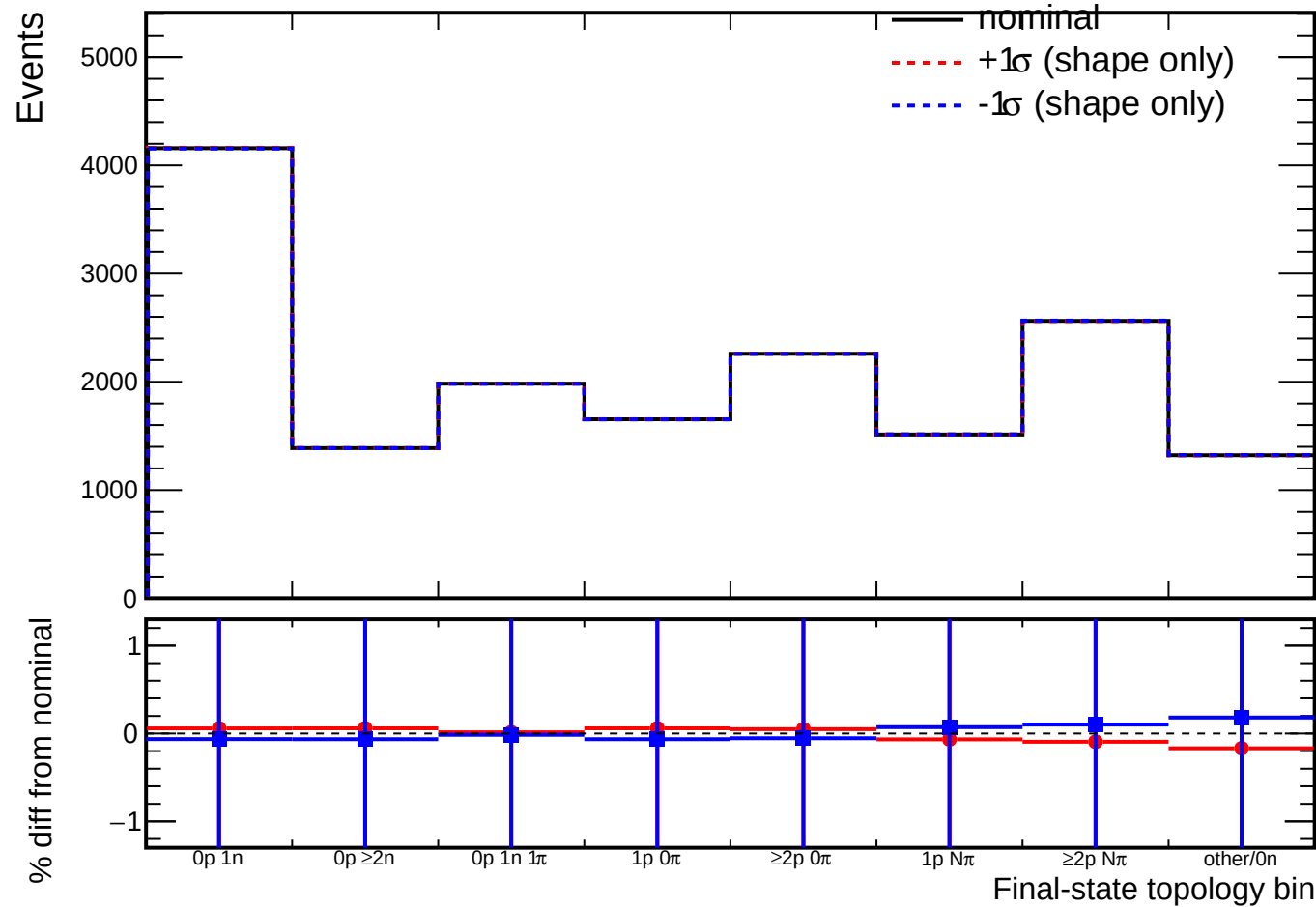
RHC [DIS]: BhtBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.03$ $\chi^2_{\text{shape},-1\sigma}=0.03$ $\Delta\langle x\rangle_{+1\sigma}=-0.0024$ $\Delta\langle x\rangle_{-1\sigma}=+0.0025$
 $\chi^2_{\text{raw},+1\sigma}=0.04$ $\chi^2_{\text{raw},-1\sigma}=0.05$ rate: -0.10% / +0.10% $\Delta\sigma^2$: -0.0033 / +0.0035



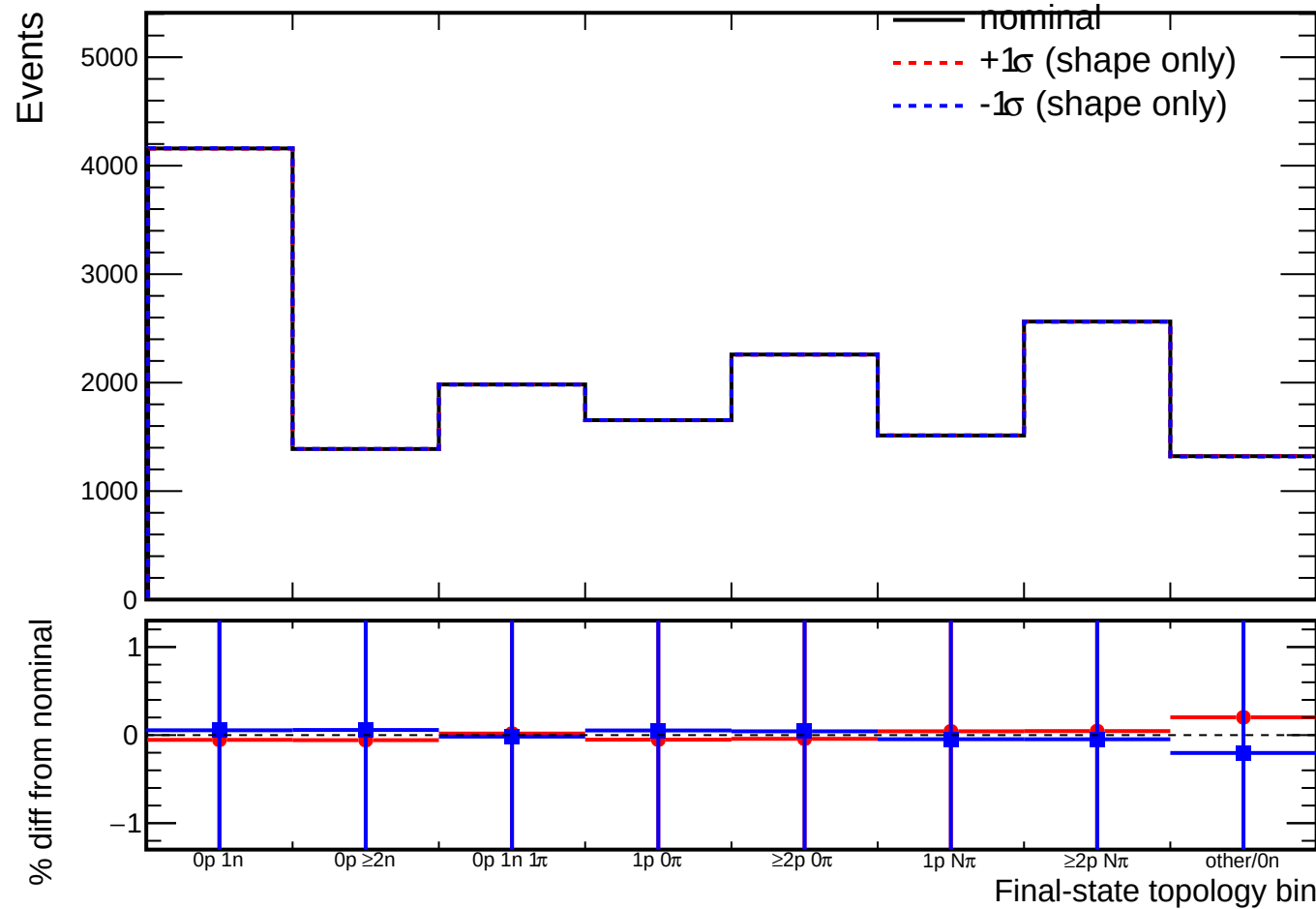
RHC [DIS]: DISvpNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.02$ $\chi^2_{\text{shape},-1\sigma}=0.02$ $\Delta\langle x\rangle_{+1\sigma}=-0.0002$ $\Delta\langle x\rangle_{-1\sigma}=+0.0002$
 $\chi^2_{\text{raw},+1\sigma}=0.03$ $\chi^2_{\text{raw},-1\sigma}=0.03$ rate: +0.06% / -0.06% $\Delta\sigma^2$: -0.0028 / +0.0028



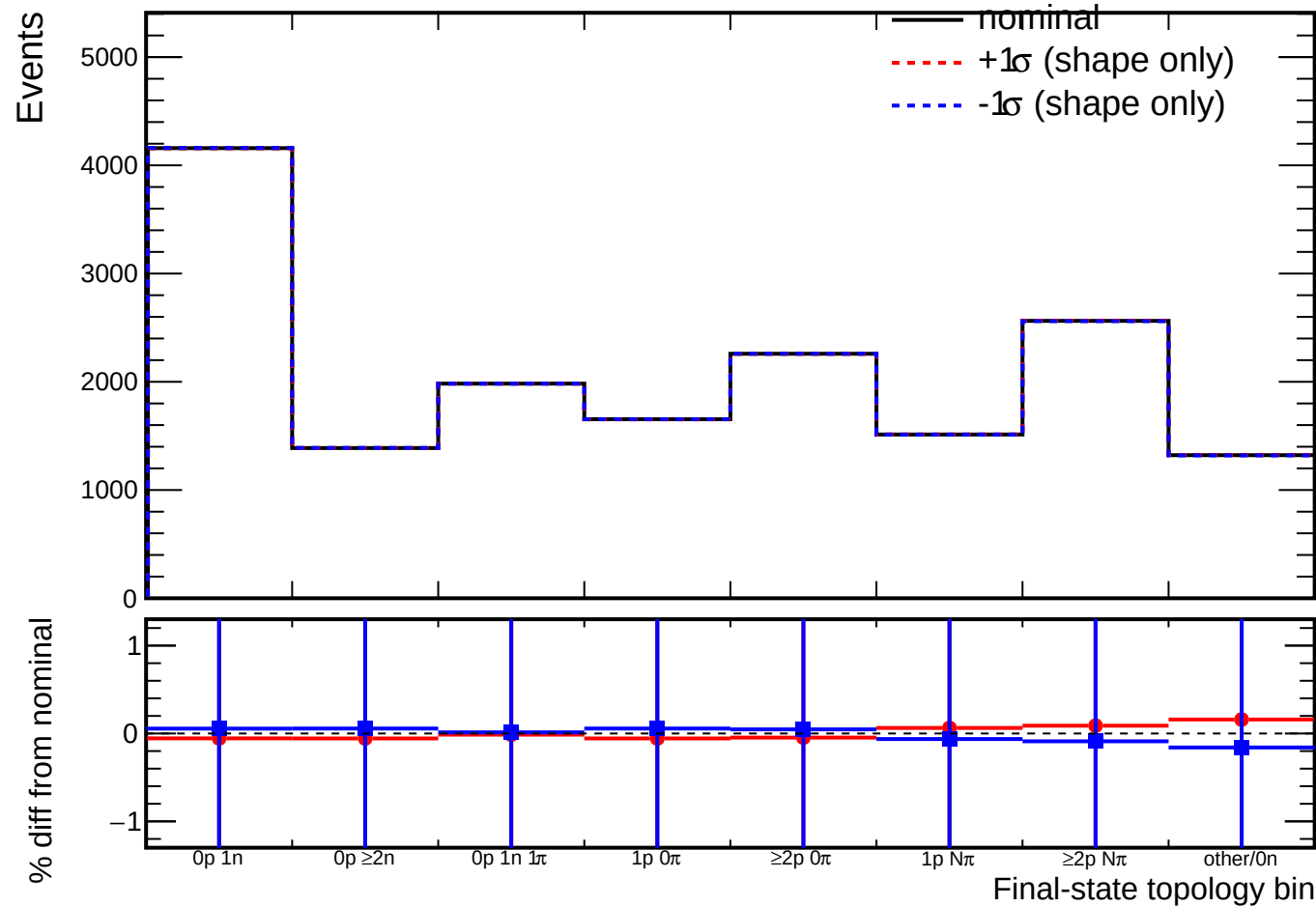
RHC [DIS]: CV1uBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=-0.0016$ $\Delta\langle x\rangle_{-1\sigma}=+0.0017$
 $\chi^2_{\text{raw},+1\sigma}=0.02$ $\chi^2_{\text{raw},-1\sigma}=0.02$ rate: -0.06% / +0.06% $\Delta\sigma^2$: -0.0019 / +0.0020



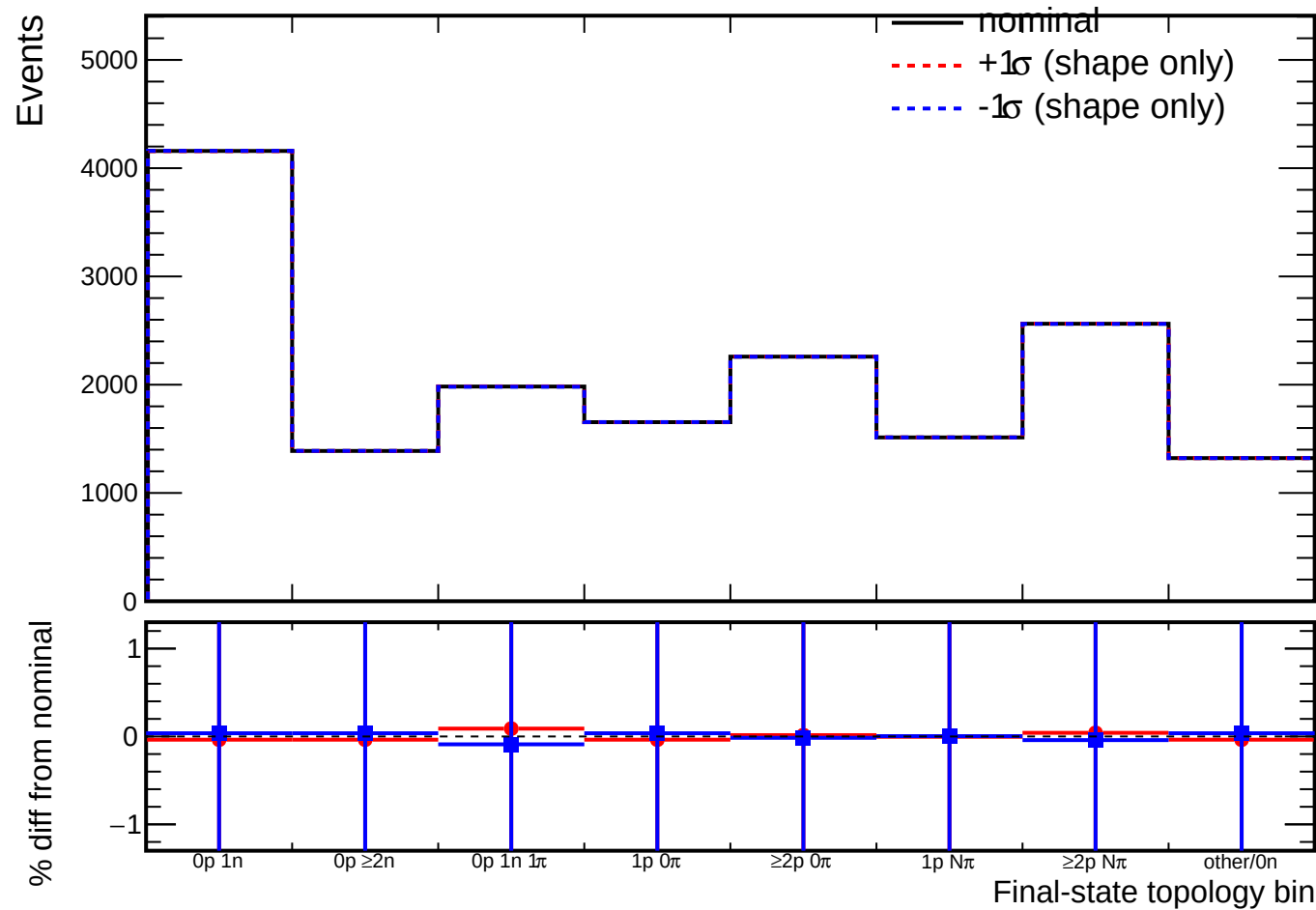
RHC [DIS]: AntiB (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=+0.0013$ $\Delta\langle x\rangle_{-1\sigma}=-0.0014$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.06% / -0.06% $\Delta\sigma^2$: +0.0017 / -0.0017



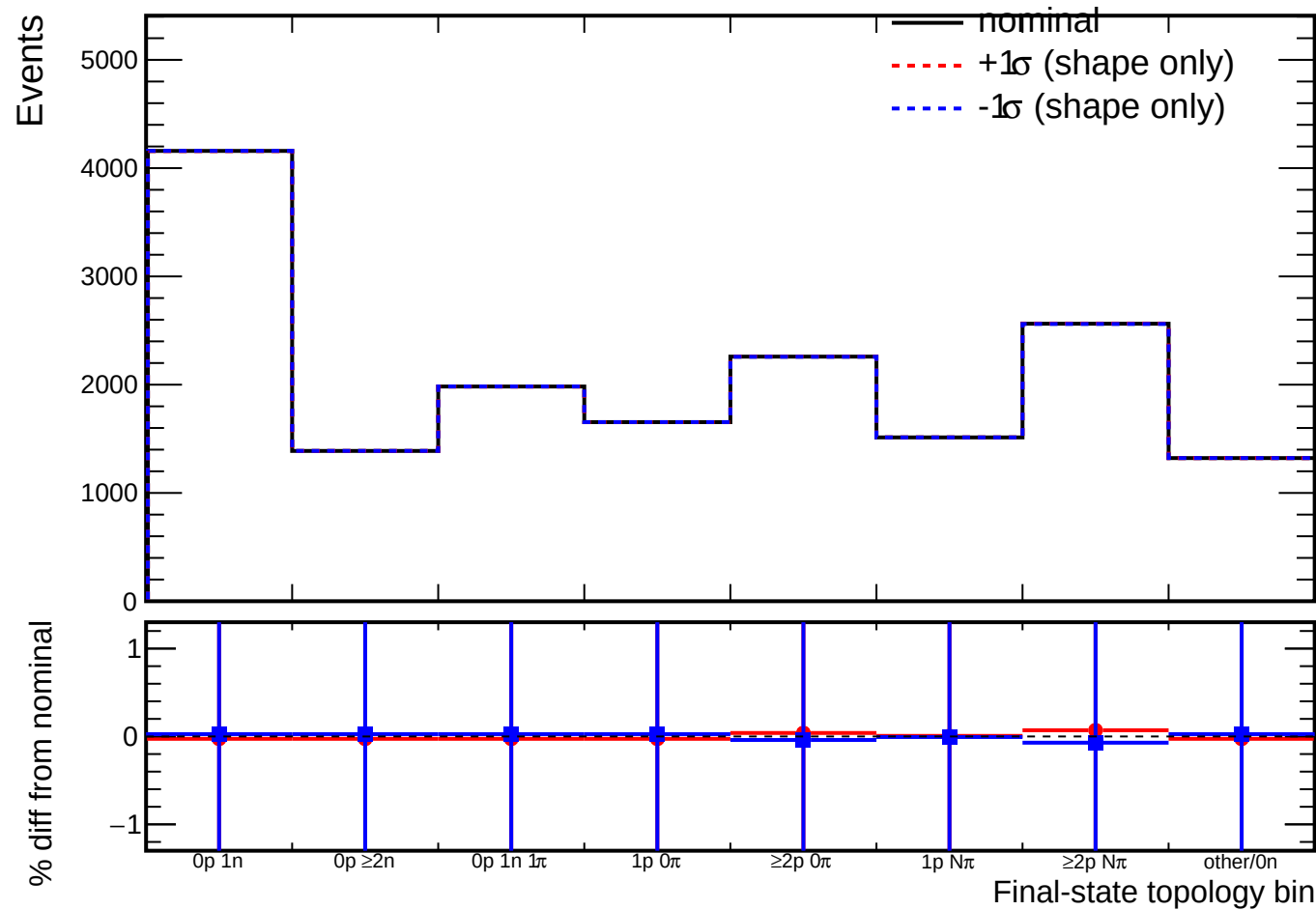
RHC [DIS]: CV2uBY (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=+0.0015$ $\Delta\langle x\rangle_{-1\sigma}=-0.0015$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.06% / -0.06% $\Delta\sigma^2$: +0.0018 / -0.0018



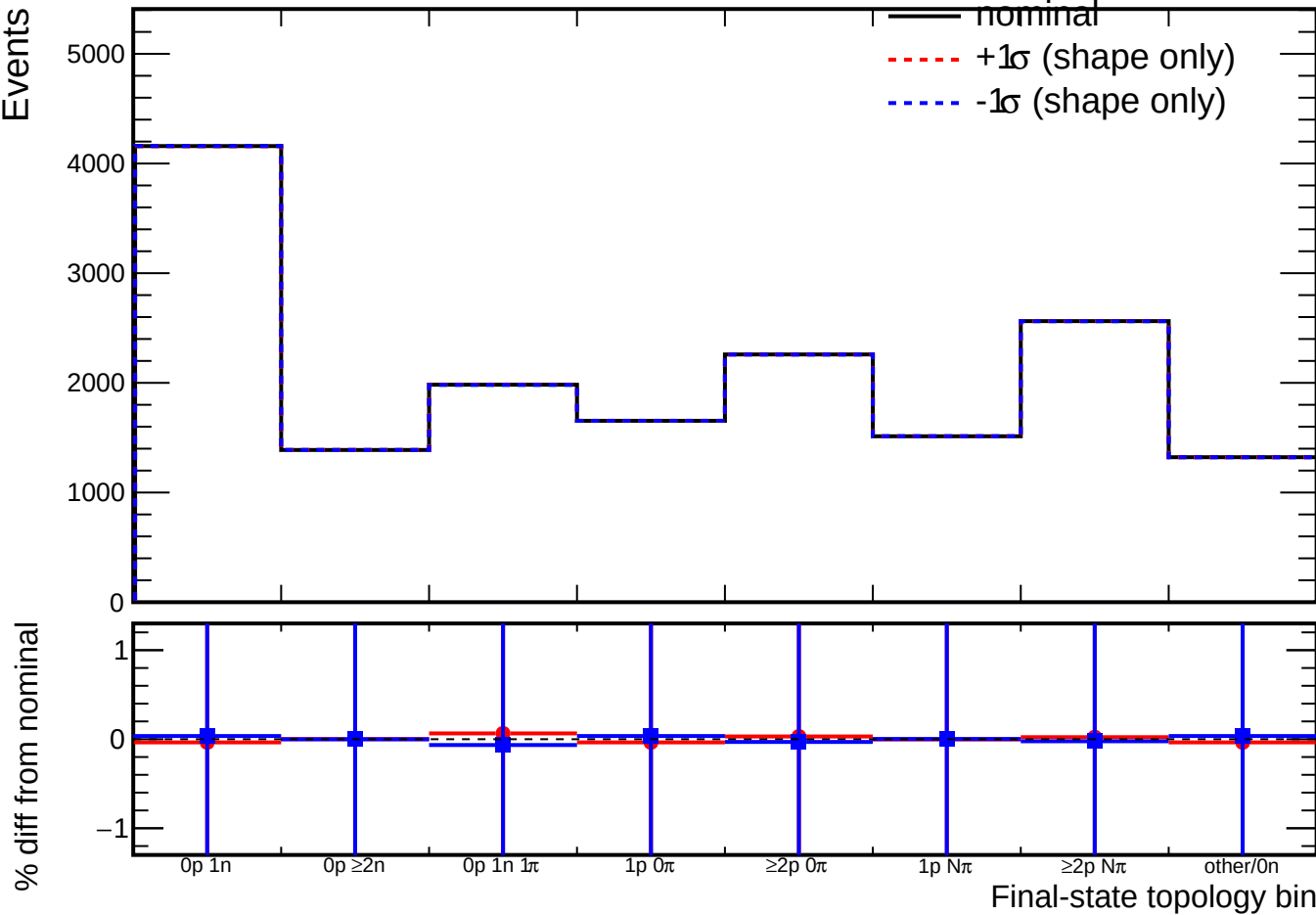
RHC [DIS]: DISvNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0003$ $\Delta\langle x\rangle_{-1\sigma}=-0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.01$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.04% / -0.04% $\Delta\sigma^2$: -0.0008 / +0.0008



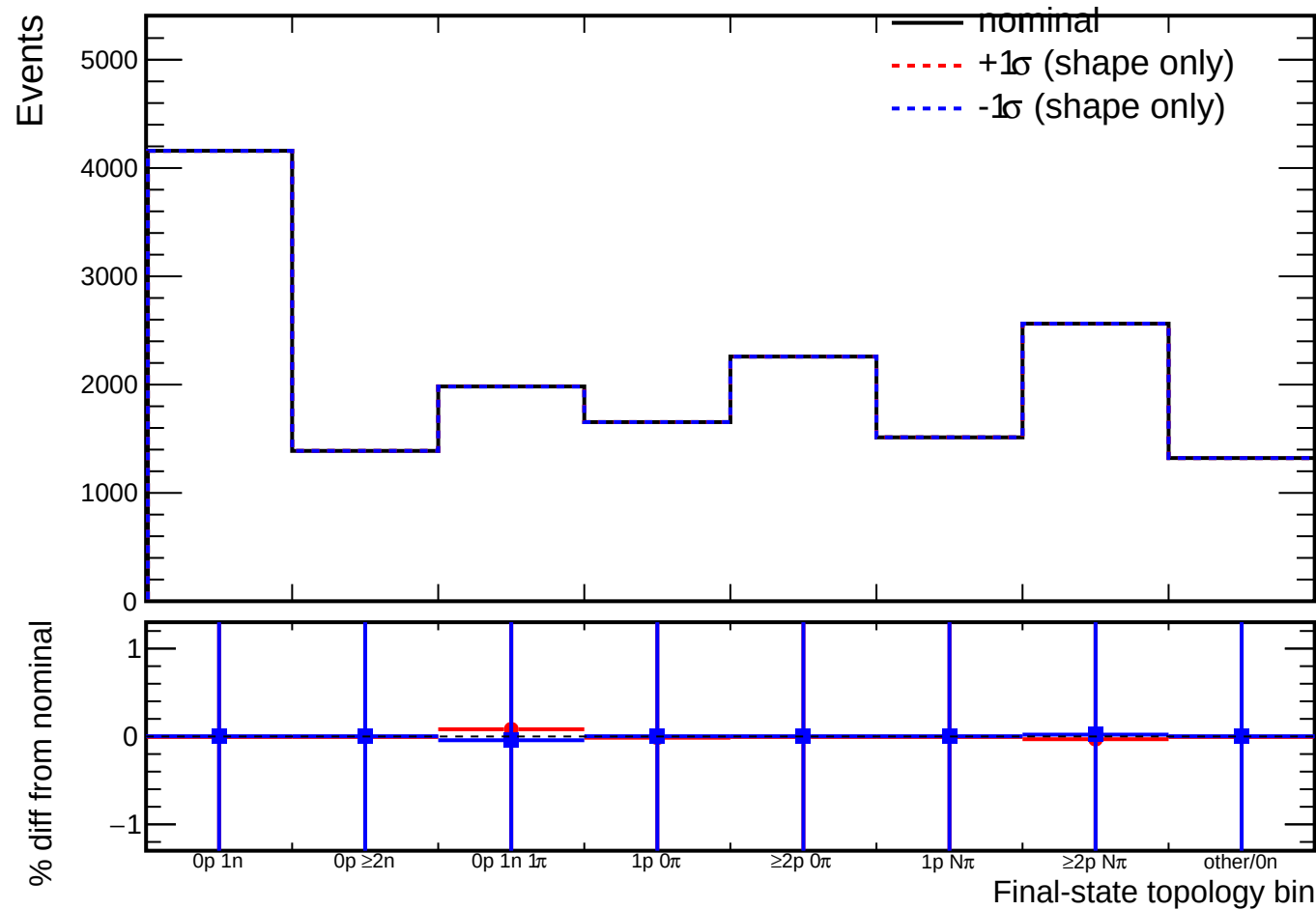
RHC [DIS]: DISvpCC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0006$ $\Delta\langle x\rangle_{-1\sigma}=-0.0006$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.03% / -0.03% $\Delta\sigma^2$: -0.0001 / +0.0001



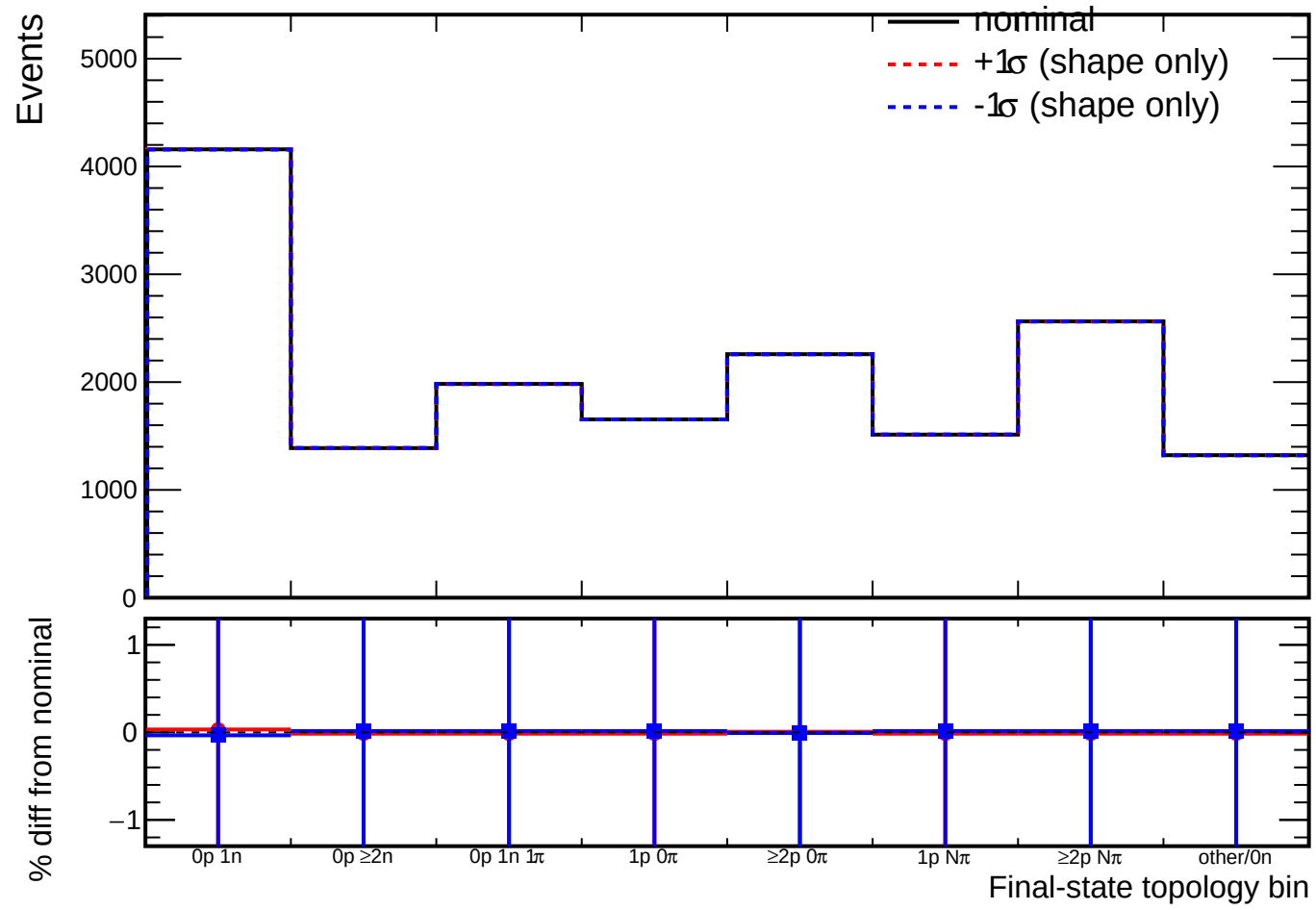
RHC [DIS]: DISvbnNC1pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0002$ $\Delta\langle x\rangle_{-1\sigma}=-0.0002$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.04% / -0.04% $\Delta\sigma^2$: -0.0008 / +0.0008



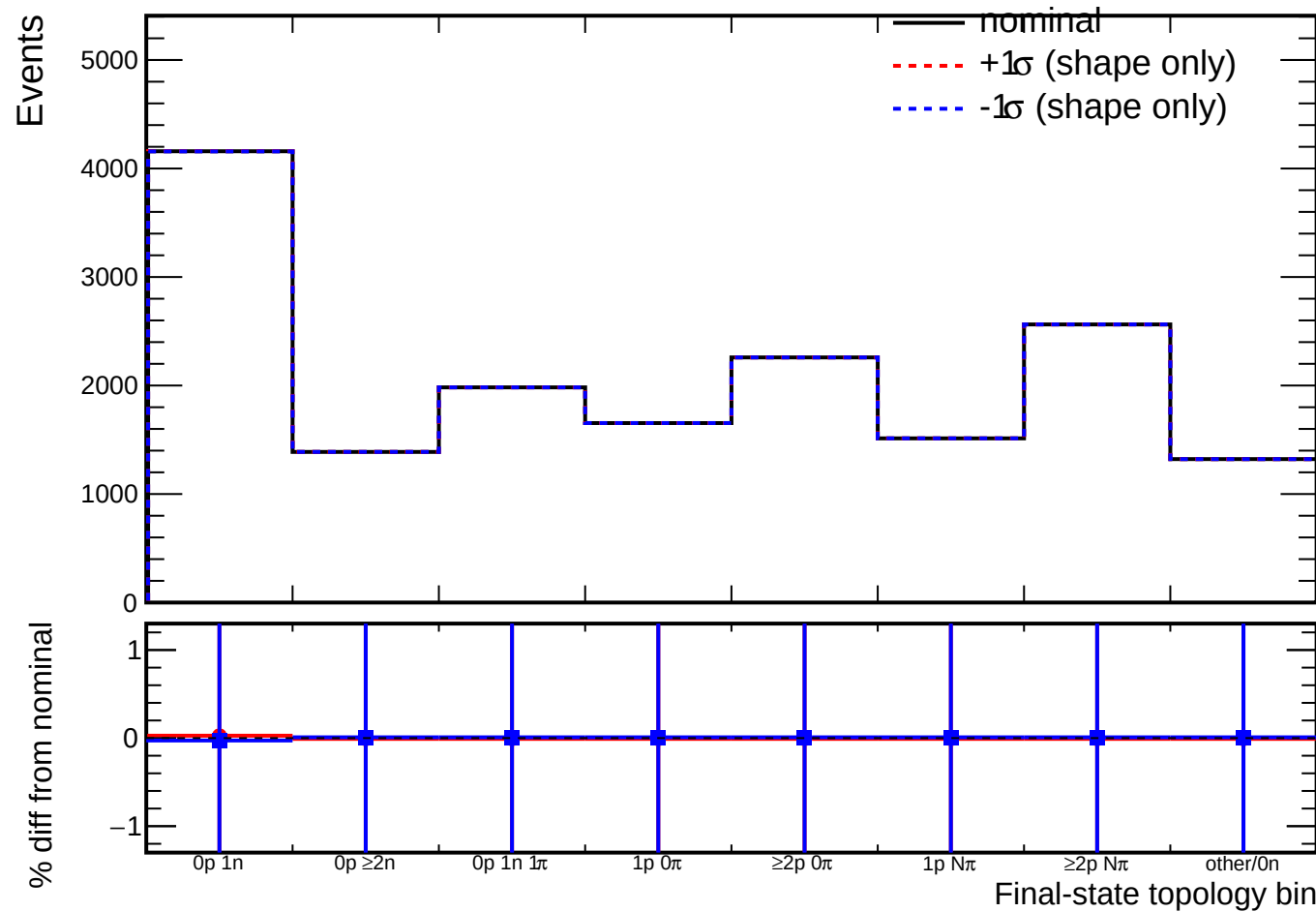
RHC [DIS]: AGKYpT1pi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0002$ $\Delta\langle x\rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / -0.00% $\Delta\sigma^2$: -0.0005 / +0.0003



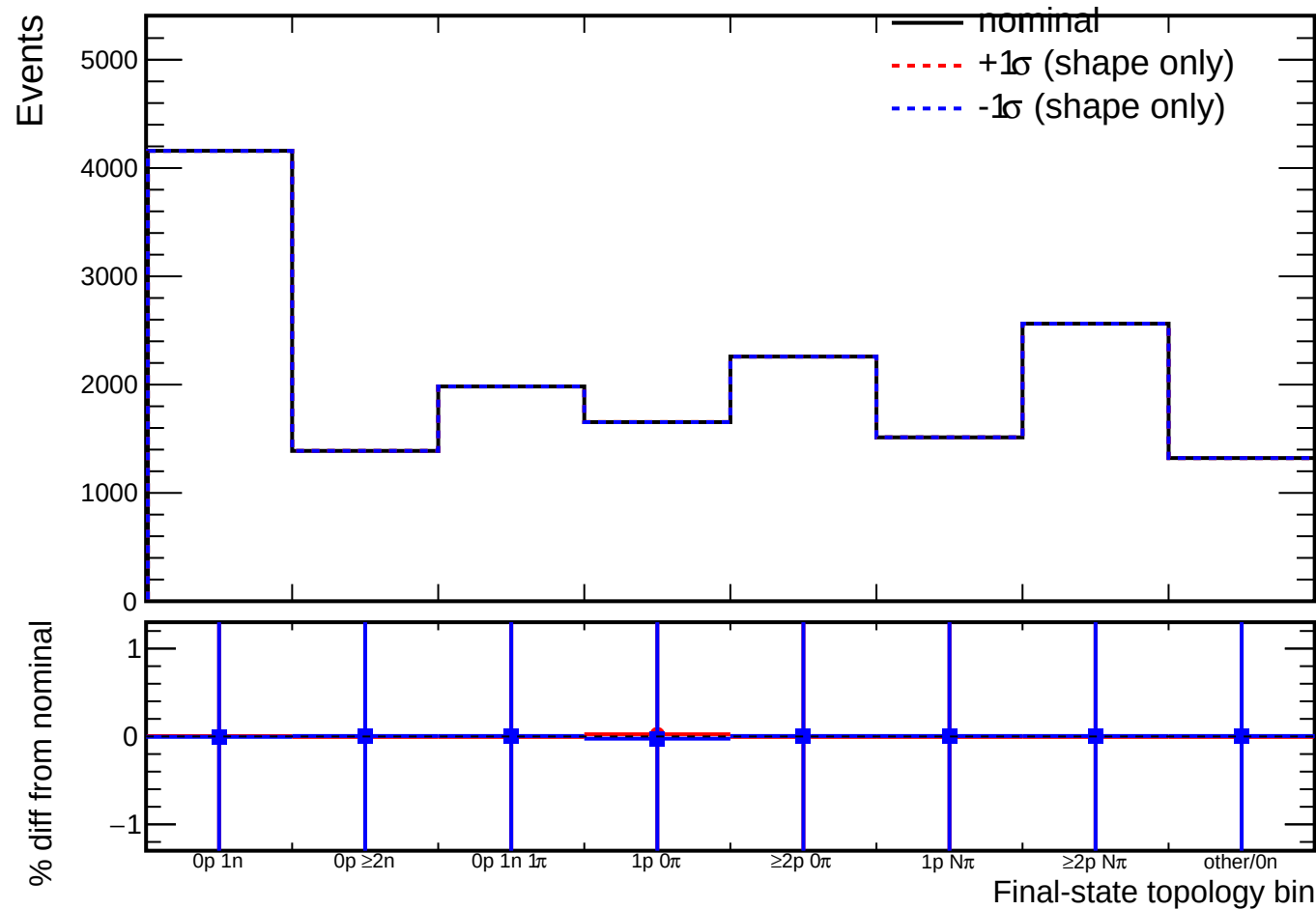
RHC [DIS]: DISvbarpCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0003$ $\Delta\langle x\rangle_{-1\sigma}=+0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / -0.01% $\Delta\sigma^2$: +0.0003 / -0.0003



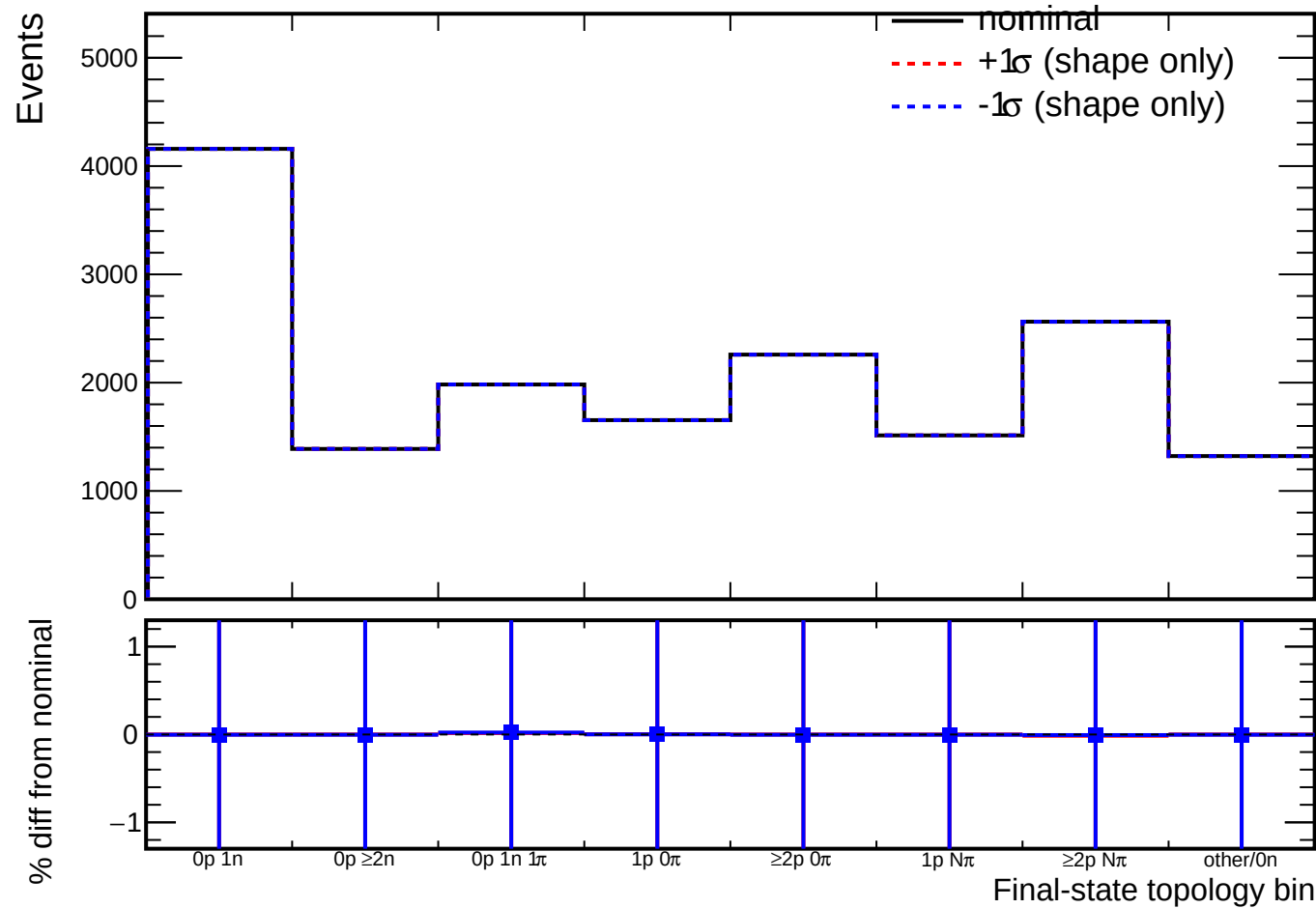
RHC [DIS]: DISvNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0003$ $\Delta\langle x\rangle_{-1\sigma}=+0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / -0.01% $\Delta\sigma^2$: +0.0003 / -0.0003



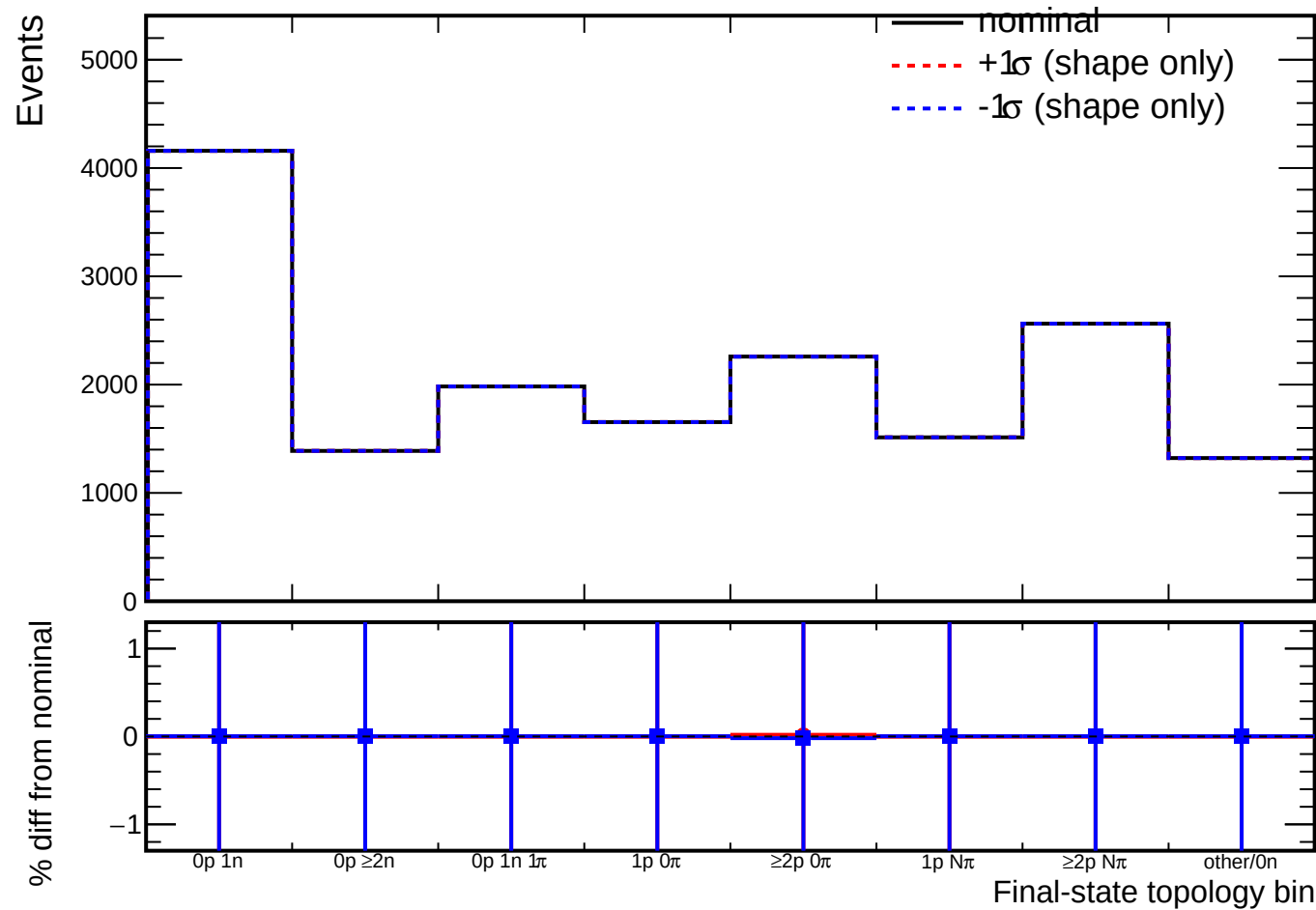
RHC [DIS]: DISvpNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0001$ $\Delta\langle x\rangle_{-1\sigma}=+0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.01% / -0.01% $\Delta\sigma^2$: -0.0001 / +0.0001



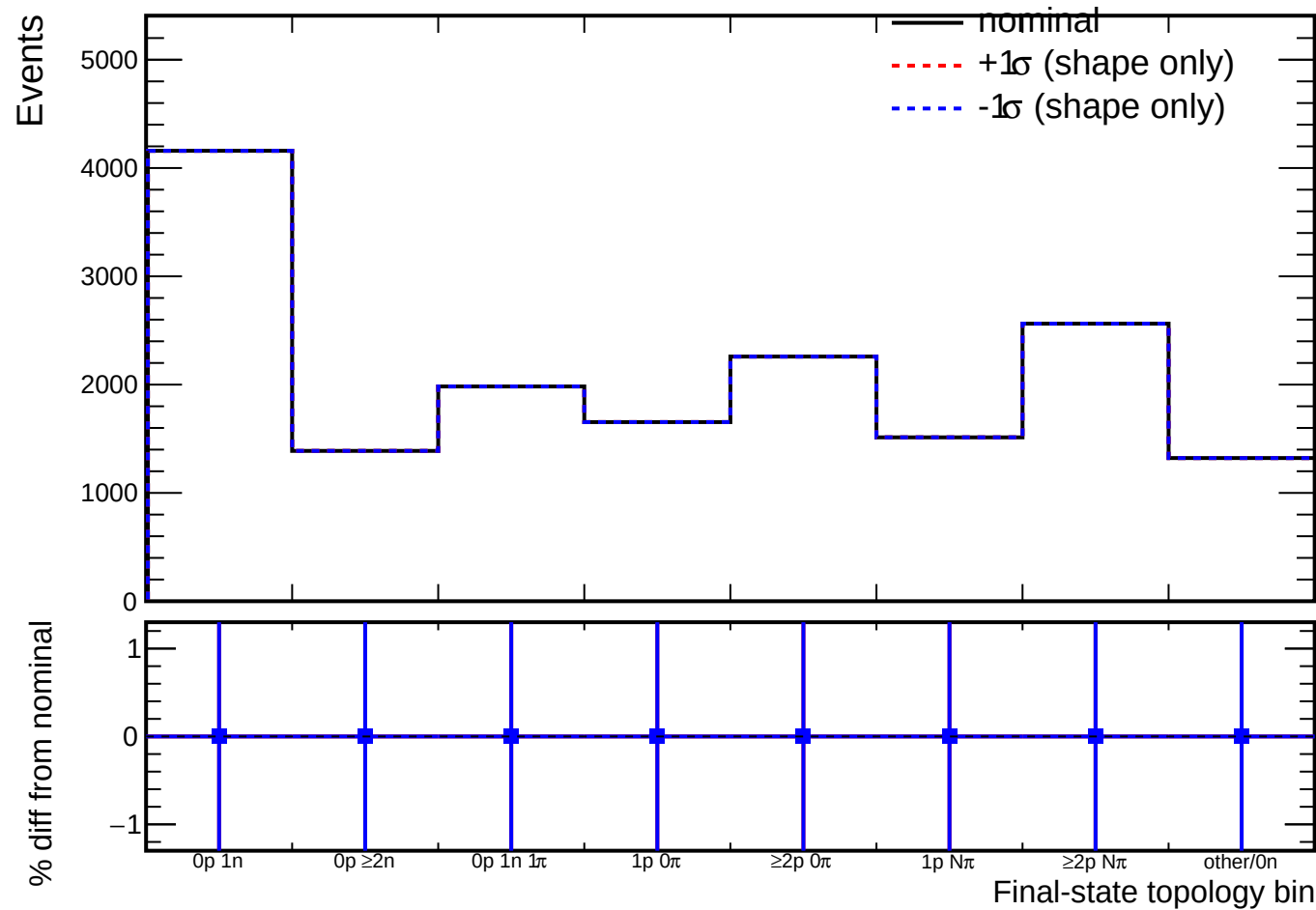
RHC [DIS]: AGKYxF1pi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x \rangle_{+1\sigma}=-0.0001$ $\Delta\langle x \rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -0.00% / +0.00% $\Delta\sigma^2$: -0.0002 / -0.0002



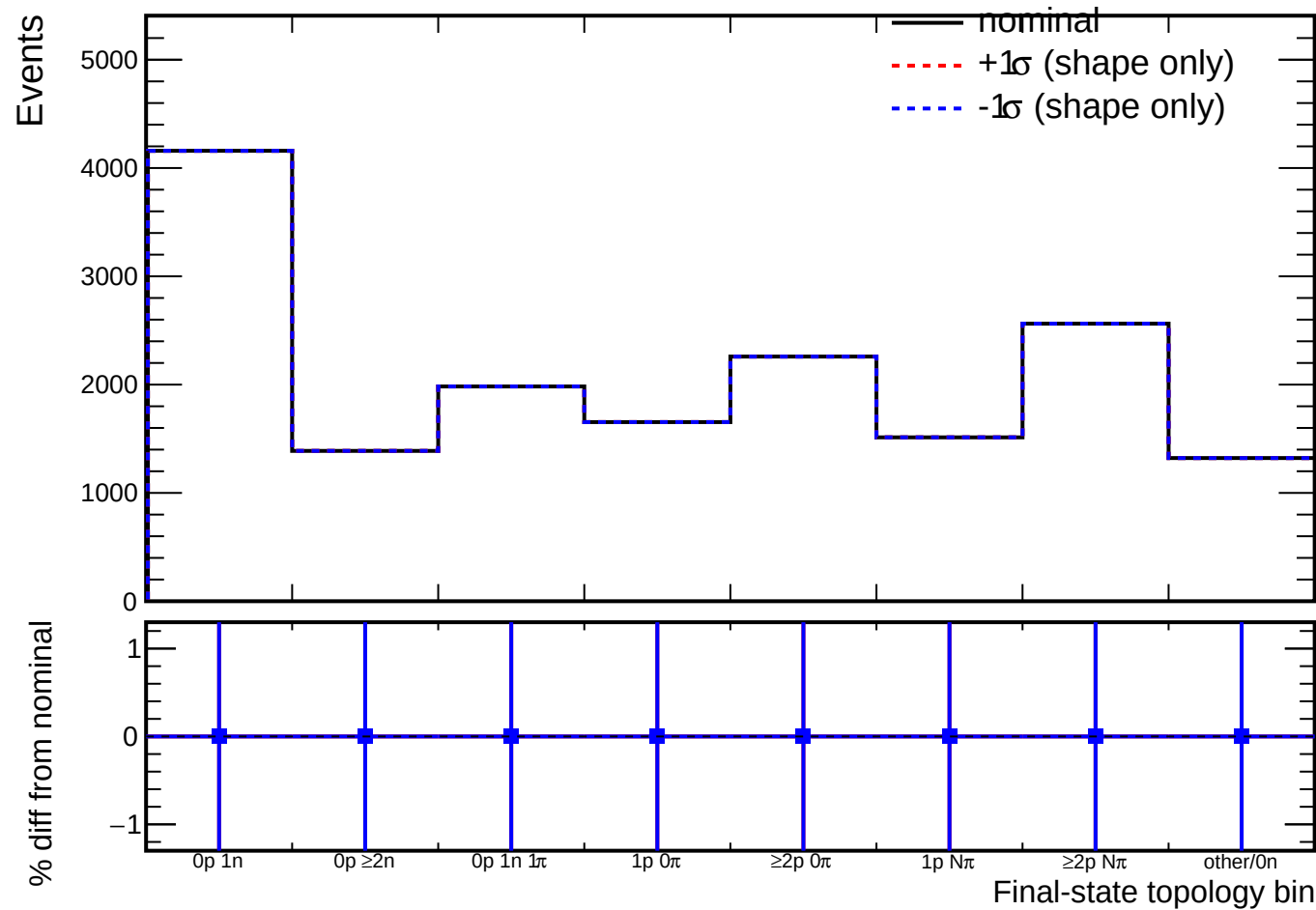
RHC [DIS]: DISvnCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / -0.00% $\Delta\sigma^2$: -0.0001 / +0.0001



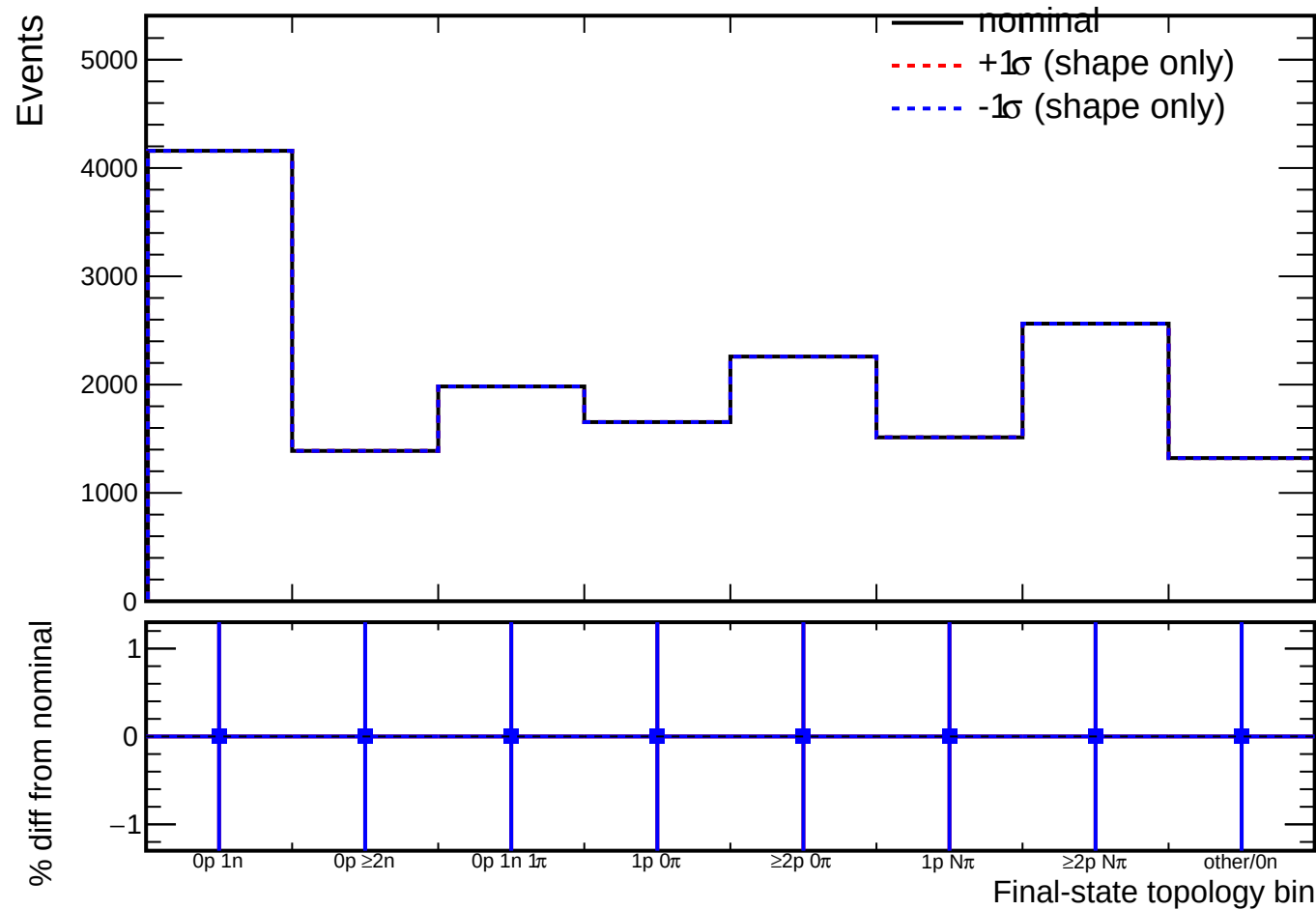
RHC [DIS]: DISvpCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=+0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



RHC [DIS]: DISvbarpNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



RHC [DIS]: DISvbnNC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000



RHC [DIS]: DISvbnCC0pi_2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0000$ $\Delta\langle x\rangle_{-1\sigma}=-0.0000$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: +0.00% / +0.00% $\Delta\sigma^2$: +0.0000 / +0.0000

